

CORPORATE GOVERNANCE



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Book: Corporate Governance

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This text was compiled on 07/05/2026

TABLE OF CONTENTS

Licensing

1: Ethical Leadership

- 1.1: Integrating the Statement of Values into Course Syllabi
- 1.2: Theory Building Activities - Mountain Terrorist Exercise
- 1.3: Ethics of Teamwork
- 1.4: Moral Exemplars in Business and Professional Ethics
- 1.5: Theory-Building Activities - Virtue Ethics
- 1.6: Ethical Leadership Using "Incident at Morales"

2: Ethical Leadership -Making

- 2.1: Theory-Building Activities - Rights
- 2.2: Three Frameworks for Ethical Problem-Solving in Business and the Professions
- 2.3: Toysmart Case Exercises - Student Module

3: Social Responsibility

- 3.1: Socio-Technical Systems in Professional Decision Making
- 3.2: Ethics and Laptops - Identifying Social Responsibility Issues in Puerto Rico

4: Changing Organizational

- 4.1: Pirate Code for Engineering Ethics
- 4.2: Developing Ethics Codes and Statements of Values
- 4.3: Gray Matters for the Hughes Aircraft Case

5: Business Ethics Bowl

- 5.1: Ethics Bowl Rules and Procedures
- 5.2: Practical and Professional Ethics Bowl Activity - Follow-Up In-Depth Case Analysis
- 5.3: Ethics Bowl - Cases and Score Sheets
- 5.4: EAC Toolkit - UPRM Ethics Bowl - IIT Summer Institute Follow-up

6: Course Procedures

- 6.1: Rubrics for Exams and Group Projects in Ethics
- 6.2: Realizing Responsibility Through Class Participation

Index

Glossary

Detailed Licensing

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Ethical Leadership

- 1.1: Integrating the Statement of Values into Course Syllabi
- 1.2: Theory Building Activities - Mountain Terrorist Exercise
- 1.3: Ethics of Teamwork
- 1.4: Moral Exemplars in Business and Professional Ethics
- 1.5: Theory-Building Activities - Virtue Ethics
- 1.6: Ethical Leadership Using "Incident at Morales"

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1.1: Integrating the Statement of Values into Course Syllabi

1.1.1 Introduction

On May 11, 2007, the College of Business Administration approved a Statement of Values designed to serve as a guideline for the moral development of this community. But this Statement of Values is as much a process as a product. Just as Johnson and Johnson sought to promote the moral development of its corporate community by "challenging" its corporate credo, ADEM also seeks to promote its moral development by challenging, interpreting, and realizing its Statement of Values.

Your task in this module is to read the appended Statement of Values (SOV) and provide two responses. First, can you think of any problems that have arisen in the past that could have been avoided if this SOV had been adopted and implemented? In other words, assess how effective you think the SOV is as a means for preventing moral harm. Second, challenge the SOV: (1) Are there any interpretation problems you see that would lead to misunderstanding and improper use? And (2) what SOV gaps do you see, that is, what ethical issues remain that are not fully treated under the SOV?

Carrying out these tasks will help you identify creative ways to integrate moral values into your academic endeavors in this class and in other activities this semester. The goal of the SOV is to set the ADEM community on a course of continual improvement. These respective exercises will help start the process.

1.1.2 What you need to know...

Include information that you expect your students to study and learn in this module as well as information that will help them carry out the module activities.

Statement of Values: http://legacy.cnx.org/content/m14850...t/SOV_Copy.doc

Figure 1.2: Clicking on this figure will open the UPRM College of Business Administration's Statement of Value. The SOV can also be accessed by clicking on the links above provided with this module.

SOV Preamble:

As a result of an ongoing process of reflection and assessment, the College of Business Administration (its students, faculty, staff, and administrators) arms its commitment and loyalty to the following values: justice and fairness, responsibility, respect, trust, and integrity. This statement sets forth these values in order to educate and inspire as well as to promote dialogue and continual improvement. In particular, these values serve to describe this community's identity and express their aspirations. It is meant to complement existing laws, regulations, professional standards, and codes of ethics by enhancing the pursuit of excellence consistent with the College's Vision and Mission. In all of its activities, the College of Business Administration will:

sov values

1. Justice/Fairness: Be impartial, objective and refrain from discrimination or preferential treatment in the administration of rules and policies and in its dealings with students, faculty, staff, administration, and other stakeholders.
2. Responsibility: Recognize and fulfill its obligations to its constituents by caring for their essential interests, by honoring its commitments, and by balancing and integrating conflicting interests. As responsible agents, the faculty, employees, and students of the College of Business Administration are committed to the pursuit of excellence, devotion to the community's welfare, and professionalism.
3. Respect: Acknowledge the inherent dignity present in its diverse constituents by recognizing and respecting their fundamental rights. These include rights to property, privacy, the free exchange of ideas, academic freedom, due process, and meaningful participation in decision making and policy formation.
4. Trust: Recognize that trust solidifies communities by creating an environment where each can expect ethically justifiable behavior from all others. While trust is tolerant of and even thrives in an environment of diversity, it also must operate within the parameters set by established personal and community standards.
5. Integrity: Promote integrity as characterized by sincerity, honesty, authenticity, and the pursuit of excellence. Integrity shall permeate and color all its decisions, actions and expressions. It is most clearly exhibited in intellectual and personal honesty in learning, teaching, mentoring and research.

Compliance Strategy:

- The traditionally most prevalent method for interpreting codes of ethics and statements of values is the compliance method. This method sets forth minimal standards and implements incentives for meeting these standards. It is based on three interrelated components::
- Rules: Compliance strategies are centered around strict codes of ethics composed of rules that set forth minimum thresholds of acceptable behavior. The use of rules to structure employee action does run into problems due to the gap between rule and application, the appearance of novel situations, and the impression that it gives to employees that obedience is based on conformity to authority.
- Monitoring: The second component consists of monitoring activities designed to ensure that employees are conforming to rules and to identify instances of non-compliance. Monitoring is certainly effective but it requires that the organization expend time, money, and energy. Monitoring also places stress upon employees in that they are aware of constantly being watched. Those under observation tend either to rebel or to automatically adopt behaviors they believe those doing the monitoring want. This considerably dampens creativity, legitimate criticism, and innovation.
- Disciplining Misconduct: The last key component to a compliance strategy is punishment. Punishment can be effective especially when establishing and enforcing conduct that remains above the criminal level. But reliance on punishment for control tends to impose solidarity on an organization rather than elicit it. Employees conform because they fear sanction. Organizations based on this fear are never really free to pursue excellence.

Values Orientation:

- The SOV can also be read as the identification and [armation] of a community's aspirations. By taking on this values orientation, the SOV replaces the reactive compliance perspective with a proactive stance oriented toward excellence. The emphasis here is on how the community can support its members by identifying best practices toward realizing these aspirations and especially how it can provide support to those who fall short. This values-based orientation is built upon the following three components
- Development of Shared Values: Using a process similar to the one described above, a community develops a Statement of Shared Values. These provide guidelines that replace the hard and fast rules of a compliance code. Statements in values-oriented codes play a different logical function than statements in compliance codes. "Principles of Professional/Organizational Conduct" in compliance codes specify circumstances of compliance: time, agent, place, purpose, manner, etc. These circumstances provide sufficient content to allow principles of professional conduct to function as rules that can be violated. This gives them "teeth," that is, makes it possible to enforce them by sanctions and punishments. "Ideals of the Profession/Organization state a community's shared aspirations. They set forth levels of behavior well beyond the minimum. Because they chart out directions for continuous improvement, Ideals of the Profession/Organization profess a community's commitment to excellence rather than the moral minimum.
- Support for Employees: Since Statements of Values can set forth excellences or aspirations, the role of the community changes from monitoring and punishing to helping community members realize key values in their day to day activities. In other words, the role of the organization changes from punitive to supportive.
- Ethical Aspirations: In summary, values orientations can be interpreted as setting forth higher standards for behavior. Going well beyond the moral and legal minimum, these values when clarified in a community's statement of values serve as aspirations. A values orientation requires that a community design strategies that reinterpret and realize basic values as excellences. Hence, it is most compatible with a virtue orientation and virtue ethical theory.

1.1.3 What you will do...

Suppose the SOV has been adopted and implemented for several years now. Exercise your moral imagination and envision problems that the pursuit of these excellences would have avoided.

Exercises

Question 1: What kind of moral harms could it have prevented had it been in effect?

Question 2: Do the adaptation and implementation of the SOV promise to make us (ADEM stakeholders) a better community?

- If so, how?
- If not, what are its weaknesses?

- Nota Bene: If you feel that the adoption of the SOV will not make us a better community, feel free to state this and then explain your position.

Challenging the Statement of Values as in successful corporate compliance and values programs, the following exercise encourages you to challenge the SOV by identifying interpretation problems and SOV gaps.

Exercises

Question 1: Can you anticipate any interpretation problems that may arise with the adaptation and implementation of the SOV? How should these be addressed?

Question 2: Can you identify important moral problems that are not covered or anticipated by the Statement of Values? How could the SOV be modified to cover these problems and fill the gaps?

1.1.4 What did you learn?

Reflecting on what you have done is an absolutely essential part of the learning process. In this section of the module, the class will be divided into small groups, and each group member begins by presenting his or her responses to the above four questions. Explaining your responses to others in terms that they understand and with reasons that you share with them helps you to see your own views in a different, more comprehensive way. Listening to what others say helps to integrate new information and perspectives into your thinking on an issue. In other words, it expands and deepens your own position.

After you explain your responses to the other members of your group, discuss how the SOV can be embedded in everyday academic activities. How can SOV values be realized in...

- group work
- course syllabi
- College Administrative procedures such as complaint processing and matriculation
- class attendance
- Choose one of these issues for discussion. If you have time, go to another.

Sample issues for discussion

- How can teachers realize justice in their evaluation procedures?
- How can students participate responsibly in their classes?
- How does cheating affect relations of trust between students, especially between those who cheat and those who don't?
- How can the practice of setting and holding office hours lead to or undermine relations of respect between teachers and students?
- How can academic integrity be interpreted as an aspiration? What would constitute an academic integrity compliance program?
- Does the SOV pertain to recent changes in the class schedule at UPRM? Which values pertain and why? Is an example of an SOV gap?
- Again, choose one of these for group discussion. If you have time, go to another.

Meta-Discussion

- A meta-discussion is a discussion about a discussion. Reflecting on the discussion your group has just had...
- Did you agree on most issues? Why do you think you all agreed? What did you do to prevent groupthink, i.e., a group atmosphere where disagreement is covered over by various methods or means.
- Did you disagree?
- How did you respond to disagreement? For example, did you try to impose consensus?
- State as clearly as possible the different positions held by group members and how they differed

1.1.5 Appendix

SOV Module Word 97 Version: http://legacy.cnx.org/content/m14850...ng_SOV_W97.doc

Figure 1.3: This module is also available in a Word 97 handout. Clicking on this figure will download the le including handouts for each of the discussion activities outlined above.

This student module was carried out in classes at UPRM in Business Ethics 8/10/07 and 8/13/07. All three sections including the Meta-Discussion were completed by close to 60 students. An informal summary of the students' responses and the issues they raised can be found in the corresponding Instructor Module which is under construction and will be published shortly.

1.1.6 My College's Values and Me: An exercise for ESOR 4019

The exercise, "My College's Values and Me," developed by Marta Colón de Toro, provides an excellent instrument for disseminating the Statement of Values to students, collecting reactions and feedback from them to incorporate into future developments, and to start reflecting on how the SOV can be realized in the classroom and the ADEM community at large. The following media file contains the classroom exercise carried out in the fall semester, 2007, at UPRM.

My College's Values and Me: legacy.cnx.org/content/m14850/latest/MyCollegeValuesandMe.doc

Figure 1.4: This exercise has been developed by Marta Colón de Toro for the integration of the SOV into the class, "Wages and Salary Administration." A revised version will be substituted shortly.

1.1.7 EAC Toolkit Project

1.1.7.1 This module is a WORK-IN-PROGRESS; the author(s) may update the content as needed. Others are welcome to use this module or create a new derived module. You can COLLABORATE to improve this module by providing suggestions and/or feedback on your experiences with this module.

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1.1.7.2 Funded by the National Science Foundation: "Collaborative Development of Ethics Across the Curriculum Resources and Sharing of Best Practices," NSF-SES-0551779

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1.2: Theory Building Activities - Mountain Terrorist Exercise

1.2.1 Module Introduction

This module poses an ethical dilemma, that is, a forced choice between two bad alternatives. Your job is to read the scenario and choose between the two horns of the dilemma. You will make your choice and then justify it in the first activity. In the second activity, you will discuss your choice with others. Here, the objective is to reach consensus on a course of action or describe the point at which your group's progress toward consensus stopped. The Mountain Terrorist Exercise almost always generates lively discussion and helps us to reflect on our moral beliefs. Don't expect to reach an agreement with your fellow classmates quickly or effortlessly. (If you do, then your instructor will find ways of throwing a monkey wrench into the whole process.) What is more important here is that we learn how to state our positions clearly, how to listen to others, how to justify our positions, and how to assess the justifications offered by others. In other words, we will all have a chance to practice the virtue of reasonableness. And we will learn reasonableness not when it's easy (as it is when we agree) but when it becomes difficult (as it is when we disagree).

The second half of this module requires that you reflect carefully on your moral reasoning and that of your classmates. The Mountain Terrorist Exercise triggers the different moral schemas that make up our psychological capacity for moral judgment. Choosing one horn of the dilemma means that you tend to favor one kind of schema while choosing the other horn generally indicates that you favor another. The dominant moral theories that we will study this semester provide detailed articulations and justifications of these moral schemas. Reflecting on your choice, the reasons for your choice, and how your choice differs from that of your classmates will help you get started on the path of studying and effectively utilizing moral theory.

The following scenario comes originally from the philosopher, Bernard Williams. It is also presented in introductory ethics textbooks (such as Geoffrey Thomas' *An Introduction to Ethics*). The first time this module's author became aware of its use in the classroom was in a workshop on Agriculture Ethics led by Paul Thompson, then of Texas A&M University, in 1992.

1.2.2 Moral Theories Highlighted

1. Utilitarianism: the moral value of an action lies in its consequences or results
2. Deontology: the moral value of an action lies, not in its consequences, but in the formal characteristics of the action itself.
3. Virtue Ethics: Actions sort themselves out into virtuous or vicious actions. Virtuous actions stem from a virtuous character while vicious actions stem from a vicious or morally awed character. Who we are is revealed through what we do.

1.2.3 Mountain Terrorist Scenario

You are in a remote mountain village. A group of terrorists has lined up 20 people from the village; they plan on shooting them for collaborating with the enemy. Since you are not from the village, you will not be killed. Taking advantage of your position, you plead with the terrorists not to carry out their plan. Finally, you convince the leader that it is not necessary to kill all 20. He takes a gun, empties it of all its bullets except one, and then hands it to you. He has decided to kill only one villager to set an example to the rest. As an honored guest and outsider, you will decide who will be killed, and you will carry out the deed. The terrorists conclude with a warning; if you refuse to kill the villager, then they will revert back to the original plan of killing all 20. And if you try any funny business, they will kill the 20 villagers and then kill you. What should you do?

your options

1. Take the gun, select a villager, and kill him or her.
2. Refuse the terrorists' offer and walk away from the situation.

activity 1

In a short essay of one to two pages, describe what you would do if you were in the position of the tourist. Then justify your choice.

activity 2

Bring your essay to class. You will be divided into small groups. Present your choice and justification to the others in your group. Then listen to their choices and justifications. Try to reach a group consensus on choice and justification. (You will be given 10-15 minutes.) If you succeed, present your results to the rest of the class. If you fail, present to the class the disagreement that blocked consensus and what you did (within the time limit) to overcome it.

1.2.4 Taxonomy of Ethical Approaches

There are many ethical approaches that can be used in decision making. The Mountain Terrorist Exercise is based on an artificial scenario designed to separate these theoretical approaches along the lines of the different "horns" of a dilemma. Utilitarians tend to choose to shoot a villager "in order to save 19." In other words, they focus their analysis on the consequences of an action alternative and choose the one that produces the least harm. Deontologists generally elect to walk away from the situation. This is because they judge an action on the basis of its formal characteristics. A deontologist might argue that killing the villager violates natural law or cannot be made into a law or rule that consistently applies to everybody. A deontologist might say something like, "What right do I have to take another person's life?" A virtue ethicists might try to imagine how a person with the virtue of courage or integrity would act in this situation. (Williams claims that choosing to kill the villager, a duty under utilitarianism, would undermine the integrity of a person who abhorred killing.)

Table Connecting Theory to Domain

1. Row 1: Utilitarianism concerns itself with consequences. It claims that the moral value of an action is "colored" by its results. The harm test, which asks us to choose the least harmful alternative, encapsulates or summarizes this theoretical approach. The basic principle of utilitarianism is the principle of utility: choose that action that produces the greatest good for the greatest number. Utilitarians Available for free at Connexions 9 would shoot a villager in order to save 19. But Utilitarianism, like other forms of consequentialism, has prediction challenges. What are the short-, middle-, and long-term consequences of an action? These become harder to determine the further we are from the present.

2. Row 2: Npn-consequentialism turns away from consequences to focus on the formal characteristics of an action. (For example, Kant says the good action is one that does duty for duty's sake.) Deontology, a kind of non-consequentialism, helps us to identify and justify rights along with their correlative duties. The reversibility test summarizes deontology by asking the question, "Does your action still work if you switch (=reverse) roles with those on the receiving end? Deontology has two formulations of its fundamental principle. The Categorical Imperative exhorts us to act only on that maxim which can be converted into a universal law. The Formula of the End proscribes that we "treat others always as ends, never merely as means." The rights that represent special cases of treating people as ends and not merely as means include (a) informed consent, (b) privacy, (c) due process, (d) property, (e) free speech, and (f) conscientious objection. The deontologist would choose not to kill a villager because the act of killing is formally wrong.

3. Row 3: Virtue ethics turns away from the action and focuses on the agent, the person performing the action. The word, "Virtue," refers to different sets of skills and habits cultivated by agents. These skills and habits, consistently and widely performed, support, sustain, and advance different occupational, social, and professional practices. (See MacIntyre, *After Virtue*, and Solomon, *Ethics and Excellence*, for more on the relation of virtues to practices.) The public identification test summarizes this approach: an action is morally acceptable if it is one with which I would willingly be publicly associated given my moral convictions. Individual virtues that we will use this semester include integrity, justice, responsibility, reasonableness, honesty, trustworthiness, and loyalty.

- These different approaches are meant to work together. Each gives us insight into different dimensions of the problematic situation. Utilitarianism and deontology both focus on the action. Utilitarianism uses consequences to evaluate the action while deontology evaluates an action in terms of its underlying motive and its formal characteristics.

- Virtue ethics turns away from the action to focus on the agent. It asks us to determine what the action says about the character or person of the agent. If the action is irresponsible, then the agent is irresponsible. Virtue ethics can be implemented by projecting a moral exemplar into the situation. You might ask, "What would so-and-so do in this situation?" if this person were your mentor, a person you admire, or a moral exemplar. Or you might examine virtues that are realized through your action. For example, Williams says that taking the life of a villager might seriously disrupt or corrupt your integrity.

- The capability approach takes a still different focus on the situation by having us bring into view those factors in the situation which could empower or impede the expression of human capabilities like thought, imagination, movement, health, and life.

Covering All the Bases

Theory Category	Ethical Approach	Ethics Test	Basic Principles	Action in MT Scenario
Consequentialism	Utilitarianism	Harm Test	Principle of Utility: greatest good for greatest number	Shoot one villager to save nineteen
Non-consequentialism	Deontology: right theory or duty theory	Reversibility Test: view action from receiving end	Categorical Imperative: act on maxim which can be universal law; Formula of End: treat persons as ends, not merely as means	Do not take a gun; leave the village
Character-Based	Virtue Ethics	Publicity Test	Virtue is the means between extremes of excess and defect	Do the honorable thing
Human Functioning	Capability Approach	Check if action expands or contracts substantive freedoms	Substantive freedoms composing a life of dignity; beings and doings essential to eudaimonia	Choose that action that expands freedom and secures dignity

Table 1.1: Table 1

1.2.5 Comments on the Relation Between Ethical Approaches

The Mountain Terrorist Exercise has, in the past, given students the erroneous idea that ethical approaches are necessarily opposed to one another. As one student put it, "If deontology tells us to walk away from the village, then utilitarianism must tell us to stay and kill a villager because deontology and utilitarianism, as different and opposed theories, always reach different and opposed conclusions on the actions they recommend." The Mountain Terrorist dilemma was specially constructed by Bernard Williams to produce a situation that ordered only a limited number of alternatives. He then tied these alternatives to different ethical approaches to separate them precisely because in most real-world situations they are not so readily distinguishable. Later this semester, we will turn from these philosophical puzzles to real-world cases where ethical approaches function in a very different and mostly complementary way. As we will see, ethical approaches, for the most part, converge on the same solutions. For this reason, this module concludes with 3 meta-tests. When approaches converge on a solution, this strengthens the solution's moral validity. When approaches diverge on a solution, this weakens their moral validity. A third meta-test tells us to avoid framing all ethical problems as dilemmas (=forced choices between undesirable alternatives) or what Carolyn Whitbeck calls "multiple-choice" problems. You will soon learn that effective moral problem solving requires moral imagination and moral creativity. We do not "find" solutions "out there" ready-made but design them to harmonize and realize ethical and practical values.

Meta-Tests:

- **Divergence Test:** When two ethical approaches differ on a given solution, then that difference counts against the strength of the solution. Solutions on which ethical theories diverge must be revised towards convergence.
- **Convergence Test:** Convergence represents a meta-test that attests to solution strength. Solutions on which different theoretical approaches converge are, by this fact, strengthened. Convergence demonstrates that a solution is strong, not just over one domain, but over multiple domains.
- **Avoid Framing a Problem as a Dilemma.** A dilemma is a no-win situation that orders only two alternatives of action both of which are equally bad. (A trilemma orders three bad alternatives, etc.) Dilemmas are better dissolved than solved. Reframe the dilemma into something that admits of more than two no-win alternatives. Dilemma framing (framing a situation as an ethical dilemma) discourages us from designing creative solutions that integrate the conflicting values that the dilemma poses as incompatible.

1.2.6 Module Wrap-Up

1. Reasonableness and the Mountain Terrorist Exercise. It may seem that this scenario is the last place where the virtue of reasonableness should prevail, but look back on how you responded to those of your classmates who chose differently in this exercise and who ordered arguments that you had not initially thought of. Did you "listen and respond thoughtfully" to them? Were you "open to new ideas" even if these challenged your own? Did you "give reasons for" your views, modifying and shaping them to respond to your classmates' arguments? Did you "acknowledge mistakes and misunderstandings" such as responding critically and personally to a classmate who put forth a different view? Finally, when you turned to working with your group, were you able to "compromise (without compromising personal integrity)"? If you did any or all of these things, then you practiced the virtue of reasonableness as characterized by Michael Pritchard in his book, *Reasonable Children: Moral Education and Moral Learning* (1996, University of Kansas Press, p. 11). Congratulate yourself on exercising reasonableness in an exercise designed to challenge this virtue. You passed the test.

2. Recognizing that we are already making ethical arguments. In the past, students have made the following arguments on this exercise: (a) I would take the gun and kill a villager in order to save nineteen; (b) I would walk away because I don't have the right to take another's life; (c) While walking away might appear cowardly it is the responsible thing to do because staying and killing a villager would make me complicit in the terrorists' project. As we discussed in class, these and other arguments make use of modes of thought captured by ethical theories or approaches. The first employs the consequentialist approach of utilitarianism while the second makes use of the principle of respect that forms the basis of our rights and duties. The third works through a conflict between two virtues, courage and responsibility. This relies on the virtue approach. One accomplishment of this exercise is to make you aware of the fact that you are already using ethical arguments, i.e., arguments that appeal to ethical theory. Learning about the theories behind these arguments will help you to make these arguments more effective.

3. Results from Muddy Point Exercises The Muddy Point Exercises you contributed kept coming back to two points. (a) Many of you pointed out that you needed more information to make a decision in this situation. For example, who were these terrorists, what causes were they fighting for, and were they correct in accusing the village of collaborating with the enemy? Your request for more information was quite appropriate. But many of the cases we will be studying this semester require decisions in the face of uncertainty and ignorance. These are unavoidable in some situations because of factors such as the cost and time of gathering more information. Moral imagination skillfully exercised can do a lot to compensate when all of the facts are not in. (b) Second, many of you felt overly constrained by the dilemma framing of the scenario. Those of you who entered the realm of "funny business" (anything beyond the two alternatives of killing the villager or walking away) took a big step toward effective moral problem-solving. By rejecting the dilemma framing of this scenario, you were trying to reframe the situation to allow for more and more ethically viable alternatives. Trying to negotiate with the Terrorists is a good example of reframing the scenario to admit of more ethical alternatives of action than killing or walking away.

4. Congratulations on completing your first ethics module! You have begun recognizing and practicing skills that will help you to tackle real-life ethical problems. (Notice that we are going to work with "problems", not "dilemmas".) We will now turn, in the next module, to look at those who managed to do good in the face of difficulty. Studying moral exemplars will provide the necessary corrective to the "no-win" Mountain Terrorist Exercise.

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1.3: Ethics of Teamwork

Module Introduction

Much of your future work will be organized around group or team activities. This module is designed to prepare you for this by getting you to reflect on ethical and practical problems that arise in small groups like work teams. Four issues, based on well-known ethical values, are especially important. How do groups achieve justice (in the distribution of work), responsibility (in specifying tasks, assigning blame, and awarding credit), reasonableness (ensuring participation, resolving conflict, and reaching consensus), and honesty (avoiding deception, corruption, and impropriety)? This module asks that you develop plans for realizing these moral values in your group work this semester. Furthermore, you are provided with a list of some of the more common pitfalls of group work and then asked to devise strategies for avoiding them. Finally, at the end of the semester, you will review your goals and strategies, reflect on your successes and problems, and carry out an overall assessment of the experience.

Module Activities

1. Groups are provided with key ethical values that they describe and seek to realize through group activity.
 2. Groups also study various obstacles that arise in collective activity: the Abilene Paradox, Groupthink, and Group Polarization.
 3. Groups prepare initial reports consisting of plans for realizing key values in their collective activity. They also develop strategies for avoiding associated obstacles.
 4. At the end of the semester, groups prepare a self-evaluation that assesses success in realizing ethical values and avoiding obstacles.
 5. Text boxes in this module describe pitfalls in group activities and offer general strategies for preventing or mitigating them.
- There is also a text box that provides an introductory orientation on key ethical values or virtues.

A Framework for Value-Integration

The objective of this module is to teach you to teach yourselves how to work in small groups. You will develop and test procedures for realizing value goals and avoiding group pitfalls. You will also use Socio-Technical System Analysis to help you understand better how to take advantage of the way in which different environments enable group activities and to anticipate and minimize the way in which other environments can constrain or even oppose group activities.

- Discovery: "The goal of this activity is to 'discover' the values that are relevant to, inspire, or inform a given design project, resulting in a list of values and bringing into focus what is often implicit in a design project." [Flanagan et al. 323]. The discovery of group values is a trial and error process. To get started, use the ADEM Statement of Values or the short value profiles listed below.
- Translation: "[T]ranslation is the activity of embodying or expressing...values in a system design. Translation is further divided into operationalization, which involves defining or articulating values in concrete terms, and implementation which involves specifying corresponding design features" [Flanagan et al., 338]. You will operationalize your values by developing profiles. (See below or the ADEM Statement of Values for examples.) Then you will implement your values by developing realization procedures. For example, to realize justice in carrying out a group task, first we will discuss the task as a group, second we will divide it into equal parts, third, fourth, etc.
- Verification: "In the activity of verification, designers assess to what extent they have successfully implemented target values in a given system. [Strategies and methods] may include internal testing among the design team, user testing in controlled environments, formal and informal interviews and surveys, the use of prototypes, traditional quality assurance measures such as automated and regression-oriented testing and more" [Flanagan et al., 344-5]. You will document your procedures in the face of different obstacles that may arise in your efforts at value-realization. At the end of your semester, you will verify your results by showing how you have refined procedures to more effectively realize values.

The framework on value realization and the above-quoted passages can be found in the following resource: M. Flanagan, D. Howe, and H. Nissenbaum, "Embodying Values in Technology: Theory and Practice," in *Information Technology and Moral Philosophy*, Jeroen van den Hoven and John Weckert, Eds. Cambridge, UK: Cambridge University Press, 2008, pp. 322-353.

Value Profiles for Professional Ethics

1. Definition - A value "refers to a claim about what is worthwhile, what is good. A value is a single word or phrase that identifies something as being desirable for human beings." Brincat and Wike, *Morality and the Professional Life: Values at Work*

2. Reasonableness - Defusing disagreement and resolving conflicts through integration. Characteristics include seeking relevant information, listening and responding thoughtfully to others, being open to new ideas, giving reasons for views held, and acknowledging mistakes and misunderstandings. (From Michael Pritchard, Reasonable Children)
3. Responsibility - The ability to develop moral responses appropriate to the moral issues and problems that arise in one's day-to-day experience. Characteristics include avoiding blame-shifting, designing overlapping role responsibilities to fill responsibility "gaps", expanding the scope and depth of general and situation-specific knowledge, and working to expand control and power.
4. Respect - Recognizing and working not to circumvent the capacity of autonomy in each individual. Characteristics include honoring rights such as privacy, property, free speech, due process, and participatory rights such as informed consent. Disrespect circumvents autonomy by deception, force, or manipulation.
5. Justice - Giving each his or her due. Justice breaks down into kinds such as distributive (dividing benefits and burdens fairly), retributive (fair and impartial administration of punishments), administrative (fair and impartial administration of rules), and compensatory (how to fairly recompense those who have been wrongfully harmed by others).
6. Trust - According to Solomon, trust is the expectation of moral behavior from others.
7. Honesty - Truthfulness as a mean between too much honesty (bluntness which harms) and dishonesty (deceptiveness, misleading acts, and mendaciousness).
8. Integrity - A meta-value that refers to the relation between particular values. These values are integrated with one another to form a coherent, cohesive and smoothly functioning whole. This resembles Solomon's account of the virtue of integrity.

Exercise 1: Developing Strategies for Value Realization

Directions

1. Identify value goals. Start with two or three. You can add or subtract from these as the semester progresses.
2. Give a brief description of each using terms that reflect your group's shared understandings. You may use the descriptions in this module or those in the ADEM Statement of Values but feel free to modify these to fit your group's context. You could also add characteristics and sample rules and aspirations.
3. For each value goal, identify and spell out a procedure for realizing it. See the examples just below for questions that can help you develop value procedures for values like justice and responsibility.

Examples

- Design a plan for realizing key moral values of teamwork. Your plan should address the following value-based tasks
- How does your group plan on realizing justice? For example, how will you assign tasks within the group that represent a fair distribution of the workload and, at the same time, recognize differences in individual strengths and weaknesses? How does your group plan on dealing with members who fail to do their fair share?
- How does your group plan on realizing responsibility? For example, what are the responsibilities that members will take on in the context of collective work? Who will be the leader? Who will play devil's advocate to avoid groupthink? Who will be the spokesperson for the group? How does your group plan to make clear to each individual his or her task or role responsibilities?
- How does your group plan on implementing the value of reasonableness? How will you guarantee that each individual participates fully in group decisions and activities? How will you deal with the differences, non-agreements, and disagreements that arise within the group? What process will your group use to reach agreement? How will your group insure that every individual has input, that each opinion will be heard and considered, and that each individual will be respected?
- How does your group plan on implementing the value of (academic) honesty? For example, how will you avoid cheating or plagiarism? How will you detect plagiarism from group members, and how will you respond to it?
- Note: Use your imagination here and be specific on how you plan to realize each value. Think preventively (how you plan on avoiding injustice, irresponsibility, injustice, and dishonesty) and proactively (how you can enhance these values). Don't be afraid to outline specific commitments. Expect some of your commitments to need reformulation. At the end of the semester, this will help you write the final report. Describe what worked, what did not work, and what you did to fix the latter.

Obstacles to Group Work (Developed by Chuck Huff for Good Computing: A Virtue Approach to Computer Ethics)

1. The Abilene Paradox. "The story involves a family who would all rather have been at home that ends up having a bad dinner in a lousy restaurant in Abilene, Texas. Each believes the others want to go to Abilene and never questions this by giving their own view that doing so is a bad idea. In the Abilene paradox, the group winds up doing something that no individual wants to

do because of a breakdown of intra-group communication." (From Huff, Good Computing, an unpublished manuscript for a textbook in computer ethics. See materials from Janis; complete reference below.)

2. Groupthink. The tendency for very cohesive groups with strong leaders to disregard and defend against information that goes against their plans and beliefs. The group collectively and the members individually remain loyal to the party line while happily marching off the cliff, all the while blaming "them" (i.e., outsiders) for the height and situation of the cliff. (Also from Huff, Good Computing, an unpublished manuscript for a textbook in computer ethics.)
3. Group Polarization. Here, individuals within the group choose to frame their differences as disagreements. Framing a difference as non-agreement leaves open the possibility of working toward agreement by integrating the differences or by developing a more comprehensive standpoint that dialectically synthesizes the differences. Framing a difference as disagreement makes it a zero-sum game; one's particular side is good, all the others bad, and the only resolution is for the good (one's own position) to win out over the bad (everything else). (Weston provides a nice account of group polarization in Practical Companion to Ethics. This is not to be confused with Cass Sunstein's different account of group polarization in Infotopia.)
4. Note: All of these are instances of a social psychological phenomenon called conformity. But there are other processes at work too, like group identification, self-serving biases, self-esteem enhancement, self-fulfilling prophecies, etc.

More Obstacles to Group Work

- Free Riders: Free riders are individuals who attempt to "ride for free" on the work of the other members of the group. Some free-riders cynically pursue their selfish agenda while others fall into this pitfall because they are unable to meet all their obligations. (See conflict of effort.)
- Outliers: These are often mistaken for free riders. Outliers want to become participants but fail to become fully integrated into the group. This could be because they are shy and need encouragement from the other group members. It could also be because the other group members know one another well and have habitual modes of interaction that exclude outsiders. One sign of outliers; they do not participate in group social activities but they still make substantial contributions working by themselves. ("No, I can't come to the meeting--just tell me what I have to do.")
- Hidden Agendas: Cass Sunstein introduces this term. A group member with a "hidden agenda" has something he or she wants to contribute but, for some reason or other, hold back. For example, this individual may have tried to contribute something in the past and was "shot down" by the group leader. The next time he or she will think, "Let them figure it out without me."
- Conflict of Effort: conflict of Effort often causes an individual to become a free rider or an outlier. These group members have made too many commitments and come unraveled when they all come due at the same time. Students are often overly optimistic when making out their semester schedules. They tightly couple work and class schedules while integrating home responsibilities. Everything goes well as long as nothing unusual happens. But if a coworker gets sick and your supervisor asks you to come in during class times to help out, or you get sick, it becomes impossible to keep the problem from "spilling out" into other areas of your schedule and bringing down the whole edifice. Developing a schedule with periods of slack and flexibility can go a long way toward avoiding conflict of effort. Groups can deal with this by being supportive and flexible. (But it is important to draw the line between being supportive and carrying a free rider.)

Best Practices for Avoiding Abilene Paradox

- At the end of the solution generating process, carry out an anonymous survey asking participants if anything was left out they were reluctant to put before group.
- Designate a Devil's Advocate charged with criticizing the group's decision.
- Ask participants to reaffirm group decision--perhaps anonymously.

Best Practices for Avoiding Groupthink (Taken from Janis, 262-271)

- "The leader of a policy-forming group should assign the role of critical evaluator to each member, encouraging the group to give high priority to airing objections and doubts."
- "The leaders in an organization's hierarchy, when assigning a policy-planning mission to a group, should be impartial instead of stating preferences and expectations at the outset."
- "Throughout the period when the feasibility and effectiveness of policy alternatives are being surveyed, the policy-making group should from time to time divide into two or more subgroups to meet separately...."
- One or more outside experts or qualified colleagues within the organization who are not core members of the policy-making group should be invited to each meeting ...and should be encouraged to challenge the views of the core members."
- "At every meeting devoted to evaluating policy alternatives, at least one member should be assigned the role of devil's advocate."

Best Practices for Avoiding Polarization (Items taken from "Good Computing: A Virtue Approach to Computer Ethics" by Chuck Huff, William Frey and Jose Cruz (Unpublished Manuscript))

- Set Quotas. When brainstorming, set a quota and postpone criticism until after quota has been met.
- Negotiate Interests, not Positions. Since it is usually easier to integrate basic interests than specific positions, try to frame the problem in terms of interests.
- Expanding the Pie. Conflicts that arise from situational constraints can be resolved by pushing back those constraints through negotiation or innovation.
- Nonspecific Compensation. One side makes a concession to the other but is compensated for that concession by some other coin.
- Logrolling. Each party lowers their aspirations on items that are of less interest to them, thus trading off a concession on a less important item for a concession from the other on a more important item.
- Cost-Cutting. One party makes an agreement to reduce its aspirations on a particular thing, and the other party agrees to compensate the party for the specific costs that reduction in aspirations involves.
- Bridging. Finding a higher-order interest on which both parties agree, and then constructing a solution that serves that agreed-upon interest.

Exercise 2 - Avoiding the Pitfalls of Group Work

- Design a plan for avoiding the pitfalls of group work enumerated in the text box above.
- How does your group plan on avoiding the Abilene Paradox?
- How does your group plan on avoiding Group Polarization?
- How does your group plan on avoiding Groupthink?
- Note: Use imagination and creativity here. Think of specific scenarios where these obstacles may arise, and what your group can do to prevent them or minimize their impact.

Exercise 3: Socio-Technical System

Your group work this semester will take place within a group of nested or overlapping environments. Taken separately and together, these will structure and channel your activity, facilitating action in certain circumstances while constraining, hindering, or blocking it in others. Prepare a socio-technical system table for your group to help structure your group self-evaluation. Include hardware/software, physical surroundings, stakeholders (other groups, teacher, other classes, etc.), procedures (realizing values, avoiding pitfalls), university regulations (attendance), and information structures (collecting, sharing, disseminating)

Some things about Socio-Technical Systems

1. Socio-Technical System Analysis provides a tool to uncover the different environments in which business activity takes place and to articulate how these constrain and enable different business practices.
2. A socio-technical system can be divided into different components such as hardware, software, physical surroundings, people/groups/roles, procedures, laws/statutes/regulations, and information systems.
3. But while these different components can be distinguished, they are in the final analysis inseparable. STSs are, first and foremost, systems composed of interrelated and interacting parts.
4. STSs also embody values such as moral values (justice, responsibility, respect, trust, integrity) and non-moral values (efficiency, satisfaction, productivity, effectiveness, and profitability). These values can be located in one or more of the system components. They come into conflict with one another causing the system to change.
5. STSs change and this change traces out a path or trajectory. The normative challenge of STS analysis is to find the trajectory of STS change and work to make it as value-realizing as possible.

Socio-Technical System Table for Groups

Hardware/Software	Physical Surroundings	Stakeholders	Procedures	University Regulations	Information Structures

Think about the new role of your smartphones in group work in class. Will you be using Google Docs to exchange documents?	How do the classroom and the arrangement of objects within it constrain and enable group activities?	Think about other teachers, classes, supervisors, jobs, and other individuals that can have an impact on your ability to carry out group assignments.	Name but don't describe in detail, the value-realizing procedures your group is adopting.	What are university regulations that will have an impact on your group work? For example, switches between MWF and TTH schedules.	There is a wealth of information and skill locked in each of your group's members. How will you unleash these and telescope them into group work and activities? How, in other words, will you work to maximize group synergies and minimize group disadvantages?
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Exercises 1-3 compose the Preliminary Self-Evaluation which is due shortly after semester-long groups are formed. Exercise 4 is the close-out group self-evaluation which is due at the end of the semester.

Exercise 4: Prepare a Final, Group Self-Evaluation

- Due Date: One week after the last class of the semester when your group turns in all its materials.
- Length: A minimum of five pages not including Team Member Evaluation Forms
- Contents:
 1. Restate the Ethical and Practical Goals that your group developed at the beginning of its formation.
 2. Provide a careful, documented assessment of your group's success in meeting these goals. (Don't just assert that "Our group successfully realized justice in all its activities this semester." How did your group characterize justice in the context of its work? What specific activities did the group carry out to realize this value? What, among these activities, worked and what did not work?)
 3. Identify obstacles, shortcomings or failures that your group experienced during the semester. How did these arise? Why did they arise? How did you respond to them? Did your response work? What did you learn from this experience?
 4. Assess the plans you set forth in your initial report on how you intended to realize values and avoid pitfalls. How did these work? Did you stick to your plans or did you find it necessary to change or abandon them in the face of challenges?
 5. Discuss your group's procedures and practices? How did you divide and allocate work tasks? How did you reach a consensus on difficult issues? How did you ensure that all members were respected and allowed significant and meaningful participation? What worked and what did not work with respect to these procedures? Will you repeat them in the future? Would you recommend these procedures as best practices for future groups?
 6. What did you learn from your experience working as a team this semester? What will require further reflection and thought? In other words, conclude your self-evaluation with a statement that summarizes your experience working together as a team this semester.

Appendix for ADMI 4016, Falkl 2013 and following

- What are the results of your group's challenge to the College of Business Administration's Statement of Values? (This can be found in Developing Ethics Codes and Statements of Value. See exercise 2. <http://cnx.org/content/m14319/1.11/>)
- What is your group's CID Structure? See presentation two at the bottom of the module, A Short History of the Corporation. <http://cnx.org/content/m17314/1.7/>

Wrap Up: Some further points to consider...

1. Don't gloss over your work with generalizations like, "Our group was successful and achieved all of its ethical and practical goals this semester." Provide evidence for successful claims. Detail the procedures designed by your group to bring about these results. Are they "best practices"? What makes them the best practices?
2. Sometimes—especially if difficulties arose—it is difficult to reflect on your group's activities for the semester. Make the effort. Schedule a meeting after the end of the semester to finalize this reflection. If things worked well, what can you do to repeat

these successes in the future? If things didn't work out, what can you do to avoid similar problems in the future? Be honest, be descriptive and avoid blame language.

3. This may sound harsh but get used to it. Self-evaluations—group and individual—are an integral part of professional life. They are not easy to carry out but properly done they help to secure success and avoid future problems.
4. Student groups—perhaps yours—often have problems. This self-evaluation exercise is designed to help you face them rather than push them aside. Look at your goals. Look at the strategies you set forth for avoiding Abilene, groupthink, and group polarization. Can you modify them to deal with problems? Do you need to design new procedures?

Ethics of Team Work Presentations

Values in Team Work (Thought Experiments)

[Ethics of Team Work.pptx](#)

Pitfalls to Avoid in Group Work

[Pitfalls to Avoid in Group Work.pptx](#)

Thought Experiments on Group Work

[Thought Experiments on Group Work.docx](#)

Team Member Evaluation Forms (Required)

[TEAM MEMBER RATING SHEET-3.docx](#)

New Ethics of Teamwork Presentation (Spring 2012)

[Ethics of Teamwork.pptx](#)

Ethics of Teamwork Jeopardy

[Team Jeopardy.pptx](#)

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1.4: Moral Exemplars in Business and Professional Ethics

Module Introduction

Through the activities of this module, you will learn to balance cautionary tales in business and professional ethics with new stories about those who consistently act in a morally exemplary way. While cautionary tales teach us what to avoid, narratives from the lives of moral exemplars show us how to be good. A study of moral best practices in business and professional ethics shows that moral exemplars exhibit positive and learnable skills. This module, then, looks at moral exemplars in business and the professions, outlines their outstanding accomplishments, and helps you to unpack the strategies they use to overcome obstacles to doing good.

You will begin by identifying outstanding individuals in business and associated practices who have developed moral "best practices." Your task is to look at these individuals, retell their stories, identify the skills that help them do good, and build a foundation for a more comprehensive study of virtue in occupational and professional ethics.

Moral Exemplar Terms

Moral Exemplar

- An individual who demonstrates outstanding moral conduct often in the face of difficult or demanding circumstances. (Beyond the "call of duty")
- Often moral exemplars perform actions that go beyond what is minimal, required, ordinary, or even extraordinary.
- Moral exemplars perform actions that are "above and beyond the call of duty."
- Most importantly, they perform these actions repeatedly across a career or even a lifetime. In some way, their exemplary conduct has become "second nature."

Supererogatory

- "A supererogatory act is an act that is beyond the call of duty. It is something that is morally good to do but not obligatory. Examples of supererogatory acts are donating blood, volunteering on a rape crisis hotline, babysitting (without accepting recompense) a friend's two-year-old triplets for the afternoon, or throwing oneself on a live hand grenade in order to save one's buddies' lives." (Baron, 1997: 614)
- Baron's definition (found in the Encyclopedia of Business Ethics) captures how the supererogatory occupies a moral space well above that of the minimally decent or even the ordinary. Supererogatory actions are outstanding, extraordinary, and exemplary in both moral and practical senses.
- Urmson, a moral philosopher, remarks how the supererogatory has been neglected (up to the mid-twentieth century) by moral philosophy, dominated as it was in the previous century by the debate between Utilitarianism and Deontology.
- Two quotations from Urmson show this clearly: (1) "But it does seem that these facts have been neglected in their general, systematic accounts of morality. It is indeed easy to see that on some of the best-known theories there is no room for such facts" (Urmson, 1958, p. 206). (2) "[s]imple utilitarianism, Kantianism, and intuitionism, then, have no obvious theoretical niche for the saint and the hero" (Urmson, 1958: 207).
- Baron, M. (1997). "Supererogation", Blackwell Encyclopedic Dictionary of Business Ethics, Patricia H. Werhane and R. Edward Freeman, eds., New York: Blackwell: 614-7.
- Urmson, J.O. (1958). "Saints and Heroes." Essays in Moral Philosophy, A.I. Melden, ed., Seattle: University of Washington Press: 198-216.

Moral Minimum

- Compare and distinguish the idea of the supererogatory with that of the moral minimum.
- The difference is between that which is morally exemplary versus that which is just over the threshold of wrongdoing.
- "I suggest that moral minimums are best understood as negative standards, universally agreed upon "bottom lines" beyond which it is morally questionable to act. For example, it is almost always wrong to deliberately harm or contribute to harming another person or persons; to deliberately violate their rights to freedom, life, or property; to treat individuals or classes of individuals with disrespect; to compete or cooperate unfairly; not to honor promises or contract; or to be dishonest or deceitful. Whereas these moral minimums do not define goodness, fairness, or benefit, or define the positive content of rights, they set minimum guidelines for behavior that most people everywhere might agree on..." (Werhane, 1999: 122).
- Werhane, P. (1999). Moral Imagination and Management Decision-Making, Oxford, UK: Oxford University Press.

Moral Exemplar Criteria in Computing

1. Either a sustained commitment to moral ideals or ethical principles in computing that include a generalized respect for humanity or sustained evidence of moral virtue in the practice of computing.
2. A disposition to make computing decisions in accord with one's moral ideals or ethical principles, implying a consistency between one's actions and intentions and between the means and ends of one's actions
3. A willingness to risk one's self-interest for the sake of one's moral values.
4. A tendency to be inspiring to other computer professional and thereby to move them to moral action
5. A sense of realistic humility about one's own importance relative to the world at large, implying a relative lack of concern for one's own ego.
6. Huff, C. and Barnard, L. (2009). "Good Computing: Moral Exemplars in the Computing Profession", IEEE Technology and Society Magazine: 47-54.

Responders and Idealists

- This quotation from Blum provides a nice characterization of "moral responders."
- "the 'responder' moral exemplar does not, prior to confronting situations in which she manifests moral excellence, possess a set of moral principles which she has worked out explicitly, committed herself to, and attempted to guide her life by."
- "the responder responds to the situations she faces and to individuals in a 'morally excellent way.'"
- Blum, L. (1994). "Moral Exemplars: reflections on Schindler, the Trochmés, and others", Moral Perception and Particularity, Cambridge, Mass.: Cambridge University Press: 65-97.

Idealists

- According to Blum (and Hailie), Magda Trochme is a responder while her husband, Andre Trochme is an idealist. Both perform morally exemplary and supererogatory actions but out of different motivations.
- "To be an idealist [one] must see these ideals as more than merely personal goals or a personal conception of the good. They must be formulated as general values, and regarded by the agent as having some kind of intrinsic worth or general validity.
- Blum, L. (1994). "Moral Exemplars: reflections on Schindler, the Trochmés, and others", Moral Perception and Particularity, Cambridge, Mass.: Cambridge University Press: 65-97.

Moral Heroes and Moral Saints

- Moral heroes achieve their good and excellent goals only by making substantial sacrifices. The notion of self-sacrifice is the key distinguishing characteristic of this kind of exemplar.
- What distinguishes moral saints from other kinds of moral exemplar is the criterion of moral faultlessness; these exemplars achieve their excellences by means of conduct that is free from any moral flaw.
- Moral saints are often used to argue for the unity of virtues thesis, namely, that the virtues work together forming a system where each is necessary and mutually supports the others.

Table of Moral Exemplars Table of Moral Exemplars

	Moral Heroes	Moral Saints
Idealists	MLK and Andre Trochme	Thomas More
Responders	Magda Trochme and Oscar Schindler	Mother Teresa and Saint Francis

Exercise 1: Choose a moral exemplar

- Identify a moral exemplar and provide a narrative description of his or her life story.
- To get this process started, look at the list of moral exemplars provided in this module. The links in the upper left-hand corner of this module will help you to explore their accomplishments in detail. Feel free to choose your own exemplar. Make sure you identify someone in the occupational and professional areas such as business and engineering. These areas have more than their share of exemplars, but they tend to escape publicity because their actions avoid publicity generating disasters rather than bring them about.

Moral Exemplars

- 1. William LeMesseur. LeMesseur designed the Citicorp Building in New York. When a student identified a critical design flaw in the building during a routine class exercise, LeMesseur responded, not by shooting the messenger, but by developing an

intricate and effective plan for correcting the problem before it issued in drastic real-world consequences. Check out LeMesseur's profile at online ethics and see how he turned a potential disaster into a good deed.

- 2. Fred Cuny, starting in 1969 with Biafra, carried out a series of increasingly effective interventions in international disasters. He brought effective methods to disaster relief such as engineering know-how, political savvy, good business sense, and aggressive advocacy. His timely interventions saved thousands of Kurdish refugees in the aftermath of the Persian Gulf War in 1991. He also helped design and implement an innovative water filtration system in Sarajevo during the Bosnia-Serb conflict in 1993. For more details, consult the biographical sketch at online ethics.
- 3. Roger Boisjoly worked on a team responsible for developing o-ring seals for fuel tanks used in the Challenger Shuttle. When his team noticed evidence of gas leaks he made an emergency presentation before officials of Morton Thiokol and NASA recommending postponing the launch scheduled for the next day. When decision-makers refused to change the launch date, Boisjoly watched in horror the next day as the Challenger exploded seconds into its flight. Find out about the courageous stand Boisjoly took in the aftermath of the Challenger explosion by reading the biographical sketch at online ethics.
- 4. Muhammad Yunus won the Nobel Prize for Peace in 2006. His effort in setting up "micro-businesses" funded through "micro-lending" has completely changed the paradigm on how to extend business practices to individuals at the bottom of the pyramid. Learn about his strategies for creating micro-businesses and how those strategies have been extended throughout the world, including Latin America, by listening to an interview with him broadcast by the Online News Hour. (See link included in this module.)
- 5. Bill Gates has often been portrayed as a villain, especially during the anti-trust suit against Microsoft in the mid-1990s. Certainly, his aggressive and often ruthless business practices need to be evaluated openly and critically. But recently Gates stopped participating in the day-to-day management of his company, Microsoft, and has set up a charitable foundation to oversee international good works projects. Click on the link included in this module to listen to and read an interview recently conducted with him and his wife, Melinda, on their charitable efforts.
- 6. Jeffrey Skilling, the former CEO of Enron, can hardly be called a moral exemplar. Yet when Enron was at its peak, its CEO, Jeffrey Skilling, was considered among the most innovative, creative, and brilliant of contemporary corporate CEOs. View the documentary, *The Smartest Guys in the Room*, read the book of the same title, and learn about the configuration of character traits that led to Skilling's initial successes and ultimate failure. A link included in this module will lead you to an interview with Skilling conducted on March 28, 2001.
- Inez Austin worked to prevent contamination from nuclear wastes produced by a plutonium production facility. Visit Online Ethics by clicking on the link above to find out more about her heroic stand.
- Rachael Carson's book, *The Silent Spring*, was one of the key events inaugurating the environmental movement in the United States. For more on the content of her life and her own personal act of courage, visit the biographical profile at Online Ethics. You can click on the Supplemental Link provided above.

Exercise Two: Moral Exemplar Profiles

- What are the positive and negative influences you can identify for your moral exemplar?
- What good deeds did your exemplar carry out?
- What obstacles did your moral exemplar face and how did he or she overcome them?
- What skills, attitudes, beliefs, and emotions helped to orient and motivate your moral exemplar?

Exercise Three

Prepare a short dramatization of a key moment in the life of your group's moral exemplar.

Text Box: Two different Types of Moral Exemplar

- Studies carried out by Chuck Huff into moral exemplars in computing suggest that moral exemplars can operate as craftspersons or reformers. (Sometimes they can combine both these modes.)
- Craftspersons (1) draw on pre-existing values in computing, (2) focus on users or customers who have needs, (3) take on the role of providers of a service/product, (4) view barriers as inert obstacles or puzzles to be solved, and (5) believe they are effective in their role.
- Reformers (1) attempt to change organizations and their values, (2) take on the role of moral crusaders, (3) view barriers as active opposition, and (4) believe in the necessity of systemic reform
- These descriptions of moral exemplars have been taken from a presentation by Huff at the STS colloquium at the University of Virginia in October 2006. Huff's presentation can be found at the link provided in the upper left-hand corner of this module.

Elements of a Life Story Interview

- Major Influences
- Peak and Nadir Experiences
- Challenges and Opportunities.
- Goals, Values, and Objectives
- Commentary: The life story interview collects the subject's life in narrative form. Those conducting the interview along with those studying it are skilled in identifying different patterns and structures in the interview. (Identifying and classifying the patterns is called "coding".)
- Huff, Rogerson, and Barnard interviewed moral exemplars in computing in Europe and coded for the following: "social support and antagonism, the use of technical or social expertise, the description of harm to victims or need for reform, actions taken toward reform, designs undertaken for users or clients, effectiveness and ineffectiveness of action, and negative and positive emotion" (Huff and Barnard, 2009: 50).
- They identified two kinds of moral exemplars in computing: helpers (or craftpersons) and reformers.

Helpers and Reformers

- Craftpersons work to preserve existing values, see themselves as providers of a service, frame problems as overcoming barriers, and seek ethical ends (Huff and Barnard, 2009: 50).
- Reformers focus on social systems, see themselves as moral crusaders, work to change values, view individuals as victims of injustice, and take system reform as their goal (Huff and Barnard, 2009: 50).

What Makes a Moral Exemplar? PRIMES Explained

General Comments on Exemplars

- Moral exemplars have succeeded in integrating moral and professional attitudes and beliefs into their core identity. Going against these considerations for moral exemplars is tantamount to acting against self. Acting in accordance with them becomes second nature.
- Moral exemplars often achieve their aims with the support of "support groups." In fact, moral exemplars are often particularly adept at drawing support from surrounding individuals, groups and communities. This goes against the notion that exemplars are isolated individuals who push against the current. (Not all exemplars need fit as heroes into Ayn Rand novels.)
- Moral exemplars often do not go through periods of intensive and prolonged deliberation in order to hit upon the correct action. If we want a literary example, we need to replace the tortured deliberations of a Hamlet with the quick and intuitive insight of an Esther Summerson. (Summerson is a character in Charles Dickens' novel, Bleak House. See both William Shakespeare and Charles Dickens for more examples of villains and exemplars.) Some have situated moral exemplars within virtue ethics. They have cultivated moral habits that allow them to do good as second nature. They have also found ways to integrate moral reasoning with emotion (as motive), perception (which helps them zero in on moral relevance), and skill (which helps implement moral value). In this sense, moral expertise functions much as athletic or technical expertise; all are difficult to acquire but once acquired lead to highly skilled actions performed almost effortlessly.

PRIMES Primes stands for Personality, Integrating value into self-system, Moral Ecology, and Moral Skills Sets. These are the elements composing moral expertise that have been identified by Huff and Rogerson based on interviews they conducted with exemplars in the areas of computing.

Personality

- Moral exemplars exhibit different configurations of personality traits based on the big five. Locate the moral exemplar you have chosen in terms of the following five continuums (or continua):
- Neuroticism to Lack of Neuroticism (Stability?)
- Agreeableness to Disagreeableness
- Extraversion to Introversion
- Openness to Closedness
- Conscientiousness to Lack of Conscientiousness
- Examine your exemplar on each of these scales. In and of themselves, these qualities are neither good nor bad. They can be integrated to form bad characters or good characters. In many cases, moral exemplars stand out through how they have put their personality characteristics to "good use." (They have used them as vehicles or channels to excellence.)

Integrating Moral Value into Self-System

- As said above, moral exemplars stand out by the way in which (and the extent to which) they have integrated moral value into their self-system. Because of this, they are strongly motivated to do good and avoid doing bad. Both (doing good and refraining from doing bad) express who they are. If they slip into bad deeds, this motivational system pushes them to improve to avoid repeating bad deeds.
- One way of integrating moral value into self-system is by looking at stories and narratives of those who have displayed moral excellence. Many of the individuals portrayed above (Carson, Boisjoly, LeMesseur, Cuny, Austin, and Yunus) provide concrete models of outstanding moral careers.
- Literature also provides its models of moral exemplars. Charles Dickens paints especially powerful portraits of both moral heroes (Esther Summerson and "Little Dorritt") and villains (Heep and Skimpole).
- Other vehicles for integrating moral value centrally into the self-system lie in affiliations, relationships, and friendships. Aristotle shows the importance of good friendships in developing virtues. Moral exemplars most often can point to others who have served as mentors or strong positive influences. For example, Roger Boisjoly tells of how he once went to a senior colleague for advice on whether to sign off on a design that was less than optimal. His colleague's advice: would you be comfortable with your wife or child using a product based on this design?
- The ethicist, Bernard Williams, has argued forcefully for the importance of personal projects in establishing and maintaining integrity. Personal projects, roles, and life tasks all convey value; when these hold positive moral value and become central unifying factors in one's character, then they also serve to integrate moral value into the self-system.
- Augusto Blasi, a well known moral psychologist, gives a particularly powerful account (backed by research) of the integration of moral value into self-system and its motivational effect.

Moral Ecology

- Moral Ecologies: "The term moral ecology encourages us to consider the complex web of relationships and influences, the long persistence of some factors and the rapid evolution of others, the variations in strength and composition over time, the micro-ecologies that can exist within larger ones, and the multidirectional nature of causality in an ecology." From Huff et. al.
- Moral ecologies refer to social surrounds, that is, the different groups, organizations, and societies that surround us and to which we are continually responding.
- We interact with these social surrounds as organisms interact with their surrounding ecosystems. In fact, moral ecologies offer us roles (like ecological niches) and envelop us in complex organizational systems (the way ecosystems are composed of interacting and interrelated parts). We inhabit and act within several moral ecologies; these moral ecologies, themselves, interact. Finally, moral ecologies, like natural ecosystems, seek internal and external harmony and balance. Internally, it is important to coordinate different the constituent individuals and the roles they play. Externally, it is difficult but equally important to coordinate and balance the conflicting aims and activities of different moral ecologies.
- Moral ecologies shape who we are and what we do. This is not to say that they determine us. But they do channel and constrain us. For example, your parents have not determined who you are. But much of what you do responds to how you have experienced them; you agree with them, refuse to question their authority, disagree with them, and rebel against them. The range of possible responses is considerable but these are all shaped by what you experienced from your parents in the past.
- The moral ecologies module (see the link provided above) describes three different moral ecologies that are important in business: quality-, customer-, and finance-driven companies. (More "kinds" could be generated by combining these in different ways: for example, one could characterize a company as customer-driven but transforming into a quality-driven company.) Roles, strategies for dissent, assessment of blame and praise, and other modes of conduct are shaped and constrained by the overall character of the moral ecology.
- Moral ecologies, like selves, can also be characterized in terms of the "centrality" of moral value. Some support the expression of moral value or certain kinds of moral value (like loyalty) while undermining or suppressing the expression of others (like courage or autonomy).
- Finally, think in terms of how personality traits integrated around moral value interact with different types of moral ecology. If a moral ecology undermines virtuous conduct, what strategies are available for changing it? Or resisting it? If there are different kinds of moral exemplar, which pair best with which moral ecology? (How would a helper or craftsman prevail in a finance-driven moral ecology like those characterized by Robert Jackall in Moral Mazes?

Moral Skills Sets

- Moral expertise is not reducible to knowing what constitutes good conduct and doing your best to bring it about. Realizing good conduct, being an effective moral agent, bringing value into the work, all require skills in addition to a "good will." PRIMES studies have uncovered four skill sets that play a decisive role in the exercise of moral expertise.
- Moral Imagination: The ability to project into the standpoint of others and view the situation at hand through their lenses. Moral imagination achieves a balance between becoming lost in the perspectives of others and failing to leave one's own perspective. Adam Smith terms this balance "proportionality" which we can achieve in empathy when we feel with them but do not become lost in their feelings. Empathy consists of feeling with others but limiting the intensity of that feeling to what is proper and proportionate for moral judgment.
- Moral Creativity: Moral Creativity is close to moral imagination and, in fact, overlaps with it. But it centers in the ability to frame a situation in different ways. Patricia Werhane draws attention to a lack of moral creativity in the Ford Pinto case. Key Ford directors framed the problem with the gas tank from an economical perspective. Had they considered other framings they might have appreciated the callousness of refusing to recall Pintos because the costs of doing so (and retrofitting the gas tanks) were greater than the benefits (saving lives). They did not see the tragic implications of their comparison because they only looked at the economic aspects. Multiple framings open up new perspectives that make possible the design of non-obvious solutions.
- Reasonableness: Reasonableness balances openness to the views of others (one listens and impartially weighs their arguments and evidence) with commitment to moral values and other important goals. One is open but not to the extent of believing anything and failing to keep fundamental commitments. The Ethics of Team Work module (see link above) discusses strategies for reaching a consensus that is employed by those with the skill set of reasonableness. These help avoid the pitfalls of group-based deliberation and action.
- Perseverance: Finally, perseverance is the "ability to plan moral action and continue on that course by responding to circumstances and obstacles while keeping ethical goals intact." Huff et. al.

Presentation on Moral Exemplars

[Brief Comments on Moral Exemplars.pptx](#)

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1.5: Theory-Building Activities - Virtue Ethics

Module Introduction

This module uses materials being prepared for Good Computing: A Virtue Approach to Computer Ethics, to set up an exercise in which you will identify and spell out virtues relevant to your professional discipline. After identifying these virtues, you will work to contextualize them in everyday practice. Emphasis will be placed on the Aristotelian approach to virtues which describes a virtue as the disposition toward the mean located between the extremes of excess and defect. You will also be asked to identify common obstacles that prevent professionals from realizing a given virtue and moral exemplars who demonstrate consistent success in realizing these virtues and responding to obstacles that stand in the way of their realization. In a variation on this module, you could be asked to compare the virtues you have identified for your profession with virtues that belong to other moral ecologies such as those of the Homeric warrior.

Three Versions of Virtue Ethics: Virtue 1, Virtue 2, and Virtue 3

Virtue ethics has gone through three historical versions. The first, Virtue 1, was set forth by Aristotle in ancient Greece. While tied closely to practices in ancient Greece that no longer exist today, Aristotle's version still has a lot to say to us in this day and age. In the second half of the twentieth century, British philosophical ethicists put forth a related but different theory of virtue ethics (virtue 2) as an alternative to the dominant ethical theories of utilitarianism and deontology. Virtue 2 promised a new foundation of ethics consistent with work going on at that time in the philosophy of mind. Proponents felt that turning from the action to the agent promised to free ethical theory from the intractable debate between utilitarianism and deontology and offered a way to expand scope and relevance of ethics. Virtue 3 reconnects with Aristotle and virtue 1 even though it drops the doctrine of the mean and Aristotle's emphasis on character. Using recent advances in moral psychology and moral pedagogy, it seeks to rework key Aristotelian concepts in modern terms. In the following, we will provide short characterizations of each of these three versions of virtue ethics.

Virtue 1: Aristotle's Virtue Ethics

- Eudaimonia. Happiness, for Aristotle, consists of a life spent fulfilling the intellectual and moral virtues. These modes of action are auto-telic, that is, they are self-justifying and contain their own ends. By carrying out the moral and intellectual virtues for a lifetime, we realize ourselves fully as humans. Because we are doing what we were meant to do, we are happy in this special sense of eudaimonia.
- Arete. Arete is the Greek word we usually translate as "virtue". But arete is more faithfully translated as excellence. For Aristotle, the moral and intellectual virtues represent excellences. So the moral life is more than just staying out of trouble. Under Aristotle, it is centered in pursuing and achieving excellence for a lifetime.
- Virtue as the Mean. Aristotle also characterizes virtue as a settled disposition to choose the mean between the extremes of excess and defect, all relative to person and situation. Courage (the virtue) is the mean between the extremes of excess (too much courage or recklessness) and defect (too little courage or cowardice). Aristotle's claim that most or all of the virtues can be specified as the mean between extremes is controversial. While the doctrine of the mean is dropped in Virtue 2 and Virtue 3, we will still use it in developing virtue tables. (See exercise 1 below.) You may not find both extremes for the virtues you have been assigned but make the effort nonetheless.
- Ethos. "Ethos" translates as character which, for Aristotle, composes the seat of the virtues. Virtues are well settled dispositions or habits that have been incorporated into our characters. Because our characters are manifested in our actions, the patterns formed by these over time reveal who we are. This can be formulated as a decision-making test, the public identification test. Because we reveal who we are through our actions we can ask, when considering an action, whether we would care to be publicly identified with this action. "Would I want to be publicly known as the kind of person who would perform that kind of action? Would I, through my cowardly action, want to be publicly identified as a coward? Would I, through my responsible action, want to be publicly identified as a responsible person? Because actions provide others with a window into our characters, we must make sure be sure that they portray us as we want to be portrayed.
- Aisthesis of the Phronimos. This Greek phrase, roughly translated as the perception of the morally experienced agent, reveals how important practice and experience are to Aristotle in his conception of moral development. One major difference between Aristotle and other ethicists (utilitarians and deontologists) is the emphasis that Aristotle places on developing into or becoming a moral person. For Aristotle, one becomes good by first repeatedly performing good actions. So morality is more like an acquired skill than a mechanical process. Through practice we develop sensitivities to what is morally relevant in a situation, we

learn how to structure our situations to see moral problems and possibilities, and we develop the skill of "hitting" consistently on the mean between the extremes. All of these are skills that are cultivated in much the same way as a basketball player develops through practice the skill of shooting the ball through the hoop.

- **Bouleusis.** This word translates as "deliberation." For Aristotle, moral skill is not the product of extensive deliberation (careful, exhaustive thinking about reasons, actions, principles, concepts, etc.) but of practice. Those who have developed the skill to find the mean can do so with very little thought and effort. Virtuous individuals, for Aristotle, are surprisingly unreflective. They act virtuously without thought because it has become second nature to them.
- **Akrasia.** Ross translates this word as "incontinence" which is outmoded. A better translation is weakness of will. For Aristotle, knowing where virtue lies is not the same as doing what virtue demands. There are those who are unable to translate knowledge into resolution and then into action. Because akrasia (weakness of will) is very real for Aristotle, he also places emphasis in his theory of moral development on the cultivation of proper emotions to help motivate virtuous action. Later ethicists seek to oppose emotion and right action; Aristotle sees properly trained and cultivated emotions as strong motives to doing what virtue requires.
- **Logos** Aristotle's full definition of virtue is "a state of character concerned with choice, lying in a mean, i.e. the mean relative to us, this being determined by a rational principle, and by that principle by which [a person] of practical wisdom would determine it." (Ross's translation in *Nicomachean Ethics*, 1106b, 36.) We have talked about character, the mean, and the person of practical wisdom. The last key term is "logos" which in this definition is translated by reason. This is a good translation if we take reason in its fullest sense so that it is not just the capacity to construct valid arguments but also includes the practical wisdom to assess the truth of the premises used in constructing these arguments. In this way, Aristotle expands reason beyond logic to include a fuller set of intellectual, practical, emotional, and perceptual skills that together form a practical kind of wisdom.

Virtue 2

- The following summary of Virtue 2 is taken largely from Rosalind Hursthouse. While she extensively qualifies each of these theses in her own version of virtue ethics, these points comprise an excellent summary of Virtue 2 which starts with G.E.M. Anscombe's article, "Modern Moral Philosophy," and continues on into the present. Hursthouse presents this characterization of Virtue 2 in her book, *On Virtue Ethics* (2001) U.K.: Oxford University Press: 17.
- Virtue 2 is agent-centered. Contrary to deontology and utilitarianism which focus on whether actions are good or right, V2 is agent-centered in that it sees the action as an expression of the goodness or badness of the agent. Utilitarianism focuses on actions which bring about the greatest happiness for the greatest number; deontology seeks those actions that respect the autonomy of individuals and carry out moral obligations, especially duties. These theories emphasize doing what is good or right. Virtue 2, on the other hand, focuses on the agent's becoming or being good.
- Can Virtue 2 tell us how to act? Because V2 is agent-centered, critics claim that it cannot provide insight into how to act in a given situation. All it can say is, "Act the way a moral exemplar would act." But what moral standards do moral exemplars use or embody in their actions? And what moral standards do we use to pick out the moral exemplars themselves? Hursthouse acknowledges that this criticism hits home. However, she points out that the moral standards come from the moral concepts that we apply to moral exemplars; they are individuals who act courageously, exercise justice, and realize honesty. The moral concepts "courage," "justice," and "honesty" all have independent content that helps guide us. She also calls this criticism unfair: while virtue 2 may not provide any more guidance than deontology or utilitarianism, it doesn't provide any less. Virtue 2 may not provide perfect guidance, but what it does provide is favorably comparable to what utilitarianism and deontology provide.
- Virtue 2 replaces Deontic concepts (right, duty, obligation) with Aretaic concepts (good, virtue). This greatly changes the scope of ethics. Deontic concepts serve to establish our minimum obligations. On the other hand, aretaic concepts bring the pursuit of excellence within the purview of ethics. Virtue ethics produces a change in our moral language that makes the pursuit of excellence an essential part of moral inquiry.
- Finally, there is a somewhat different account of virtue 2 (call it virtue 2a) that can be attributed to Alisdair MacIntyre. This version "historicizes" the virtues, that is, looks at how our concepts of key virtues have changed over time. (MacIntyre argues that the concept of justice, for example, varies greatly depending on whether one views justice in Homeric Greece, Aristotle's Greece, or Medieval Europe.) Because he argues that skills and actions are considered virtuous only in relation to a particular historical and community context, he redefines virtues as those skill sets necessary to realize the goods or values around which social practices are built and maintained. This notion fits in well with professional ethics because virtues can be derived from the habits, attitudes, and skills needed to maintain the cardinal ideals of the profession.

Virtue 3

Virtue 3 can best be outlined by showing how the basic concepts of Virtue 1 can be reformulated to reflect current research in moral psychology.

1. Reformulating Happiness (Eudaimonia). Mihaly Csikszentmihalyi has described flow experiences (see text box below) in which autotelic activities play a central role. For Aristotle, the virtues also are autotelic. They represent faculties whose exercise is key to realizing our fullest potentialities as human beings. Thus, virtues are self-validating activities carried out for themselves as well as for the ends they bring about. Flow experiences are also important in helping us to conceptualize the virtues in a professional context because they represent a well practiced integration of skill, knowledge, and moral sensitivity.
2. Reformulating Values (Into Arete or Excellence). To carry out the full project set forth by virtue 3, it is necessary to reinterpret as excellence key moral values such as honesty, justice, responsibility, reasonableness, and integrity. For example, moral responsibility has often been described as carrying out basic, minimal moral obligations. As an excellence, responsibility becomes refocused on extending knowledge and power to expand our range of effective, moral action. Responsibility reformulated as an excellence also implies a high level of care that goes well beyond what is minimally required.
3. De-emphasizing Character. The notion of character drops out to be replaced by more or less enduring and integrated skills sets such as moral imagination, moral creativity, reasonableness, and perseverance. Character emerges from the activities of integrating personality traits, acquired skills, and deepening knowledge around situational demands. The unity character represents is always complex and changing.
4. Practical Skill Replaces Deliberation. Moral exemplars develop skills which, through practice, become second nature. These skills obviate the need for extensive moral deliberation. Moral exemplars resemble more skillful athletes who quickly develop responses to dynamic situations than Hamlets stepping back from action for prolonged and agonizing deliberation.
5. Greater Role for Emotions. Nancy Sherman discusses how, for Aristotle, emotion is not treated as an irrational force but as an effective tool for moral action once it has been shaped and cultivated through proper moral education. To step beyond the controversy of what Aristotle did and did not say about the emotions (and where he said it) we place this enhanced role for emotions within virtue 3. Emotions carry out four essential functions: (a) they serve as modes of attention; (b) they also serve as modes of responding to or signaling value; (c) they fulfill a revelatory function; and (d) they provide strong motives to moral action. Nancy Sherman, *Making a Necessity of Virtue: Aristotle and Kant on Virtue* (1997), U.K.: Cambridge University Press: 39-50.

Flow Experiences

- The psychologist, Mihaly Csikszentmihalyi, has carried out fascinating research on what he terms "flow experiences." Mike Martin in *Meaningful Work* (2000) U.K.: Oxford,; 24, summarizes these in the following bullets:
- "clear goals as one proceeds"
- "immediate feedback about progress"
- "a balance between challenges and our skills to respond to them"
- "immersion of awareness in the activity without disruptive distractions"
- "lack of worry about failure"
- loss of anxious self-consciousness"
- time distortions (either time flying or timeslowing pleasurably)"
- the activity becomes autotelic: an end in itself, enjoyed as such"

Virtue Tables

The table just below provides a format for spelling out individual virtues through (1) a general description, (2) the correlative vices of excess and defect, (3) the skills and mental states that accompany and support it, and (4) real and fictional individuals who embody it. Following the table are hints on how to identify and characterize virtues. We start with the virtue of integrity:

Virtue	Description	Excess	Defect	Obstacles to realizing the virtue in professional practices	Moral Exemplar

Integrity	A meta-virtue in which the holder exhibits unity of character manifested in holding together even in the face of strong disruptive pressures or temptations	Excess: Rigidity--sticking to one's guns even when one is obviously wrong(2,3)	Defect: Wantonness. A condition where one exhibits no stability or consistency in character	Individual corruption: Individuals can be tempted by greed toward the vice of defect. Lack of moral courage can also move one to both extremes	Saint Thomas More as portrayed in Robert Bolt's A Man for All Seasons. More refuses to take an oath that goes against the core beliefs in terms of which he defines himself.
				Institutional Corruption: One may work in an organization where corruption is the norm. This generates dilemmas like following an illegal order or getting fired.	

Exercise 1: Construct Virtue Tables for Professional Virtues

1. Discuss in your group why the virtue you have been assigned is important for the practice of your profession. What goods or values does the consistent employment of this virtue produce?
2. Use the discussion in #1 to develop a general description of your virtue. Think along the following lines: people who have virtue X tend to exhibit certain characteristics (or do certain things) in certain kinds of situations. Try to think of these situations in terms of what is common and important to your profession or practice.
3. Identify the corresponding vices. What characterizes the points of excess and defect between which your virtue as the mean lies?
4. What obstacles arise that prevent professionals from practicing your virtue? Do well-meaning professionals lack power or technical skill? Can virtues interfere with the realization of non-moral values like financial values? See if you can think of a supporting scenario or case here.
5. Identify a moral exemplar for your virtue. Make use of the exemplars described in the Moral Exemplars in Business and Professional Ethics module.
6. Go back to task #2. Redefine your description of your virtue in light of the subsequent tasks, especially the moral exemplar you identified. Check for coherence.
7. Finally, does your virtue stand alone or does it need support from other virtues or skills? For example, integrity might also require moral courage.

Exercise 2: Reflect on these Concluding Issues

- Did you have trouble identifying a moral exemplar? Many turn to popular figures for their moral exemplars. Movies and fiction also offer powerful models. Why do you think that it is hard to find moral exemplars in your profession? Is it because your profession is a den of corruption? (Probably not.) Do we focus more on villains than on heroes? Why or why not?
- What did you think about the moral leaders portrayed in the Moral Exemplars in Business and Professional Ethics module?
- Did you have trouble identifying both vices, i.e., vices of excess and defect? If so, do you think this because some virtues may not have vices of excess and defect? What do you think about Aristotle's doctrine of the mean?
- Did you notice that the virtue profiles given by your group and the other groups in the class overlapped? Is this a problem for virtue theory? Why do our conceptions of the key moral values and virtues overlap?
- Did you find the virtues difficult to apply? What do you think about the utilitarian and deontological criticism of virtue ethics, namely, that it cannot provide us with guidelines on how to act in difficult situations? Should ethical theories emphasize the act or the person? Or both?

- The most tenacious obstacle to working with virtue ethics is to change focus from the morally minimal to the morally exemplary. “Virtue” is the translation of the Greek word, *arête*. But “excellence” is, perhaps, a better word. Understanding virtue ethics requires seeing that virtue is concerned with the exemplary, not the barely passable. (Again, looking at moral exemplars helps.) *Arête* transforms our understanding of common moral values like justice and responsibility by moving from minimally acceptable to exemplary models.

[Moral Leaders](#) The profiles of several moral leaders in practical and professional ethics. [Computer Ethics Cases](#) This link provides several computer ethics cases and also has a description of decision making and socio-technical systems frameworks. [Moral Exemplars in Business and Professional Ethics](#) Profiles of several moral leaders in practical and professional ethics.

Presentation on Virtue Ethics

[An Introduction to Virtue Ethics.pptx](#)

I. Why Study Virtue Ethics?

Reasons

- It provides new insights into moral education
- Involves the whole self: attitudes, knowledge, skill, emotion
- It reorients moral theory toward excellence

II. Three Definitions

Elena Lugo

- “Las virtudes son disposiciones y rasgos del carácter del agente moral a la hora de ejecutar las acciones inherentes al ser persona.
- se trata de un punto intermedio entre dos extremos, ninguno de los cuales representa un valor moral, sino que más bien puede constituir un vicio o al menos carecer de excelencia
- no son meros rasgos del carácter que se operan automáticamente, sino respuestas deliberadas ante las situaciones concretas
- Lugo, E. (2002) *Relación Médico / paciente: encuentro interpersonal ética y espiritualidad*. Pontificia Universidad Católica de Puerto Rico: 88

Rosalind Hursthouse

- “A virtue such as honesty or generosity is not just a tendency to do what is honest or generous, nor is it to be helpfully specified as a “desirable” or “morally valuable” character trait.
- It is, indeed a character trait—that is, a disposition which is well entrenched in its possessor, something that, as we say “goes all the way down”, unlike a habit such as being a tea-drinker—but the disposition in question...is multi-track.
- It is concerned with many other actions as well, with emotions and emotional reactions, choices, values, desires, perceptions, attitudes, interests expectations and sensibilities.
- To possess a virtue is to be a certain sort of person with a certain complex mindset.”
- Hursthouse, R. (2007) “Virtue Ethics” *Stanford Encyclopedia of Philosophy* <http://plato.stanford.edu/entries/ethics-virtue/>

MacIntyre

- MacIntyre, a modern theorist, brings out the communitarianism in Aristotle
- “A virtue is an acquired human quality the possession and exercise of which tend to enable us to achieve those goods which are internal to practices and the lack of which effectively prevents us from achieving any such goods.

III. Virtues and Practices

Virtues are dispositions that bring about the internal and external goods around which a social or professional practice is built.

Constituents of a Practice

- Participants: Formed of individuals whose activities, attitudes, and goals are integrated, shared, or overlap in significant ways
- Rules and Procedures: Participants occupy roles which outline tasks and procedures. Roles in a practice are coordinated so that they combine to bring about complex ends beyond the capabilities of isolated individuals
- Boundaries: Boundaries such as disciplinary and theoretical principles surround practices and serve to distinguish one from the other

- External Goals: Engineering serves public wellbeing. Medicine health. Law justice. Business commerce.
- Internal Goals: Engineering has the internal goals of faithful agency (to client), collegiality (to peers), and loyalty (to the profession or practice itself)

IV. Developing Virtues for Practices

1. Choose a virtue that is important for your occupation or profession. What goods or values does the consistent employment of this virtue produce?
2. Develop a general description of your virtue. (Think along the following lines: people who have virtue X tend to exhibit certain characteristics (or do certain things) in certain kinds of situations. Try to think of these situations in terms of what is common and important to your profession or practice.)
3. Identify the corresponding vices of excess and defect.
4. Identify the obstacles arise that prevent professionals from practicing your virtue? Do well-meaning professionals lack power or technical skill?
5. Identify a moral exemplar for your virtue. Make use of the exemplars described in the Moral Exemplars in Business and Professional Ethics module.
6. Does your virtue stand alone or does it need support from other virtues or skills? For example, integrity might also require moral courage.

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1.6: Ethical Leadership Using "Incident at Morales"

Module Introduction

In this module, you will view the DVD Incident at Morales and carry out a series of activities designed to familiarize you with issues in ethical leadership, social responsibility, and globalization. Links to interviews with major figures on globalization, to the Connexions module "Socio-Technical Systems in Decision Making" and to online material on "Incident at Morales" will help you to gather the information you need to complete this module.

Issues in Incident at Morales

- Quotes are taken from the Study Guide to "Incident at Morales"
- Confidentiality: "Although the lawyers note that Fred has no legal obligations to Chemitoil because he did not sign a non-disclosure agreement, does Fred have a moral obligation to ensure the confidentiality of the information he may have learned at Chemitoil?"
- Wally's "One Rule": What is the impact of Wally's "One Rule" on Fred's ability to do his job? More importantly, does this interfere with Fred's ability to meet his professional ethical obligations in the course of conducting his job?"
- **Lutz and Lutz** Controls: Wally claims that **Lutz and Lutz** controls are the best among the available alternatives. He also claims that the fact that Chuck's brother-in-law works with **Lutz and Lutz** is not a relevant factor. How should Fred choose in this situation regarding controls?
- Couplings: In choosing both the type of couplings and piping as well as to use a local (Mexico) supplier without a plant inspection, what factors should Fred take into account? What should be the margin of error in terms of pressure? How does Fred balance safety and reliability with the need to cut costs due to the parent company's recent acquisitions?
- Environmental Regulations--When in Rome...: Should Fred take advantage of less strict environmental regulations in Mexico to save money for Phaust corporation? What are the responsibilities of multi-national corporations that operate in countries like Mexico?

Exercise 1: Incident at Moral Socio-Technical System

Prepare a socio-technical analysis of Morales, Mexico. Your analysis will examine the insertion of the Phaust chemical plant into the Morales context. Using the following list of values, can you identify any potential value conflicts? Safety, Equity/Justice, intellectual property, confidentiality, responsibility, reasonableness.

Preparing a STS Table

- Study the two templates in the module, "Socio-Technical Systems in Professional Decision Making." See which one applies best to the Incident at Morales case.
- Redo the headings of the table substituting relevant items for those in the templates that are not relevant. For example, in preparing a STS table for a computer system, you may wish to change rate and rate structures into something like data and data structures.
- Fill in the relevant columns in your newly revised table. For example, in the Incident at Morales, the description of the physical surroundings would be based on the brief video segment where Fred is consulting with Wally and Manuel. What is the geographical area like? (It looks like a dry climate given the DVD.) What is the plant like? (It is, at the very least, small.) Attention to detail--even trivial detail--is important for these columns of the STS.
- For the second table, take the short value list we have been working with this semester and (1) look for new value mismatches, (2) identify existing value conflicts, and (3) describe any harmful long term consequences. In Incident at Morales, you may want to concentrate on justice (equity), responsibility for safety, respect, property, and free speech.
- Keep your tables simple and direct. You will have only a few minutes to debrief on them. Remember, this is a device to help you visualize value conflicts hidden in technologies and socio-technical systems.

Socio-Technical System

Hardware	Software	Physical Surroundings	People, Groups, Roles	Procedures	Laws, Statutes, Regulations	Data and Data Structures

STS and Values

	Hardware/Software	Physical Surroundings	People, Groups, Roles	Procedures	Laws, Statutes, Regulations	Data and Data Structures
Justice (Equity and Access)	Responsibility					
Responsibility						
Respect (Privacy and Due Process)						
Property						
Free Speech						

Exercise 2: Opportunities for Ethical Leadership

You will be assigned one of the topics described above. Discuss this topic with your group. Answer the questions. Then prepare a brief summary of your answers to share with the rest of the class. The topics, again, are confidentiality, Wally's "One Rule", Lutz and Lutz Controls, the quality and integrity of the couplings, and the difference in environmental regulations. Throughout your reflections look for opportunities open to Fred to demonstrate ethical leadership. What obstacles stand in his way? What can he do to overcome them?

Decision Point for Business Ethics, Fall 2007

- Generate Solutions, Test Solutions, and Develop a Solution Implementation plan from the perspective of Fred. Focus specifically on whether Fred, as an engineer, should sign off on the plant as it is being passed over to operations.
- **Decision Point:** Chuck's solution to the French company's budget cuts was to pass along long term expenses and operational problems to the plant operation group.
- At the end of the video, Fred has been asked to sign off on the plant's documents and, essentially, approve this "pass along" strategy.
- What kind of ethical problems does Chuck's solution create?
- Knowing this, should Fred have signed off on the plant at the end of the video?
- Take Fred's perspective. Generate solutions, test them, and develop an implementation from Fred's perspective. Summarize your group's work by developing a solution table, solution evaluation matrix, and a feasibility table. Be prepared to summarize (not present) these tables informally to the rest of the class.

Refined Solution Table

Decision Alternative	Description	Justification: problem fit, ethics, feasibility
Solution 1		
Solution 2		

Solution Evaluation Matrix

Solution / Test	Reversibility	Harm / Benefits	Publicity	Feasibility (Global)
Solution 1				
Solution 2				

For Feasibility Table, see m14789.

Exercise 3

Read and listen to the interviews with Shiva, who is opposed to globalization, and O'Rourke, who takes a pro-globalization. Summarize their arguments. Using these arguments, construct your own argument on globalization and apply it to the Morales

case: Is the incident that occurred at Morales an inevitable result of globalization or merely the result of bad individual and corporate decisions?

Incident at Morales in Ethics Bowl

Decision Scenario from "Incident at Morales" (Taken from Study Guide)

- "Although the lawyers note that Fred has no legal obligations to Chemitoil because he did not sign a non-disclosure agreement, does Fred have a moral obligation to ensure the confidentiality of the information he may have learned at Chemitoil?"
- Return to the moment where Wally gives Fred the preliminary plant plans. Then place yourself in the following dialogue:
- WALLY Good. Chuck is going to have a project kick-off meeting this afternoon. Your plant design will be on the agenda. It'll be at three. We don't waste time around here. We're fast at Phaust. Corporate tag line. As Fred gazes around his new workstation, smiling, Wally starts routing through a filing cabinet. He finds the preliminary plant plans and hands them to Fred. WALLY You might want to look at this. (hopeful) Tell me if this is like what you were building at your last job.
- You are Fred. Is Wally asking you to violate your (moral) confidentiality obligation with Chemitoil? Present a response to Wally's question. Show how this response respects both your former employer, Chemitoil, and your current employer, Phaust.

Decision Scenario from "Incident at Morales:" Environmental Integrity or Reliable Controls

- You are Fred. After you point out to Wally, that Lutz and Lutz controls are expensive, he advises you to "pick your fights when you can win them." (Chuck's brother-in-law is the customer representative for Lutz and Lutz.) On the other hand your wife, an EPA compliance litigator, points out how dangerous it is to put untreated toxic waste material in unlined evaporation ponds because of the possibility of drinking water contamination.
- You think about taking Wally's advice. Which fight should you choose, saving the environment while opting for cheaper controls or remaining with the expensive Lutz and Lutz controls but going ahead with the unlined evaporation ponds?
- In your presentation address this broader issue. Is Wally right? Should we trade off safety and environmental concerns when the budget is tight?

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CHAPTER OVERVIEW

2: Ethical Leadership -Making

[2.1: Theory-Building Activities - Rights](#)

[2.2: Three Frameworks for Ethical Problem-Solving in Business and the Professions](#)

[2.3: Toysmart Case Exercises - Student Module](#)

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2.1: Theory-Building Activities - Rights

Module Introduction

Preliminary Draft distributed at APPE, 2005 in San Antonio, TX

Engineers and other professionals work in large corporations under the supervision of managers who may lack their expertise, skills, and commitment to professional standards. This creates communication and ethical challenges. At the very least, professionals are put in the position of having to advocate their ethical and professional standards to those who, while not being opposed to them, may not share their understanding of and commitment to them.

This module is designed to give you the tools and the practice using them necessary to prevail in situations that require advocacy of ethical and professional standards. In this module you carry out several activities. (1) You will study the philosophical and ethical foundations of modern rights theory through a brief look at Kantian Formalism. (2) You will learn a framework for examining the legitimacy of rights claims. (3) You will practice this framework by examining several rights claims that engineers make over their supervisors. This examination will require that you reject certain elements, rephrase others, and generally recast the claim to satisfy the requirements of the rights justification framework. (4) Finally, in small groups, you will build tables around your reformulation of these rights claims and present the results to the class. This module will help you to put your results together with the rest of your classmates and collectively assemble a toolkit consisting of the legitimate rights claims that engineers and other professionals can make over their managers and supervisors.

For more background on rights theory and the relation of rights and duties see (1) Henry Shue, **Basic Rights: Subsistence, Affluence, and U.S. Foreign Policy**, 2nd edition, Princeton, 1980 and (2) Thomas Donaldson, **The Ethics of International Business**, Oxford, 1989. This exercise has been used in computer and engineering ethics classes at the University of Puerto Rico at Mayaguez from 2002 on to the present. It is being incorporated into the textbook, *Good Computing: A Virtue Approach to Computer Ethics* by Chuck Huff, William Frey, and Jose Cruz.

What you need to know...

Problematic Right Claims

1. El derecho para actuar de acuerdo a la conciencia etica y rechazar trabajos en los cuales exista una variacion de opinones morales.
2. El derecho de expresar juicio profesional, y hacer pronunciamientos publicos que sean consistentes con restricciones corporativas sobre la informacion propietaria.
3. El derecho a la lealtad corporativa y la libertad de que sea hecho un chivo expiatorio para catastrofes naturales, ineptitud de administracion u otras fuerzas mas alla del control del ingeniero.
4. El derecho a buscar el mejoramiento personal mediante estudios postgraduados y involucrase en asociaciones profesionales.
5. .El derecho a participar en actividades de partidos politicos fuera de las horas de trabajo.
6. El derecho a solicitar posiciones superiores con otras companias sin que la companis en la que trabaje tome represalias contra el ingeniero.
7. El derecho al debido proceso de ley y la libertad de que se le apliquen penalidades arbitrarias o despidos.
8. El derecho a apelar por revision ante una asociacion profesional, ombudsman o arbitro independiente.
9. El derecho a la privacidad personal.
10. Rights claims come from: Bill W. Baker. (2004) "Engineering Ethics: An Overview," in **Engineering Ethics: Concepts, Viewpoints, Cases and Codes**, eds. Jimmy H. Smith and Patricia M. Harper. Compiled and Published by the National Institute for Engineering Ethics: 21-22.
11. Translated into Spanish and published in: **Etica en la Practica Profesional de la Ingenieria** by Wilfredo Munoz Roman published in 1998 by the Colegio de Ingenieros y Agrimensores de Puerto Rico and Universidad Politecnica de Puerto Rico

Problematic Rights Claims quoted directly from Bill Baker, *Engineering Ethics: An Overview*. Claims form a "Bill of Rights" set forth by Murray A. Muspratt of Chisholm Institute of Technology, Victoria, Australia (*American society of Civil Engineers' Journal of Professional Issues in Engineering*, October 1985)

1. "The right to act in according to ethical conscience and to decline assignments where a variance of moral opinion exists.
2. The right to express professional judgment, and to make public pronouncements that are consistent with corporate constraints on proprietary information.
3. The right to corporate loyalty and freedom from being made a scapegoat for natural catastrophes, administrative ineptitude or other forces beyond the engineer's control.
4. The right to seek self-improvement by further education and involvement in professional associations.
5. The right to participate in political party activities outside of working hours.
6. The right to apply for superior positions with other companies without being blacklisted.
7. The right to due process and freedom from arbitrary penalties or dismissal.
8. The right to appeal for ethical review by a professional association, ombudsman or independent arbitrator.
9. The right to personal privacy."

Kantian Formalism, Part I: Aligning the moral motive and the moral act

- Kant's moral philosophy has exercised substantial influence over our notions of right and duty. We begin with a brief summary of this theory based on the work, **The Foundations of the Metaphysics of Morals**.

- Kant states that the only thing in this world that is good without qualification is a good will. He characterizes this will in terms of its motive, "duty for duty's sake."
- Consider the following example. You see a boy drowning. Even though the water is rough and the current strong you are a good enough swimmer to save him. So while your inclination may be to give way to fear and walk away, you are duty-bound to save the drowning boy.
- An action (saving or not saving the drowning boy) has moral worth depending on the correct correlation of right action and right motive. The following table shows this.

Duty for Duty's Sake

	Motive = Inclination (desire for reward or fear)	Motive = Duty
Act Conforms to Duty	You save the drowning boy for the reward. Act conforms to duty but is motivated by inclination. Has no moral worth.	You save the drowning boy because it is your duty. Act conforms to duty and is for the sake of duty. Your act has moral worth.
Act violates a duty.	You don't save the drowning boy because you are too lazy to jump in. Act violates duty motivated by inclination.	You drown trying to save the drowning boy. He also dies. Act fails to carry out duty but is motivated by duty anyway. The act miscarries but since the motive is duty it still has moral worth.

Part II of Kantian Formalism: Giving content to Duty for Duty's Sake

- Kant sees morality as the expression and realization of the rational will. The first formulation of this rational will is to will consistently and universally.
- This leads to the Categorical Imperative: **I should act only on that maxim (=personal rule or rule that I give to myself) that can be converted into a universal law (=a rule that applies to everybody without self-contradiction).**
- This formulation is an imperative because it commands the will of all reasonable beings. It is categorical because it commands without exceptions or conditions. The CI tells me unconditionally not to lie. It does not say, do not lie unless it promotes your self interest to do so.
- The following table shows how to use the Categorical Imperative to determine whether I have a duty not to lie.

Applying the Categorical Imperative

1. Formulate your maxim (=personal rule)	Whenever I am in a difficult situation, I should tell a lie.
2. Universalize your maxim.	Whenever anybody is in a difficult situation, he or she should tell a lie.
3. Check for a contradiction (logical or practical)	When I lie, I will the opposite for the universal law. Put differently, I will that everybody (but me) be a truth-teller and that everybody believe me a truth-teller. I then make myself the exception to this universal law. Thus my maxim (I am a liar) contradicts the law (everybody else is a truth-teller)

Kantian Formalism, Part III: The Formula of the End

- When I will one thing as universal law and make myself the exception in difficult circumstances, I am treating others, in Kantian terms, merely as means.
- This implies that I subordinate or bend them to my interests and projects without their consent. I do this by circumventing their autonomy through (1) force, (2) fraud (often deception), or (3) manipulation. Treating them with respect would involve telling them what I want (what are my plans and projects) and on this basis asking them to consent to participate and help me. The extreme case for treating others merely as means is enslaving them.
- We do on occasion treat others as means (and not as mere means) when we hire them as employees. But this is consistent with their autonomy and rational consent because we explain to them what is expected (we give them a job description) and compensate them for their efforts. For this reason there is a world of difference between hiring others and enslaving them.
- **The Formula of the End = Act so as to treat others (yourself included) always as ends and never merely as means.**

Some Key Definitions for a Rights Framework

- Kantian formalism provides a foundation for respect for the intrinsic value of humans as autonomous rational beings. Using this as a point of departure, we can develop a method for identifying, spelling out, and justifying the rights and duties that go with professionalism. This framework can be summarized in four general propositions:
- 1. Definition: A **right** is an essential capacity of action that others are obliged to recognize and respect. This definition follows from autonomy. Autonomy can be broken down into a series of specific capacities. Rights claims arise when we identify these capacities and take social action to protect them. Rights are inviolable and cannot be overridden even when overriding would bring about substantial public utility.
- 2. All rights claims must satisfy three requirements. They must be (1) **essential to the autonomy** of individuals and (2) **vulnerable** so that they require special recognition and protection (on the part of both individuals and society). Moreover, the burden of recognizing and respecting a claim as a right must not deprive others of something essential. In other words, it must be (3) **feasible** for both individuals and social groups to recognize and respect legitimate rights claims.

- 3. Definition: A **duty** is a rule or principle requiring that we both recognize and respect the legitimate rights claims of others. Duties attendant on a given right fall into three general forms: (a) duties not to deprive, (b) duties to prevent deprivation, and (c) duties to aid the deprived.
- 4. **Rights and duties are correlative**; for every right there is a correlative series of duties to recognize and respect that right.
- These four summary points together form a system of professional and occupational rights and correlative duties.

Right Claim Justification Framework

- **Essential:** To say that a right is essential to autonomy is to say that it highlights a capacity whose exercise is necessary to the general exercise of autonomy. For example, autonomy is based on certain knowledge skills. Hence, we have a right to an education to develop the knowledge required by autonomy, or we have a right to the knowledge that produces informed consent. In general, rights are devices for recognizing certain capacities as essential to autonomy and respecting individuals in their exercise of these capacities.
- **Vulnerable:** The exercise of the capacity protected under the right needs protection. Individuals may interfere with us in our attempt to exercise our rights. Groups, corporations, and governments might overwhelm us and prevent us from exercising our essential capacities. In short, the exercise of the capacity requires some sort of protection. For example, an individual's privacy is vulnerable to violation. People can gain access to our computers without our authorization and view the information we have stored. They can even use this information to harm us in some way. The right to privacy, thus, protects certain capacities of action that are vulnerable to interference from others. Individual and social energy needs to be expended to protect our privacy.
- **Feasible:** Rights make claims over others; they imply duties that others have. These claims must not deprive the correlative duty-holders of anything essential. In other words, my rights claims over you are not so extensive as to deprive you of your rights. My right to life should not deprive you of your right to self-protection were I to attack you. Thus, the scope of my right claims over you and the rest of society are limited by your ability to reciprocate. I cannot push my claims over you to recognize and respect my rights to the point where you are deprived of something essential.

Types of Duty Correlative to a Right

- **Duty not to deprive:** We have a basic duty not to violate the rights of others. This entails that we must both recognize and respect these rights. For example, computing specialists have the duty not to deprive others of their rights to privacy by hacking into private files.
- **Duty to prevent deprivation:** Professionals, because of their knowledge, are often in the position to prevent others from depriving third parties of their rights. For example, a computing specialist may find that a client is not taking sufficient pains to protect the confidentiality of information about customers. Outsiders could access this information and use it without the consent of the customers. The computing specialist could prevent this violation of privacy by advising the client on ways to protect this information, say, through encryption. The computing specialist is not about to violate the customers' rights to privacy. But because of special knowledge and skill, the computing specialist may be in a position to prevent others from violating this right.
- **Duty to aid the deprived:** Finally, when others have their rights violated, we have the duty to aid them in their recovery from damages. For example, a computing specialist might have a duty to serve as an expert witness in a lawsuit in which the plaintiff seeks to recover damages suffered from having her right to privacy violated. Part of this duty would include accurate, impartial, and expert testimony.

Application of Right/Duty Framework

1. We can identify and define specific rights such as due process. Moreover, we can set forth some of the conditions involved in recognizing and respecting this right.
2. Due Process can be justified by showing that it is essential to autonomy, vulnerable, and feasible.
3. Right holders can be specified.
4. Correlative duties and duty holders can be specified.
5. Finally, the correlative duty-levels can be specified as the duties not to violate rights, duties to prevent rights violations (whenever feasible), and the duties to aid the deprived (whenever is feasible).

Example Rights Table: Due Process

Right: Due Process	Justification	Right-Holder: Engineer as employee and member of professional society.	Correlative Duty-Holder: Engineer's Supervisor, officials in professional society.	Duty Level

Definition: The right to respond to organizational decisions that may harm one in terms of a serious organizational grievance procedure. Necessary Conditions: 1. Several levels of appeal. 2. Time limits to each level of appeal. 3. Written notice of grievance. 4. Peer representation. 5. Outside arbitration.	Essential: Due Process is essential in organizations to prevent the deprivation of other rights or to provide aid in the case of their deprivation. Vulnerable: Rights in general are not recognized in the economic sphere of organizations. Feasible: Organizations, have successfully implemented due process procedures.	Human Resources, Management, Personnel Department. (Individuals with duty to design, implement, and enforce a due process policy) Corporate directors have the duty to make sure this is being done.	Not to Deprive: Individuals cannot be transferred, or demoted without cause. Prevent Deprivation: Organizations must prevent deprivation by designing and implementing a comprehensive due process policy. Aid the Deprived: Binding arbitration measures must exist to aid those who have lost due process rights.
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What you are going to do...

Exercise: Develop a Rights Table

1. You will be divided into small groups and each will be assigned a right claim taken from the above list.
2. Describe the claim (essential capacity of action) made by the right. For example, due process claims the right to a serious organizational grievance procedure that will enable the right-holder to respond to a decision that has an adverse impact on his or her interests. It may also be necessary in some situations to specify the claim's necessary conditions.
3. Justify the right claim using the rights justification framework. In other words show that the right claim is essential, vulnerable, and feasible.
4. Be sure to show that the right is essential to **autonomy**. If it is vulnerable be sure to identify the **standard threat**. (A standard threat is an existing condition that threatens autonomy.)
5. Provide an example of a situation in which the right claim becomes active. For example, an engineer may claim a right to due process in order to appeal what he or she considers an unfair dismissal, transfer, or performance evaluation.
6. Identify the correlative duty-holder(s) that need to take steps to recognize and respect the right. For example, private and government organizations may be duty-bound to create due process procedures to recognize and respect this right.
7. Further spell out the right by showing what actions the correlative duties involve. For example, a manager should not violate an employee's due process right by firing him or her without just cause. The organization's human resources department might carry out a training program to help managers avoid depriving employees of this right. The organization could aid the deprived by designing and implementing binding arbitration involving an impartial third party.

Be prepared to debrief on your right claim to the rest of the class. When other groups are debriefing, you are free to challenge them on whether their claim is essential to autonomy, whether they have identified a valid "standard threat," and whether the correlative duties are feasible or deprive others of something essential. Your goal as a class is to have a short but effective list of rights that professionals take with them to the workplace.

Makes copies of your rights table and give it to the other groups in class. Be sure to make a copy for your instructor. Together, you will build a table of rights claims that engineers and other professionals make against managers and corporations. This will provide you a useful and comprehensive decision-making tool in that you will be able to examine decision alternatives in terms of how they stand with regard to the rights you and your classmates and scrutinized and justified through this exercise.

Conclusion

Conclusion: Topics for Further Reflection

- Not every claim to a right is a legitimate or justifiable claim. The purpose of this framework is to get you into the habit of thinking critically and skeptically about the rights claims that you and others make. Every legitimate right claim is essential, vulnerable, and feasible. Correlative duties are sorted out according to different levels (not to deprive, prevent deprivation, and aid the deprived); this, in turn, is based on the capacity of the correlative duty holder to carry them out. Finally, duties correlative to rights cannot deprive the duty-holder of something essential.
- Unless you integrate your right and its correlative duties into the context of your professional or practical domain, it will remain abstract and irrelevant. Think about your right in the context of the real world. Think of everyday situations in which the right and its correlative duties will arise. Invent cases and scenarios. If you are an engineering student, think of informed consent in terms of the public's right to understand and consent to the risks associated with engineering projects. If you are a computing student think of what you can do with computing knowledge and skills to respect or violate privacy rights. Don't stop with an abstract accounting of the right and its correlative duties.
- Rights and duties underlie professional codes of ethics. But this is not always obvious. For example, the right of free and informed consent underlies much of the engineer's interaction with the public, especially the code responsibility to hold paramount public health, safety, and welfare. Look at the different stakeholder relations covered in a code of ethics. (In engineering this would include public, client, profession, and peer.) What are the rights and duties outlined in these stakeholder relations? How are they covered in codes of ethics?
- This module is effective in counter-acting the tendency to invent rights and use them to rationalize dubious actions and intentions. Think of rights claims as credit backed by a promise to pay at a later time. If you make a right claim, be ready to justify it. If someone else makes a right

claim, make them back it up with the justification framework presented in this module.

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2.2: Three Frameworks for Ethical Problem-Solving in Business and the Professions

Module Introduction

In this module, you will learn and practice three frameworks designed to integrate ethics into decision making in the areas of practical and occupational ethics. The first framework divides the decision making process into four stages: problem specification, solution generation, solution testing, and solution implementation. It is based on an analogy between ethics and design problems that is detailed in a table presented below. The second framework focuses on the process of testing solution alternatives for their ethics by deploying three ethics tests that will help you to evaluate and rank alternative courses of action. The reversibility, harm, and publicity tests each "encapsulate" or summarize an important ethical theory. Finally, a feasibility test will help you to uncover interest, resource, and technical constraints that will affect and possibly impede the realization of your solution or decision. Taken together, these three frameworks will help steer you toward designing and implementing ethical solutions to problems in the professional and occupational areas.

Two online resources provide more extensive background information. The first, www.computingcases.org, provides background information on the ethics tests, socio-technical analysis, and intermediate moral concepts. The second, onlineethics.org/essays/educa.../teaching.html, explores in more detail the analogy between ethics and design problems. Much of this information will be published in *Good Computing: A Virtue Approach to Computer Ethics*, a textbook of cases and decision-making techniques in computer ethics that is being authored by Chuck Huff, William Frey, and Jose A. Cruz-Cruz.

Problem-Solving or Decision-Making Framework: Analogy between ethics and design

Traditionally, problem-solving frameworks in professional and occupational ethics have been taken from rational decision procedures used in economics. While these are useful, they lead one to think that ethical decisions are already "out there" waiting to be discovered. In contrast, taking a design approach to ethical decision making emphasizes that ethical decisions must be created, not discovered. This, in turn, emphasizes the importance of moral imagination and moral creativity. Carolyn Whitbeck in *Ethics in Engineering Practice and Research* describes this aspect of ethical decision making through the analogy she draws between ethics and design problems in chapter one. Here she rejects the idea that ethical problems are multiple-choice problems. We solve ethical problems not by choosing between ready-made solutions given with the situation; rather we use our moral creativity and moral imagination to design these solutions. Chuck Huff builds on this by modifying the design method used in software engineering so that it can help structure the process of framing ethical situations and creating actions to bring these situations to a successful and ethical conclusion. The key points in the analogy between ethical and design problems are summarized in the table presented just below.

Analogy between design and ethics problem-solving	
Design Problem	Ethical Problem
Construct a prototype that optimizes (or satisfices) designated specifications	Construct a solution that integrates and realizes ethical values (justice, responsibility, reasonableness, respect, and safety)
Resolve conflicts between different specifications by means of integration	Resolve conflicts between values (moral vs. moral or moral vs. non-moral) by integration
Test prototype over the different specifications	Test solution over different ethical considerations encapsulated in ethics tests
Implement tested design over background constraints	Implement ethically tested solution over resource, interest, and technical constraints

Software Development Cycle: Four Stages

(1) problem specification, (2) solution generation, (3) solution testing, and (4) solution implementation.

Problem specification

Problem specification involves exercising moral imagination to specify the socio-technical system (including the stakeholders) that will influence and will be influenced by the decision we are about to make. Stating the problem clearly and concisely is essential to

design problems; getting the problem right helps structure and channel the process of designing and implementing the solution. There is no algorithm available to crank out effective problem specification. Instead, we offer a series of guidelines or rules of thumb to get you started in a process that is accomplished by the skillful exercise of moral imagination.

For a broader problem framing model see Harris, Pritchard, and Rabins, *Engineering Ethics: Concepts and Cases*, 2nd Edition, Belmont, CA: Wadsworth, 2000, pp. 30-56. See also Cynthia Brincat and Victoria Wike, *Morality and Professional Life: Values at Work*, New Jersey: Prentice Hall, 1999.

Different Ways of Specifying the Problem

- Many problems can be specified as disagreements. For example, you disagree with your supervisor over the safety of the manufacturing environment. Disagreements over facts can be resolved by gathering more information. Disagreements over concepts (you and your supervisor have different ideas of what safety means) require working toward a common definition.
- Other problems involve conflicting values. You advocate installing pollution control technology because you value environmental quality and safety. Your supervisor resists this course of action because she values maintaining a solid profit margin. This is a conflict between a moral value (safety and environmental quality) and a nonmoral value (solid profits). Moral values can also conflict with one another in a given situation. Using John Doe lawsuits to force Internet Service Providers to reveal the real identities of defamers certainly protects the privacy and reputations of potential targets of defamation. But it also places restrictions on legitimate free speech by making it possible for powerful wrongdoers to intimidate those who would publicize their wrongdoing. Here the moral values of privacy and free speech are in conflict. Value conflicts can be addressed by harmonizing the conflicting values, compromising on conflicting values by partially realizing them, or setting one value aside while realizing the other (=value trade offs).
- If you specify your problem as a disagreement, you need to describe the facts or concepts about which there is disagreement.
- If you specify your problem as a conflict, you need to describe the values that conflict in the situation.
- One useful way of specifying a problem is to carry out a stakeholder analysis. A stakeholder is any group or individual that has a vital interest at risk in the situation. Stakeholder interests frequently come into conflict and solving these conflicts requires developing strategies to reconcile and realize the conflicting stakes.
- Another way of identifying and specifying problems is to carry out a socio-technical analysis. Socio-technical systems (STS) embody values. Problems can be anticipated and prevented by specifying possible value conflicts. Integrating a new technology, procedure, or policy into a socio-technical system can create three kinds of problem. (1) Conflict between values in the technology and those in the STS. For example, when an attempt is made to integrate an information system into the STS of a small business, the values present in an information system can conflict with those in the socio-technical system. (Workers may feel that the new information system invades their privacy.) (2) Amplification of existing value conflicts in the STS. The introduction of a new technology may magnify an existing value conflict. Digitalizing textbooks may undermine copyrights because digital media is easy to copy and disseminate on the Internet. (3) Harmful consequences. Introducing something new into a socio-technical system may set in motion a chain of events that will eventually harm stakeholders in the socio-technical system. For example, giving laptop computers to public school students may produce long term environmental harm when careless disposal of spent laptops releases toxic materials into the environment.
- The following table helps summarize some of these problem categories and then outlines generic solutions.

Problem Type	Sub-Type	Solution Outline		
Disagreement	Factual	Type and mode of gathering information		
	Conceptual	Concept in dispute and method for agreeing on its definition		
Conflict	Moral vs. Moral	Value Integrative	Partially Value Integrative	Trade Off
	Non-moral vs. moral			
	Non-moral vs. non-moral			

Moral Ecologies	Finance-Driven Ecologies Customer-Driven Ecologies Quality-Driven Ecologies	Strategy for dissenting from a staff position where one is outside decision-making	Practicing ethical advocacy when "going to the mat" on ethical perspectives in group decision-making	Ability to draw attention to ethical values that form center of organization identity
Likely Concepts in Conceptual Disagreement	Public Intellectual Property, Faithful Agency, Professional Integrity, Loyalty, Public Safety and Health, Due Process, Responsible Dissent	Working from Legal Definitions	Bridging: moving from cases to concepts	Discussion: Playing on shared values and trust to reach consensus through dialogue

The materials on moral ecologies come from Huff, C., Barnard, L., and Frey, W. (2008). "Good computing: a pedagogically focused model of virtue in the practice of computing (parts 1 and 2)", *Journal of Information, Communication and Ethics in Society*, Volume 6, Issues 3 and 4: 246-316. See also, Michael Davis, *Thinking Like An Engineer*, Oxford, 1998, 119-156.

Instructions for Using Problem Classification Table

1. Is your problem a conflict? Moral versus moral value? Moral versus non-moral values? Non-moral versus non-moral values? Identify the conflicting values as concisely as possible. Example: In Toysmart, the financial values of creditors come into conflict with the privacy of individuals in the database: financial versus privacy values.
2. Is your problem a disagreement? Is the disagreement over basic facts? Are these facts observable? Is it a disagreement over a basic concept? What is the concept? Is it a factual disagreement that, upon further reflection, changes into a conceptual disagreement?
3. Does your problem arise from an impending harm? What is the harm? What is its magnitude? What is the probability that it will occur?
4. If your problem is a value conflict then can these values be fully integrated in a value integrating solution? Or must they be partially realized in a compromise or traded off against one another?
5. If your problem is a factual disagreement, what is the procedure for gathering the required information, if this is feasible?
6. If your problem is a conceptual disagreement, how can this be overcome? By consulting a government policy or regulation? (OSHA on safety for example.) By consulting a theoretical account of the value in question? (Reading a philosophical analysis of privacy.) By collecting past cases that involve the same concept and drawing analogies and comparisons to the present case?

Moral Ecologies

- "Moral Ecology" refers to the organization in which one works. Calling this organization an "ecology" conveys the idea that it is a system of interrelated parts. These "ecologies" differ depending on the content of the organization's central, identity-conferring values.
- In finance-driven companies, financial values form the core of the organization's identity. Ethical advocacy requires skills in bringing ethical issues to the attention of decision-makers and getting them to take these issues seriously. It helps to state ethical concerns in multi-disciplinary language. (For example, show that ignoring ethical concerns will cost the company money in the long run.)
- Customer-driven ecologies place customer values like usability, affordability, and efficiency, in the forefront of group deliberation and decision-making. Often, one must play the role of "ethics advocate" in deliberation and decision-making. One is expected to argue forcefully and persistently ("go to the mat") to make sure that ethical considerations are integrated into group deliberations and decision-making.
- Quality-driven companies place ethical values into the core of group deliberations and decision-making. Here one is not so much ethics advocate as ethics enabler. This new role requires that one help one's group find creative ways of integrating ethical values with other concerns like customer and financial values.

If you are having problems specifying your problem

- Try identifying the stakeholders. Stakeholders are any group or individual with a vital interest at stake in the situation at hand.

- Project yourself imaginatively into the perspectives of each stakeholder. How does the situation look from their standpoint? What are their interests? How do they feel about their interests?
- Compare the results of these different imaginative projections. Do any stakeholder interests conflict? Do the stakeholders themselves stand in conflict?
- If the answer to one or both of these questions is "yes" then this is your problem statement. How does one reconcile conflicting stakeholders or conflicting stakeholder interests in this situation?

Framing Your Problem

- We miss solutions to problems because we choose to frame them in only one way.
- For example, the Mountain Terrorist Dilemma is usually framed in only one way: as a dilemma, that is, a forced decision between two equally undesirable alternatives. (Gilbane Gold is also framed as a dilemma: blow the whistle on Z-Corp or go along with the excess pollution.)
- Framing a problem differently opens up new horizons of solution. Your requirement from this point on in the semester is to frame every problem you are assigned in at least two different ways.
- For examples of how to frame problems using socio-technical system analysis see module m14025.
- These different frames are summarized in the next box below.

Different Frames for Problems

- Technical Frame: Engineers frame problems technically, that is, they specify a problem as raising a technical issue and requiring a technical design for its resolution. For example, in the Hughes case, a technical frame would raise the problem of how to streamline the manufacturing and testing processes of the chips.
- Physical Frame: In the Laminating Press case, the physical frame would raise the problem of how the layout of the room could be changed to reduce the white powder. Would better ventilation eliminate or mitigate the white powder problem?
- Social Frame: In the "When in Aguadilla" case, the Japanese engineer is uncomfortable working with the Puerto Rican woman engineer because of social and cultural beliefs concerning women still widely held by men in Japan. Framing this as a social problem would involve asking whether there would be ways of getting the Japanese engineer to see things from the Puerto Rican point of view.
- Financial or Market-Based Frames: The DOE, in the Risk Assessment case below, accuses the laboratory and its engineers of trying to extend the contract to make more money. The supervisor of the head of the risk assessment team pressures the team leader to complete the risk assessment as quickly as possible so as not to lose the contract. These two framings highlight financial issues.
- Managerial Frame: As the leader of the Puerto Rican team in the "When in Aguadilla" case, you need to exercise leadership in your team. The refusal of the Japanese engineer to work with a member of your team creates a management problem. What would a good leader, a good manager, do in this situation? What does it mean to call this a management problem? What management strategies would help solve it?
- Legal Frame: OSHA may have clear regulations concerning the white powder produced by laminating presses. How can you find out about these regulations? What would be involved in complying with them? If they cost money, how would you get this money? These are questions that arise when you frame the Laminating Press case as a legal problem.
- Environmental Framing: Finally, viewing your problem from an environmental frame leads you to consider the impact of your decision on the environment. Does it harm the environment? Can this harm be avoided? Can it be mitigated? Can it be offset? (Could you replant elsewhere the trees you cut down to build your new plant?) Could you develop a short term environmental solution to "buy time" for designing and implementing a longer term solution? Framing your problem as an environmental problem requires that you ask whether this solution harms the environment and whether this harming can be avoided or remedied in some other way.

Solution Generation

In solution generation, agents exercise moral creativity by brainstorming to come up with solution options designed to resolve the disagreements and value conflicts identified in the problem specification stage. Brainstorming is crucial to generating nonobvious solutions to difficult, intractable problems. This process must take place within a non-polarized environment where the members of the group respect and trust one another. (See the module on the Ethics of Group Work for more information on how groups can be successful and pitfalls that commonly trip up groups.) Groups effectively initiate the brainstorming process by suspending criticism and analysis. After the process is completed (say, by meeting a quota), then participants can refine the solutions generated by combining them, eliminating those that don't fit the problem, and ranking them in terms of their ethics and feasibility. If a problem

can't be solved, perhaps it can be dissolved through reformulation. If an entire problem can't be solved, perhaps the problem can be broken down into parts some of which can be readily solved.

Having trouble generating solutions?

- One of the most difficult stages in problem-solving is to jump-start the process of brainstorming solutions. If you are stuck then here are some generic options guaranteed to get you "unstuck."
- Gather Information: Many disagreements can be resolved by gathering more information. Because this is the easiest and least painful way of reaching consensus, it is almost always best to start here. Gathering information may not be possible because of different constraints: there may not be enough time, the facts may be too expensive to gather, or the information required goes beyond scientific or technical knowledge. Sometimes gathering more information does not solve the problem but allows for a new, more fruitful formulation of the problem. Harris, Pritchard, and Rabins in *Engineering Ethics: Concepts and Cases* show how solving a factual disagreement allows a more profound conceptual disagreement to emerge.
- Nolo Contendere. Nolo Contendere is Latin for not opposing or contending. Your interests may conflict with your supervisor but he or she may be too powerful to reason with or oppose. So your only choice here is to give in to his or her interests. The problem with nolo contendere is that non-opposition is often taken as agreement. You may need to document (e.g., through memos) that your choosing not to oppose does not indicate agreement.
- Negotiate. Good communication and diplomatic skills may make it possible to negotiate a solution that respects the different interests. Value integrative solutions are designed to integrate conflicting values. Compromises allow for partial realization of the conflicting interests. (See the module, *The Ethics of Team Work*, for compromise strategies such as logrolling or bridging.) Sometimes it may be necessary to set aside one's interests for the present with the understanding that these will be taken care of at a later time. This requires trust.
- Oppose. If nolo contendere and negotiation are not possible, then opposition may be necessary. Opposition requires marshaling evidence to document one's position persuasively and impartially. It makes use of strategies such as leading an "organizational charge" or "blowing the whistle." For more on whistle-blowing consult the discussion of whistleblowing in the Hughes case that can be found in computing cases.
- Exit. Opposition may not be possible if one lacks organizational power or documented evidence. Nolo contendere will not suffice if non-opposition implicates one in wrongdoing. Negotiation will not succeed without a necessary basis of trust or a serious value integrative solution. As a last resort, one may have to exit from the situation by asking for reassignment or resigning.

Refining solutions

- Are any solutions blatantly unethical or unrealizable?
- Do any solutions overlap? Can these be integrated into broader solutions?
- Can solutions be brought together as courses of action that can be pursued simultaneously?
- Go back to the problem specification? Can any solutions be eliminated because they do not address the problem? (Or can the problem be revised to better fit what, intuitively, is a good solution.)
- Can solutions be brought together as successive courses of action? For example, one solution represents Plan A; if it does not work then another solution, Plan B, can be pursued. (You negotiate the problem with your supervisor. If she fails to agree, then you oppose your supervisor on the grounds that her position is wrong. If this fails, you conform or exit.)
- The goal here is to reduce the solution list to something manageable, say, a best, a second best, and a third best. Try adding a bad solution to heighten strategic points of comparison. The list should be short so that the remaining solutions can be intensively examined as to their ethics and feasibility.

Solution Testing: The solutions developed in the second stage must be tested in various ways.

1. Reversibility: Would I still think the choice of this option good if I were one of those adversely affected by it? (Davis uses this formulation in various publications.) I identify different stakeholders and then take up their roles. Through this imaginative projection, I should consider how the action under consideration will affect them and how they will view, interpret, and experience this affect.
2. Harm: Does this option do less harm than any available alternative? Here I try to design an action that will minimize harmful effects. I should factor in the likely results of the action under consideration but I should also evaluate how justly these results will be distributed among stakeholders.
3. Publicity: What kind of person will I become if I choose this action? This is Davis' formulation of this test as a virtue test. The key to this test is that you associate the agent with the action. If I (the agent) am publicly judged as a person in terms of this

action, what does this say about me as a person? Am I comfortable being judged an irresponsible person on the basis of my being identified with my irresponsible action?

4. Meta-Test - Convergence: Do a quick inventory here. Do the ethics tests come together and agree on ranking this solution as a strong one? Then this solution satisfies the convergence meta-test and this provides independent evidence of the strength of the solution.
5. Meta-Test - Divergence: Again, do a quick inventory of your solution evaluation matrix results to this point. Do the tests differ or diverge on this point? This is independent evidence of the weakness of this solution. Think about why this solution may be strong under one test but weak under the others.
6. The solution evaluation matrix presented just below models and summarizes the solution testing process.

Solution Evaluation Matrix

Solution/Test	Reversibility	Harm	Publicity	Meta-Test: Convergence	Meta-Test: Divergence
Description	Would I still think the choice of this option good if I were one of those adversely affected by it? (Davis)	Does this option do less harm than any available alternative?	What person would I become were I to choose and perform this action? (Associating my character with the moral color of the action.)	Do the three ethics tests (reversibility, harm, publicity) come together on this solution?	Do the three ethics tests (reversibility, harm, publicity) differ on this solution?
Your best solution					
A good (but not the best) solution					
Your worst solution or a really bad solution					

Solution Implementation

The chosen solution must be examined in terms of how well it responds to various situational constraints that could impede its implementation. What will be its costs? Can it be implemented within necessary time constraints? Does it honor recognized technical limitations or does it require pushing these back through innovation and discovery? Does it comply with legal and regulatory requirements? Finally, could the surrounding organizational, political, and social environments give rise to obstacles to the implementation of the solution? In general this phase requires looking at interest, technical, and resource constraints or limitations. A Feasibility Matrix helps to guide this process.

The Feasibility Tests focuses on situational constraints. How could these hinder the implementation of the solution? Should the solution be modified to ease implementation? Can the constraints be removed or remodeled by negotiation, compromise, or education? Can implementation be facilitated by modifying both the solution and changing the constraints?

Feasibility Matrix		
Resource Constraints	Technical Constraints	Interest Constraints
		Personalities
Time		Organizational
Cost	Applicable Technology	Legal
Materials	Manufacturability	Social, Political, Cultural

Different Feasibility Constraints

1. The Feasibility Test identifies the constraints that could interfere with realizing a solution. This test also sorts out these constraints into resource (time, cost, materials), interest (individuals, organizations, legal, social, political), and technical limitations. By identifying situational constraints, problem-solvers can anticipate implementation problems and take early steps to prevent or mitigate them.
2. Time. Is there a deadline within which the solution has to be enacted? Is this deadline fixed or negotiable?
3. Financial. Are there cost constraints on implementing the ethical solution? Can these be extended by raising more funds? Can they be extended by cutting existing costs? Can agents negotiate for more money for implementation?
4. Technical. Technical limits constrain the ability to implement solutions. What, then, are the technical limitations to realizing and implementing the solution? Could these be moved back by modifying the solution or by adopting new technologies?
5. Manufacturability. Are there manufacturing constraints on the solution at hand? Given time, cost, and technical feasibility, what are the manufacturing limits to implementing the solution? Once again, are these limits fixed or flexible, rigid or negotiable?
6. Legal. How does the proposed solution stand with respect to existing laws, legal structures, and regulations? Does it create disposal problems addressed in existing regulations? Does it respond to and minimize the possibility of adverse legal action? Are there legal constraints that go against the ethical values embodied in the solution? Again, are these legal constraints fixed or negotiable?
7. Individual Interest Constraints. Individuals with conflicting interests may oppose the implementation of the solution. For example, an insecure supervisor may oppose the solution because he fears it will undermine his authority. Are these individual interest constraints fixed or negotiable?
8. Organizational. Inconsistencies between the solution and the formal or informal rules of an organization may give rise to implementation obstacles. Implementing the solution may require support of those higher up in the management hierarchy. The solution may conflict with organization rules, management structures, traditions, or financial objectives. Once again, are these constraints fixed or flexible?
9. Social, Cultural, or Political. The socio-technical system within which the solution is to be implemented contains certain social structures, cultural traditions, and political ideologies. How do these stand with respect to the solution? For example, does a climate of suspicion of high technology threaten to create political opposition to the solution? What kinds of social, cultural, or political problems could arise? Are these fixed or can they be altered through negotiation, education, or persuasion?

Ethics Tests For Solution Evaluation

Three ethics tests (reversibility, harm/beneficence, and public identification) encapsulate three ethical approaches (deontology, utilitarianism, and virtue ethics) and form the basis of stage three of the SDC, solution testing. A fourth test (a value realization test) builds upon the public identification/virtue ethics test by evaluating a solution in terms of the values it harmonizes, promotes, protects, or realizes. Finally a code test provides an independent check on the ethics tests and also highlights intermediate moral concepts such as safety, health, welfare, faithful agency, conflict of interest, confidentiality, professional integrity, collegiality, privacy, property, free speech, and equity/access). The following section provides advice on how to use these tests. More information can be found at www.computingcases.org.

Setting Up the Ethics Tests: Pitfalls to avoid

Set-Up Pitfalls: Mistakes in this area lead to the analysis becoming unfocused and getting lost in irrelevancies. (a) Agent-switching where the analysis falls prey to irrelevancies that crop up when the test application is not grounded in the standpoint of a single agent, (b) Sloppy action-description where the analysis fails because no specific action has been tested, (c) Test-switching where the analysis fails because one test is substituted for another. (For example, the public identification and reversibility tests are often reduced to the harm/beneficence test where harmful consequences are listed but not associated with the agent or stakeholders.)

Set up the test

1. Identify the agent (the person who is going to perform the action)
2. Describe the action or solution that is being tested (what the agent is going to do or perform)
3. Identify the stakeholders (those individuals or groups who are going to be affected by the action), and their stakes (interests, values, goods, rights, needs, etc.
4. Identify, sort out, and weigh the consequences (the results the action is likely to bring about)

Harm/Beneficence Test

- What harms would accompany the action under consideration? Would it produce physical or mental suffering, impose financial or non-financial costs, or deprive others of important or essential goods?
- What benefits would this action bring about? Would it increase safety, quality of life, health, security, or other goods both moral and non-moral?
- What is the magnitude of each these consequences? Magnitude includes likelihood it will occur (probability), the severity of its impact (minor or major harm) and the range of people affected.
- Identify one or two other viable alternatives and repeat these steps for them. Some of these may be modifications of the basic action that attempt to minimize some of the likely harms. These alternatives will establish a basis for assessing your alternative by comparing it with others.
- Decide on the basis of the test which alternative produces the best ratio of benefits to harms?
- Check for inequities in the distribution of harms and benefits. Do all the harms fall on one individual (or group)? Do all of the benefits fall on another? If harms and benefits are inequitably distributed, can they be redistributed? What is the impact of redistribution on the original solution imposed?

Pitfalls of the Harm/Beneficence Test

1. "Paralysis of Analysis" comes from considering too many consequences and not focusing only on those relevant to your decision.
2. Incomplete Analysis results from considering too few consequences. Often it indicates a failure of moral imagination which, in this case, is the ability to envision the consequences of each action alternative.
3. Failure to compare different alternatives can lead to a decision that is too limited and one-sided.
4. Failure to weigh harms against benefits occurs when decision-makers lack the experience to make the qualitative comparisons required in ethical decision making.
5. Finally, justice failures result from ignoring the fairness of the distribution of harms and benefits. This leads to a solution which may maximize benefits and minimize harms but still give rise to serious injustices in the distribution of these benefits and harms.

Reversibility Test

1. Set up the test by (i) identifying the agent, (ii) describing the action, and (iii) identifying the stakeholders and their stakes.
2. Use the stakeholder analysis to identify the relations to be reversed.
3. Reverse roles between the agent (you) and each stakeholder: put them in your place (as the agent) and yourself in their place (as the one subjected to the action).
4. If you were in their place, would you still find the action acceptable?

Cross Checks for Reversibility Test (These questions help you to check if you have carried out the reversibility test properly.)

- Does the proposed action treat others with respect? (Does it recognize their autonomy or circumvent it?)
- Does the action violate the rights of others? (Examples of rights: free and informed consent, privacy, freedom of conscience, due process, property, freedom of expression)
- Would you recommend that this action become a universal rule?
- Are you, through your action, treating others merely as means?

Pitfalls of the Reversibility Test

- Leaving out a key stakeholder relation
- Failing to recognize and address conflicts between stakeholders and their conflicting stakes
- Confusing treating others with respect with capitulating to their demands ("Reversing with Hitler")
- Failing to reach closure, i.e., an overall, global reversal assessment that takes into account all the stakeholders the agent has reversed with.

Steps in Applying the Public Identification Test

- Set up the analysis by identifying the agent, describing the action, and listing the key values or virtues at play in the situation.
- Association the action with the agent.
- Describe what the action says about the agent as a person. Does it reveal him or her as someone associated with a virtue or a vice?

Alternative Version of Public Identification

- Does the action under consideration realize justice or does it pose an excess or defect of justice?
- Does the action realize responsibility or pose an excess or defect of responsibility?
- Does the action realize reasonableness or pose too much or too little reasonableness?
- Does the action realize honesty or pose too much or too little honesty?
- Does the action realize integrity or pose too much or too little integrity?

Pitfalls of Public Identification

- Action not associated with agent. The most common pitfall is failure to associate the agent and the action. The action may have bad consequences and it may treat individuals with respect but these points are not as important in the context of this test as what they imply about the agent as a person who deliberately performs such an action.
- Failure to specify moral quality, virtue, or value. Another pitfall is to associate the action and agent but only ascribe a vague or ambiguous moral quality to the agent. To say, for example, that willfully harming the public is bad fails to zero in on precisely what moral quality this ascribes to the agent. Does it render him or her unjust, irresponsible, corrupt, dishonest, or unreasonable? The virtue list given above will help to specify this moral quality.

Code of Ethics Test

- Does the action hold paramount the health, safety, and welfare of the public, i.e., those affected by the action but not able to participate in its design or execution?
- Does the action maintain faithful agency with the client by not abusing trust, avoiding conflicts of interest, and maintaining confidences?
- Is the action consistent with the reputation, honor, dignity, and integrity of the profession?
- Does the action serve to maintain collegial relations with professional peers?

Meta Tests

- The ethics and feasibility tests will not always converge on the same solution. There is a complicated answer for why this is the case but the simple version is that the tests do not always agree on a given solution because each test (and the ethical theory it encapsulates) covers a different domain or dimension of the action situation. Meta tests turn this disadvantage to your advantage by feeding the interaction between the tests on a given solution back into the evaluation of that solution.
- When the ethics tests converge on a given solution, this convergence is a sign of the strength and robustness of the solution and counts in its favor.
- When a given solution responds well to one test but does poorly under another, this is a sign that the solution needs further development and revision. It is not a sign that one test is relevant while the others are not. Divergence between test results is a sign that the solution is weak.

Application Exercise

You will now practice the four stages of decision making with a real-world case. This case, Risk Assessment, came from a retreat on Business, Science, and Engineering Ethics held in Puerto Rico in December 1998. It was funded by the National Science Foundation, Grant SBR 9810253.

Risk Assessment Scenario Case Scenario: You supervise a group of engineers working for a private laboratory with expertise in nuclear waste disposal and risk assessment. The DOE (Department of Energy) awarded a contract to your laboratory six years ago to do a risk assessment of various nuclear waste disposal sites. During the six years in which your team has been doing the study, new and more accurate calculations in risk assessment have become available. Your laboratory's study, however, began with the older, simpler calculations and cannot integrate the newer without substantially delaying completion. You, as the leader of the team, propose a delay to the DOE on the grounds that it is necessary to use the more advanced calculations. Your position is that the laboratory needs more time because of the extensive calculations required; you argue that your group must use state of the art science in doing its risk assessment. The DOE says you are using overly high standards of risk assessment to prolong the process, extend the contract, and get more money for your company. They want you to use simpler calculations and finish the project; if you are unwilling to do so, they plan to find another company that thinks differently. Meanwhile, back at the laboratory, your supervisor (a high level company manager) expresses to you the concern that while good science is important in an academic setting, this is the real world and the contract with the DOE is in jeopardy. What should you do?

Part One: Problem Specification

1. Specify the problem in the above scenario. Be as concise and specific as possible
2. Is your problem best specifiable as a disagreement? Between whom? Over what?
3. Can your problem be specified as a value conflict? What are the values in conflict? Are the moral, nonmoral, or both?

Part Two: Solution Generation

1. Quickly and without analysis or criticism brainstorm 5 to ten solutions
2. Refine your solution list. Can solutions be eliminated? (On what basis?) Can solutions be combined? Can solutions be combined as plan a and plan b?
3. If you specified your problem as a disagreement, how do your solutions resolve the disagreement? Can you negotiate interests over positions? What if your plan of action doesn't work?
4. If you formulated your problem as a value conflict, how do your solutions resolve this conflict? By integrating the conflicting values? By partially realizing them through a value compromise? By trading one value off for another?

Part Three: Solution Testing

1. Construct a solution evaluation matrix to compare two to three solution alternatives.
2. Choose a bad solution and then compare to it the two strongest solutions you have.
3. Be sure to avoid the pitfalls described above and set up each test carefully.

Part Four: Solution Implementation

1. Develop an implementation plan for your best solution. This plan should anticipate obstacles and offer means for overcoming them.
2. Prepare a feasibility table outlining these issues using the table presented above.
3. Remember that each of these feasibility constraints is negotiable and therefore flexible. If you choose to set aside a feasibility constraint then you need to outline how you would negotiate the extension of that constraint.

Decision-Making Presentation

[Decision Making Manual V4.pptx](#)

Clicking on this figure will allow you to open a presentation designed to introduce problem solving in ethics as analogous to that in design, summarize the concept of a socio-technical system, and provide an orientation in the four stages of problem solving. This presentation was given February 28, 2008 at UPRM for ADMI 6005 students, Special Topics in Research Ethics.

Problem Solving Presentation

[Decision Making Manual V5.pptx](#)

Shortened Presentation for Fall 2012

[Decision Making Manual V6.pptx](#)

Vigo Socio-Technical System Table and Problems

[Vigo STS.docx](#)

Decision Making Worksheet

[Decision Making Worksheet.docx](#)

This exercise is designed to give you practice with the three frameworks described in this module. It is based on the case, "When in Aguadilla."

Test Rubric Fall 2009: Problem-Solving

[PE Rubric EO S09.docx](#)

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2.3: Toysmart Case Exercises - Student Module

Introduction

In this module, you will study a real-world ethical problem, the Toysmart case, and employ frameworks based on the software development cycle to (1) specify ethical and technical problems, (2) generate solutions that integrate ethical value, (3) test these solutions, and (4) implement them over situation-based constraints. This module will provide you with an opportunity to practice integrating ethical considerations into real-world decision-making and problem-solving in business and computing. This whole approach is based on an analogy between ethics and design (Whitbeck).

Large real-world cases like Toysmart pivot around crucial decision points. You will take on the role of one of the participants in the Toysmart case and problem-solve in teams from one of three decision points. Problem-solving in the real world requires perseverance, moral creativity, moral imagination, and reasonableness; one appropriates these skills through practice in different contexts. Designing and implementing solutions requires identifying conflicting values and interests, balancing them in creative and dynamic solutions, overcoming technical limits, and responding creatively to real-world constraints.

Each decision point requires that you take up the position of a participant in the case and work through decision-making frameworks from his or her perspective. You may be tempted to back out and adopt an evaluative posture from which to judge the participants. Resist this temptation. This module is specifically designed to give you practice in making real-world decisions. These skills emerge when you role-play from one of the standpoints within the case. You will learn that decision-making requires taking stock of one's situation from within a clearly defined standpoint and then accepting responsibility for what arises from within that standpoint.

Cases such as Toysmart are challenging because of the large amount of information gathering and sorting they require. Moral imagination responds to this challenge by providing different framings that help to filter out irrelevant data and structure what remains. Framing plays a central role in problem specification. For example, Toysmart could be framed as the need to develop more effective software to help negotiate the exchange of information online. In this case, a software programming expert would be brought in to improve P3P programs. Or it could be framed as a legal problem that requires amending the Bankruptcy Code. What is important at this stage is that you and your group experiment with multiple framings of the case around your decision point. This makes it possible to open up avenues of solution that would not be possible under one framing.

Tackling large cases in small teams also helps develop the communication and collaboration skills that are required for group work. Take time to develop strategies for dividing the workload among your team members. The trick is to distribute equally but, at the same time, to assign tasks according to the different abilities of your team members. Some individuals are better at research while others excel in interviewing or writing. Also, make sure to set aside time when you finish for integrating your work with that of your teammates. Start by quickly reviewing the information available on the case. This is called "scoping the case." Then formulate specific questions to focus further research on information relevant to your problem-solving efforts. This includes information pertinent to constructing a socio-technical analysis, identifying key "embedded" ethical issues, and uncovering existing best and worst practices.

A case narrative, STS (socio-technical system) description, and two ethical reflections have been published at <http://computingcases.org>. This module also links to websites on bankruptcy and privacy law, the Model Business Corporation Act, consumer privacy information, and the TRUSTe website.

Toysmart Narrative

Toysmart was a Disney-supported company that sold educational toys online from December 1998 to May 2000. After disappointing Christmas sales in 1999, Disney withdrew its financial support. The greatly weakened dot-com company lasted less than a year after this. On May 22, 2000, Toysmart announced that it was closing down and brought in a consulting firm, The Recovery Group, to evaluate its assets, including a customer database of 260,000 profiles, each worth up to \$500.

Fierce opposition emerged when Toysmart placed ads in the **Wall Street Journal** and the **Boston Globe** to sell this database. Customer interest groups pointed out that Toysmart had promised not to share customer information with third parties. Toysmart also prominently displayed the TRUSTe seal which testified further to the company's obligations to respect customer privacy and security. Selling this data to third parties would break Toysmart promises, violate TRUSTe policies, and undermine consumer confidence in the security and privacy of online transactions. Toysmart's obligations to its customers came into direct conflict with its financial obligations to its investors and creditors.

TRUSTe reported Toysmart's intention to sell its database to the FTC (Federal Trade Commission) who on July 10, 2000 filed a complaint "seeking injunctive and declaratory relief to prevent the sale of confidential, personal customer information" (FTC article) Toysmart's promise never to share customer PII with third parties provided the legal foundation for this complaint. According to the FTC, Toysmart "violated Section 5 of the FTC Act by misrepresenting to customers that personal information would never be shared with third parties, then disclosing, selling, or offering that information for sale." Finally, because it collected data from children under 13 who entered various contests offered on its website, Toysmart was also cited for violating the Children's Online Privacy Protection Act or COPPA.

The FTC reached a settlement with Toysmart. The bankrupt dot-com must "file an order in the bankruptcy court prohibiting the sale of its customer data as a 'stand-alone asset'. In other words, the rights bundled in the liquidation and sale of Toysmart did not include the liberty of buyers to dispose of the asset in whatever way they saw fit. According to the negotiated settlement, buyers were bound by the commitments and promises of the original owners. Toysmart creditors "can sell electronic assets only if the purchasing company abided by the same privacy policy." In essence, the FTC asked Toysmart creditors to honor the spirit, if not the letter, of Toysmart's original promise to its customers not to sell their PII to third parties. Creditors now had to guarantee that (1) the buyer had the same basic values as Toysmart (for example, a commitment to selling quality, educational toys), (2) the buyer use the data in the same way that Toysmart had promised to use it when collecting it, and (3) the buyer would not transfer the information to third parties without customer consent. In this way, the settlement proposed to protect Toysmart customer privacy interests while allowing creditors to recover their losses through the sale of the bankrupt company's "crown jewel", its customer database.

On August 17, 2000, the Federal Bankruptcy Court declined to accept the Toysmart-FTC settlement. Instead, they argued that Toysmart and the FTC should wait to see if any parties willing to buy the database would come forward. The Bankruptcy Court felt that potential buyers would be scared off by the FTC suit and the pre-existing obligations created by Toysmart promises and TRUSTe standards. Should a buyer come forth, then they would evaluate the buyer's offer in terms of the FTC-Toysmart settlement designed to honor the privacy and security commitments made to Toysmart customers.

A final settlement was reached on January 10, 2001. When a buyer did not come forward, Buena Vista Toy Company, a Disney Internet subsidiary who was also a major Toysmart creditor, agreed to buy the database for \$50,000 with the understanding that it would be immediately destroyed. The database was then deleted and affidavits were provided to this effect.

Toysmart Chronology

Time LineChronology of Toysmart Case

1997	David Lord, a former college football player, come to work for Holt Education Outlet in Waltham, Mass.
December 1998	Lord and Stan Fung (Zero Stage Capital) buy Holt Education Outlet and rename it "Toysmart." (Lorek) Toysmart focuses on providing customers with access to 75,000 toys through an online catalog. (Nashelsky).
August 1999	Toysmart turns down a 25 million offer from an investment firm. Accepts Disney offer of 20 million in cash and 25 million in advertising,
September 1999	Toysmart posts privacy policy which promises not to release information collected on customers to third parties. At about this time, Toysmart receives permission from TRUSTe to display its seal certifying that Toysmart has adopted TRUSTe procedures for protecting privacy and maintaining information security.
Christmas 1999	After disappointing Christmas toy sales, Disney withdraws its support from Toysmart.
April 2000	COPPA goes into effect. (Childhood Online Privacy Protection Act) Prohibits soliciting information from children under 13 without parental consent.

June 2000 (approximately)	Toysmart erases 1500 to 2000 customer profiles from database to comply with COPPA (information collected after law went into effect)
May 22, 2000	Toysmart announces that it is closing its operations and selling its assets. Its initial intention is to reorganize and start over.
June 9, 2000	Toysmart creditors file an involuntary bankruptcy petition rejecting Toysmart proposal to reorganize. They petition the U.S. Trustee to form a Creditors Committee to oversee the liquidation of Toysmart assets.
June 23, 2000	Toysmart consents to involuntary bankruptcy petition. Files Chapter 11 bankruptcy. It rejects reorganization and works with lawyers and the Recovery Group to liquidate its assets.
June 2000	Recovery Group analyzes Toysmart assets and identifies its customer information database as one of its most valuable assets (a "crown jewel")
June 9, 2000	Disney subsidiary, acting as Toysmart creditor, places ads in Wall Street Journal and Boston Globe offers Toysmart customer database for sale.
After June 9, 2000	TRUSTe discovers Toysmart ad. Informs FTC (Federal Trade Commission) that selling of customer database to third parties violates TRUSTe guidelines and violates Toysmart's promises to customers(13,2)
July 10, 2000	FTC files complaint against Toysmart "seeking injunctive and declaratory relief to prevent the sale of confidential, personal customer information." District attorneys of 41 states also participate in complaint against Toysmart.
July 27, 2000	Hearing by U.S. Bankruptcy Court on Toysmart case. Includes Toysmart proposal to sell customer database.
Late July 2000	FTC and Toysmart reach settlement. Toysmart can only sell customer information to a third party who shares Toysmart values and agrees to carry out same privacy policy as Toysmart.
Late July 2000	Federal bankruptcy court rejects FTC and Toysmart settlement. Suggests waiting to see if a buyer comes forth.
January 10, 2001	Walt Disney Internet subsidiary (Buena Vista Toy Company?) pays Toysmart \$50,000 for its database. Toysmart then destroys the database and provides a confirming affidavit. (18,2)

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Supporting Documents and Tables

Toysmart Creditors Source Lorek

Creditor	Description	Debt	Impact
Zero Stage Capital	Venture Capital Firm	4 million	
Citibank		4 million	
Arnold Communications		2.5 million	
Children's Television Workshop		1.3 million	

Data Connections	Set up high-speed cable and fiber optics for Toysmart	85,000	Data Connections took out a loan to keep solvent
Integrated Handling Concepts	Set up packaging and handling system for Toysmart	40,000	Requires dot-coms to pay upfront after Toysmart experience
Blackstone	Software business	45,000	"It puts us in jeopardy as well"
PAN Communications	"Public relations agency specializing in e-business"	171,390	Turns down deals with dot-com companies and requires up-front payments

Insert paragraph text here.

Intermediate Moral Concept: Informed Consent

Concept and Definition

- **Informed Consent:** The risk bearer consents to take on the risk on the basis of a complete understanding of its nature and breadth.
- **Belmont Report:** "subjects, to the degree that they are capable, be given the opportunity to choose what shall or shall not happen to them."
- "This opportunity is provided when adequate standards for informed consent are satisfied."
- Quotes take from Belmont Report

Arguments for Free and Informed Consent as a Moral Right

- Free and informed consent is **essential** for the exercise of moral autonomy. Absence implies force, fraud, or manipulation all of which block the exercise of moral autonomy.
- The **standard threat** occurs when crucial risk information is not communicated to the risk-taker. This could be because the risk taker cannot appreciate the risk, because the mode of communication is inadequate, or because the information has been covered up. Given this standard threat, free and informed consent is **vulnerable**; it must be protected.
- Informed consent must be shaped around its feasibility, that is, the ability of the duty holder to recognize and respect this right in others. If private individuals exercise their right as a veto, then they can block socially beneficial projects. There are also serious problems concerning children, mentally challenged adults, and future generations. Finally, it may not be possible or feasible to know all risks in advance.

Conditions for Recognizing and Respecting Right

- From **Belmont Report**
- **Information:** research procedure, their purposes, risks and anticipated benefits, alternative procedures (where therapy is involved), and a statement offering the subject the opportunity to ask questions and to withdraw at any time from the research.
- **Comprehension:** manner and context in which information is conveyed is as important as the information itself.
- **Voluntariness:** an agreement to participate in research constitutes a valid consent only if voluntarily given. This element of informed consent requires conditions free of coercion and undue influence.

Other Legal and Moral Frameworks

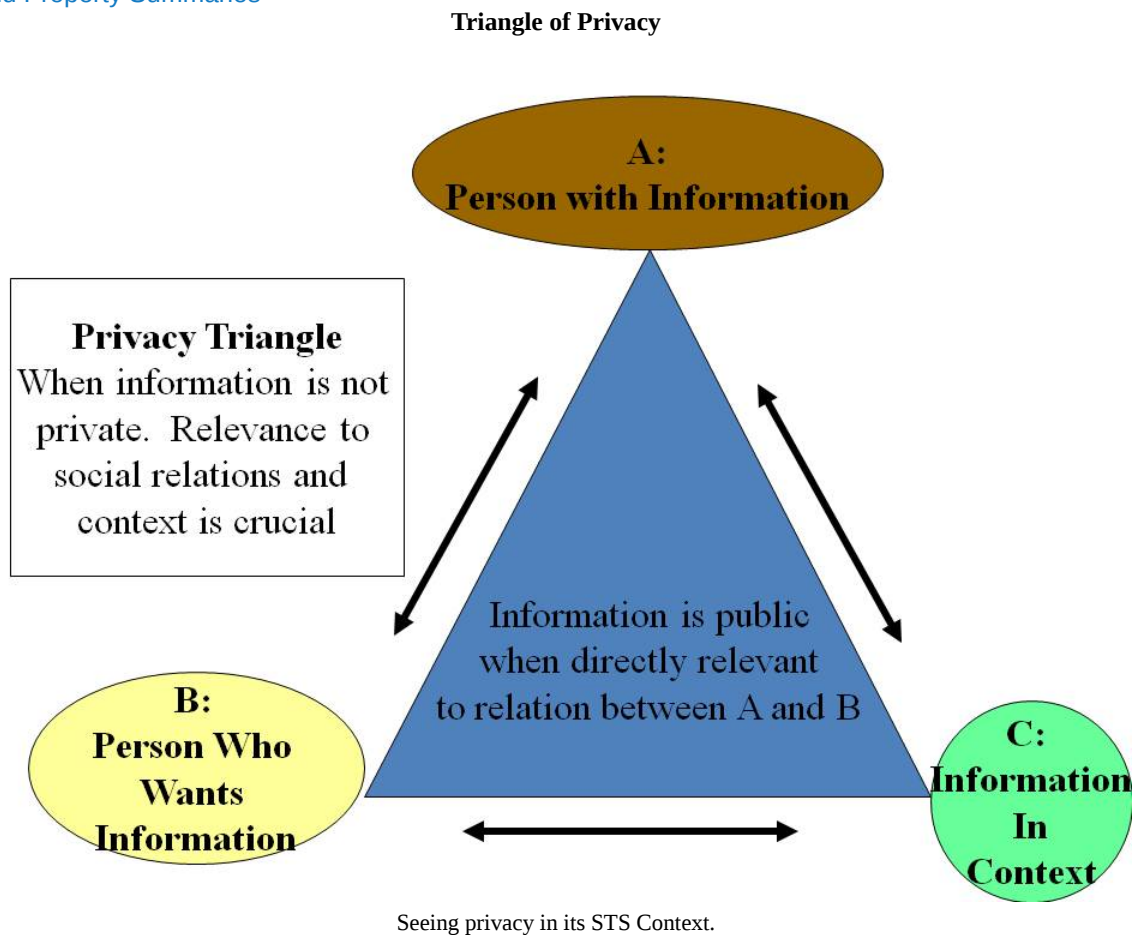
- Institutional Research Boards or IRBs now require documentation of informed consent on research projects carried out under the university's auspices. This is in response to requirements by granting agencies such as the National Institute for Health and the National Science Foundation.
- **Consenting to the transfer of PII (personal identifying information) online:** opt-in and opt-out.
- **Opt-in:** Information is transferred only upon obtaining express consent. Default is not transferring information.
- **Opt-in:** Information transfer is halted only when person to whom information applies does something positive, i.e., refuses to consent to transfer. Default is on transferring the information.
- **Liability Rules and Property Rules:** These also have to do with consent. Sagoff makes this distinction with reference to activities that have an impact on the environment. an injunction referring to liability rules stops the activity to protect the

individual who proves impact. Property rules require only that the producer of the environmental impact compensate the one who suffers the impact.

Cases Employing Informed Consent

- **Therac-25:** Patients receiving radiation therapy should be made aware of the risks involved with treatment by the machine. Free and informed consent is involved when shutting down the machines to investigate accident reports or continuing operating the machines while investigating accident reports. In both cases, it is necessary, under this right, to let patients know what is going on and their risks.
- **Toysmart Case:** Toysmart creditors are about to violate Toysmart's promise not to transfer customer information profiles to third parties. This transfer can occur, morally, but only with the express consent of the customers who have provided the information. The devil is in the details. Do opt-in or opt-out procedures best recognize and respect free and informed consent in this case?
- **Hughes Case:** Hughes customers want their chips right away and are pressuring Saia and the crowd to deliver them. Would they consent to renegotiate the conditions under which environmental tests can be skipped?

Privacy and Property Summaries



Intellectual Property

- I own that with which I mix my labor
- The US constitution attempts to balance two contradictory policies of property
 - Giving innovators and inventors control to encourage more innovation
 - Limiting this control to ensure dissemination of good ideas throughout society
- One way to do this: bundle different liberties under property
 - possess, control, use, benefit from, dispose of, and exclude others
- Another way to do this is to create and balance two things
 - Intellectual commons: shared repository of ideas
 - Legal devices to restrict dissemination and use of property: patents, copyright, trade secrets
 - Control/ Adjustment: extending-contracting terms limits
- Intellectual property is different from physical property
 - Ideas are non-rivalrous (like one candle lighting another)
 - Ideas are non-excludable (like the air which cannot be contained in one restricted area)
 - These two characteristics make ideas “ideal” for forming an “intellectual commons”

Summary of issues on Intellectual Property

Bibliographical Note The triangle of privacy is widely disseminated in the literature of business ethics. The author first became aware of it from George G Brenkert (1981) "Privacy, Polygraphs and Work," **Business and Professional Ethics** 1, Fall 1981" 19-34. Information on intellectual property comes from Lawrence Lessig (2006) **Code.2**, Basic Books: Chapter 10.

What you need to know ...

What you need to know about socio-technical systems

1. STS have seven broad components: hardware, software, physical surroundings, people/groups/roles, procedures, laws, and data/data structures.

2. Socio-technical systems embody values

- These include moral values like safety, privacy, property, free speech, equity and access, and security. Non-moral values can also be realized in and through Socio Technical Systems such as efficiency, cost-effectiveness, control, sustainability, reliability, and stability.
- Moral values present in Socio Technical Systems can conflict with other embedded moral values; for example, privacy often conflicts with free speech. Non-moral values can conflict with moral values; developing a safe system requires time and money. And, non-moral values can conflict; reliability undermines efficiency and cost effectiveness. This leads to three problems that come from different value conflicts within Socio Technical Systems and between these systems and the technologies that are being integrated into them.
- Mismatches often arise between the values embedded in technologies and the Socio Technical Systems into which they are being integrated. As UNIX was integrated into the University of California Academic Computing STS (see Machado case at Computing Cases), the values of openness and transparency designed into UNIX clashed with the needs of students in the Academic Computing STS at UCI for privacy.
- Technologies being integrated into Socio Technical Systems can magnify, exaggerate, or exacerbate existing value mismatches in the STS. The use of P2P software combined with the ease of digital copying has magnified existing conflicts concerning music and picture copyrights.

- Integrating technologies into STSs produces both immediate and remote consequences and impacts.

3. Socio-technical systems change

- These changes are brought about, in part, by the value mismatches described above. At other times, they result from competing needs and interests brought forth by different stakeholders. For example, bicycle designs, the configuration of typewriter keys, and the design and uses of cellular phones have changed as different users have adapted these technologies to their special requirements.
- These changes also exhibit what sociologists call a “trajectory”, that is, a path of development. Trajectories themselves are subject to normative analysis. For example, some STSs and the technologies integrated into them display a line of development where the STS and the integrated technology are changed and redesigned to support certain social interests. The informing capacities of computing systems, for example, provide information which can be used to improve a manufacturing processes can or to monitor workers for enhancing management power. (See Shoshanna Zuboff, **The Age of the Smart Machine**
- Trajectories, thus, outline the development of STSs and technologies as these are influenced by internal and external social forces.

In this section, you will learn about this module’s exercises. The required links above provide information on the frameworks used in each section. For example, the Socio-Technical System module provides background information on socio-technical analysis. The “Three Frameworks” module provides a further description of the ethics tests, their pitfalls, and the feasibility test. These exercises will provide step by step instructions on how to work through the decision points presented above.

For more information see Huff and Jawer below.

Decision Point One: You are David Lord, a former employee of Holt Educational Outlet, a manufacturer of educational toys located in Waltham, Mass. Recently, you have joined with Stan Fung of Zero Stage Capital, a venture capital firm to buy out Holt Educational Outline. After changing its name to Toysmart, you and Fung plan to transform this brick and mortar manufacturer of educational toys into an online firm that will link customers to a vast catalogue of educational, high quality toys. Designing a website to draw in toy customers, linking to information on available toys, setting up a toy distribution and shipping system, and implementing features that allow for safe and secure online toy purchases will require considerable financing. But, riding the crest of the dot-com boom, you have two promising options. First, a venture capital firm has offered you \$20,000,000 for website development, publicity, and other services. Second, Disney has offered the same amount for financing, but has added to it an additional \$25,000,000 in advertising support. Disney has a formidable reputation in this market, a reputation which you can use to trampoline Toysmart into prominence in the growing market in educational toys. However, Disney also has a reputation of micro-managing its partners. Develop a plan for financing your new dot-com.

Things to consider in your decision-making:

1. What are Toysmart values? What are Disney values? Would Disney respect Toysmart’s values?
2. What synergies could result from working with Disney? For example, could you share information on customers? You could feed your customer profiles to Disney in exchange for their customer profiles. What kind of data managing technology would be required for this? What ethical problems could arise from transferring customer identifying information to third parties?
3. What kind of commitment would you be willing to make to Disney in terms of product and sales? How should Disney reciprocate? For example, how long should they stick with you through sales that fall short of projections?

Decision Point Two: You work for Blackstone, “an 18-person software business.” You have been asked by Toysmart to provide software the following functions: (1) designing a webpage that would attract customers and communicate Toysmart Values, (2) advise Toysmart on its privacy and data security policy including whether to register with an online trust, security measures to protect customer data during online transactions, and measures to prevent unauthorized access to customer data while stored, and (3) a comprehensive online catalogue that would provide customers with access to educational toys from a variety of small business manufacturers. An example of small toy manufacturers to which Toysmart should be linked is Brio Corporation which manufactures wooden toys such as blocks, trains, and trucks. Develop general recommendations for Toysmart around these three areas.

Information for this scenario comes from Laura Lorek, “When Toysmart Broke,” www.zdnet.com/eweek/stories/g...612962,00.html. Accessed July 16, 2001.

Things to consider in your decision-making

- Toysmart is a fairly new dot-com. While it is supported by Disney, it is still a risky venture. Should you ask them for advance payment for whatever services you render? What kind of policies does your company have for identifying and assessing financial risk?
- What kind of privacy and data security policy should you recommend to Toysmart? What kind of values come into conflict when a company like Toysmart develops and implements privacy and data security measures? (Use your STS description to answer this question.)
- Should Toysmart become bankrupt, their data base would turn into a valuable asset. What recommendations should you make to help Toysmart plan around this possibility? What values come into conflict when planning to dispose of assets during bankruptcy proceedings? What kind of obligations does a company take on during its operation that continue even after it has become bankrupt?
- Using the link provided with this module, visit the TRUSTe website and find its white paper on developing a privacy policy. Evaluate this privacy policy for Toysmart. What benefits can a strong privacy policy bring to a dot-com? Should Toysmart work to qualify to display the TRUSTe seal on its website? Examine TRUSTe procedures for transferring confidential customer PII to third parties? What obligations will this create? Would this over-constrain Toysmart?

Decision Point Three: You work for PAN Communications and have been providing advertising services for Toysmart. Now you find out that Toysmart has filed a Chapter 11 bankruptcy, and it has an outstanding debt to your company for \$171,390. As a part of this filing procedure, Toysmart has reported its assets at \$10,500,000 with debts of \$29,000,000. Toysmart creditors, including PAN Communications, have petitioned the Office of the United States Trustee for a "Creditors' Committee Solicitation Form." This will allow for the formation of a committee composed of Toysmart creditors who decide on how the assets of the bankrupt firm will be distributed. You, because of your knowledge of bankruptcy and accounting procedures, have been asked to represent your company on this committee. This bleak situation is somewhat remedied by the customer data base that Toysmart compiled during its operation. It contains profiles of the PII (personal identifying information) of 260,000 individuals. Because selling educational toys is profitable, there is a good chance that this data base could be sold for up to \$500 a profile to a third party. Should you recommend selling this data base? Should Toysmart customers be notified of the pending transfer of their PII and, if so, how should they be notified?

Here are some constraints that outline your decision

- As a member of the Creditors' Committee, you have a fiduciary duty to Toysmart creditors in working to distribute fairly the remaining Toysmart assets. This would, all things being equal, lead to recommending selling the Toysmart customer data base
- There are some provisions in the bankruptcy code that may require or allow overriding fiduciary duties given prior legal commitments made by Toysmart. These commitments, in the form of strong privacy guarantees made to customers by Toysmart on its webpage, may constitute an "executory contract." See the Legal Trail table in the Toysmart case narrative and also Larren M. Nashelsky, "On-Line Privacy Collides With Bankruptcy Creditors," New York Law Journal, New York Law Publishing Company, August 28, 2000.
- Finally, Nashelsky makes an interesting argument. While deontological considerations would require setting aside creditor interests and honoring Toysmart privacy promises, a justice-based argument would recommend a compromise. Bankruptcy proceedings start from the fact that harm (financial) has been done. Consequently, the important justice consideration is to distribute fairly the harms involved among the harmed parties. Harm distributions are correlated with benefit distributions. Because Toysmart customers benefited from Toysmart offerings, they should also bear a share of the harms produced when the company goes bankrupt. This requires that they allow the distribution of their PII under certain conditions.

Things to consider in your decision-making

- How do you balance your obligations to PAN with those to other Toysmart creditors as a member of the Creditors' Committee?
- How should you approach the conflict between honoring Toysmart promises and carrying out Creditor Committee fiduciary duties? Do you agree with Nashelsky's argument characterized above?
- Should the Bankruptcy Code be changed to reflect issues such as these? Should privacy promises be considered an "executory contract" that overrides the duty to fairly and exhaustively distribute a company's assets?
- Finally, what do you think about the FTC's recommendation? The Bankruptcy Court's response? The final accommodation between Toysmart and Buena Vista Toy Company?

What you will do ...

In this section, you will learn about this module's exercises. The required links above provide information on the frameworks used in each section. For example, the Socio-Technical System module provides background information on socio-technical analysis.

The "Three Frameworks" module provides a further description of the ethics tests, their pitfalls, and the feasibility test. These exercises will provide step by step instructions on how to work through the decision points presented above.

Exercise One: Problem Specification

In this exercise, you will specify the problem using socio-technical analysis. The STS section of the Toysmart Case narrative (found at Computing Cases) provides a good starting point. In the first table, enter the information from the Toysmart case materials pertinent to the general components of a STS, its hardware, software, physical surroundings, people/groups/roles, procedures, laws, data. Some examples taken from the STS description at Computing Cases are provided to get you started. Then, using the second table, identify the values that are embedded in the different components of the STS. For example, PICS (platforms for internet content selection) embody the values of security and privacy. Finally, using the data from your socio-technical analysis, formulate a concise problem statement.

Exercise 1a: Read the socio-technical system analysis of the Toysmart case at <http://computingcases.org>. Fill in the table below with elements from this analysis that pertain to your decision point.

Socio-Technical System Table

Hardware	Software	Physical Surroundings	People/Groups/Roles	Procedures	Laws, Codes, Regulations	Data and Data Structures
Holt Education Outlet	Platforms for Internet Content Selection	Cyber Space	Toysmart the corporation	Buying Toys Online	COPPA	Toysmart Customer Data Base

Instructions for Table 1:

1. Go to <http://computingcases.org> and review the STS description provided for the Toysmart case.
2. Pull out the elements of the STS description that are relevant to your decision point. List them under the appropriate STS component in the above table.
3. Think about possible ways in which these components of the Toysmart STS interact. For example, what kinds of legal restrictions govern the way data is collected, stored, and disseminated?
4. Develop your STS table with an eye to documenting possible ethical conflicts that can arise and are relevant to your decision point.

Values Embedded by Relevant Software: Values embedded in key software components in the Toysmart case. Emphasis on machine/software negotiation for privacy preferences in Internet transactions.

Software / Value Embedded	PICS (Platforms for Internet Content Selection)	(Platforms for Privacy Preferences)	SSLs (Secured Socket Layers) that encrypt pages asking for SS numbers
Security	Embodies privacy and security by filtering objectionable data. Security selected over free speech.	Integrates property with security and privacy by converting information into property.	Realizes / supports security by sealing off domains of information.
Privacy	Embodies privacy and security by filtering objectionable data. Security selected over free speech.	Integrates property and security by filtering objectionable data. Security selected over free speech.	Realizes and supports privacy by sealing off domains of information.
Property		Integrates property with security and privacy by converting information into property	Realizes and supports property by restricting access (intellectual property protected by excluding non-authorized access.

Free Speech	Interferes with free speech by filtering content. Content can be filtered with recipient's awareness.	Facilitates by permitting information exchange on model of property exchange. But this limits exchange by assigning it a price.	Restricts access.
Justice (Equity and Access)	Could be used to restrict access to ideas by filtering ideas. Thus it could cut off flow of information into the intellectual commons.	Facilitates by permitting information exchange on model of property exchange. But this limits exchange by assigning it a price.	Because it restricts access to a domain, it can be used to reduce or cut off flow of information into the intellectual commons.

Exercise 1b Examine the values embedded in the STS surrounding this decision point. Locate your values under the appropriate component in the Toysmart STS. For example, according to the STS description for Toysmart found at Computing Cases, the software programs prominent in this case embody certain values; SSLs embody security and privacy, P3P property, and PICS privacy. Next, look for areas where key values can come into conflict.

Value Table

Hardware	Software	Physical Surroundings	People/Groups/Roles	Procedures	Laws/Codes/Regulations	Data/Data Structures
Security						
Privacy						
Property						
Justice (Equity/Access)						
Free Speech						

Instructions for Table 2:

1. This module links to another Connexions module, Socio-Technical Systems in Professional Decision-Making. There you will find short profiles of the values listed in the above table: security, privacy, property, justice, and free speech. These profiles will help you to characterize the values listed in the above table.
2. The second ethical reflection in the Toysmart case narrative (at Computing Cases) also contains a discussion of how property comes into conflict with privacy.
3. Identify those components of the Toysmart STS that embody or embed value. For example, list the values realized and frustrated by the software components discussed in the Toysmart case in the STS description.
4. Look for ways in which different elements of the STS that embed value can interact and produce value conflicts. These conflicts are likely sources for problems that you should discuss in your problem statement and address in your solution.

Exercise 1c: Write out the requirements (ethical and practical) for a good solution. Identify the parts of the STS that need changing. Then, develop a concise summary statement of the central problem your decision point raises. As you design solutions to this problem, you may want to revise this problem statement. Be sure to experiment with different ways of framing this problem.

Harris, Pritchard, and Rabins provide a useful approach to problem specification. See references below.

Exercise Two: Solution Generation

Generate solutions to the problem(s) you have specified in Exercise 1. This requires that...

- each member of your group develop a list of solutions,
- the group combines these individual lists into a group list, and...
- the group reduces this preliminary list to a manageable number of refined and clarified solutions for testing in the next stage.

Helpful Hints for Solution Generation

1. **Solution generation requires proficiency in the skills of moral imagination and moral creativity.** Moral imagination is the ability to open up avenues of solution by framing a problem in different ways. Toysmart could be framed as a technical problem requiring problem-solving skills that integrate ethical considerations into innovative designs. Moral creativity is the ability to formulate non-obvious solutions that integrate ethical considerations over various situational constraints.

2. Problems can be formulated as interest conflicts. In this case different solution options are available.

- **Gather Information.** Many disagreements can be resolved by gathering more information. Because this is the easiest and least painful way of reaching consensus, it is almost always best to start here. Gathering information may not be possible because of different constraints: there may not be enough time, the facts may be too expensive to gather, or the information required goes beyond scientific or technical knowledge. Sometimes gathering more information does not solve the problem but allows for a new, more fruitful formulation of the problem. Harris, Pritchard, and Rabins in *Engineering Ethics: Concepts and Cases* show how solving a factual disagreement allows a more profound conceptual disagreement to emerge.
- **Nolo Contendere.** Nolo Contendere is latin for not opposing or contending. Your interests may conflict with your supervisor but he or she may be too powerful to reason with or oppose. So your only choice here is to give in to his or her interests. The problem with nolo contendere is that non-opposition is often taken as agreement. You may need to document (e.g., through memos) that you disagree with a course of action and that your choosing not to oppose does not indicate agreement.
- **Negotiate.** Good communication and diplomatic skills may make it possible to negotiate a solution that respects the different interests. Value integrative solutions are designed to integrate conflicting values. Compromises allow for partial realization of the conflicting interests. (See the module, **The Ethics of Team Work**, for compromise strategies such as logrolling or bridging.) Sometimes it may be necessary to set aside one's interests for the present with the understanding that these will be taken care of at a later time. This requires trust.
- **Oppose.** If nolo contendere and negotiation are not possible, then opposition may be necessary. Opposition requires marshalling evidence to document one's position persuasively and impartially. It makes use of strategies such as leading an "organizational charge" or "blowing the whistle." For more on whistle-blowing consult the discussion of whistle blowing in the Hughes case that can be found at computing cases.
- **Exit.** Opposition may not be possible if one lacks organizational power or documented evidence. Nolo contendere will not suffice if non-opposition implicates one in wrongdoing. Negotiation will not succeed without a necessary basis of trust or a serious value integrative solution. **As a last resort**, one may have to exit from the situation by asking for reassignment or resigning.

3. Solutions can be generated by readjusting different components of the STS.

- **Technical Puzzle.** If the problem is framed as a technical puzzle, then solutions would revolve around developing designs that optimize both ethical and technical specifications, that is, resolve the technical issues and realize ethical value. In this instance, the problem-solver must concentrate on the hardware and software components of the STS.
- **Social Problem.** If the problem is framed as a social problem, then solutions would revolve around changing laws or bringing about systemic reform through political action. This would lead one to focus on the people/groups/roles component (working to social practices) or the legal component.
- **Stakeholder Conflict.** If the problem is framed as a conflict between different stakeholder interests, then the solution would concentrate on getting stakeholders (both individuals and groups) to agree on integrative or interest compromising solutions. This requires concentrating on the people/group/role component of the STS. (Note: A stakeholder is any group or individual with a vital interest at play in the situation.)
- **Management Problem.** Finally, if the problem is framed as a management problem, then the solution would revolve around changing an organization's procedures. Along these lines, it would address the (1) fundamental goals, (2) decision recognition procedures, (3) organizational roles, or (4) decision-making hierarchy of the organization. These are the four components of the CID (corporate internal decision) structure described in the "Ethical Reflections" section of the Toysmart case.
- **Nota Bene:** Financial issues are covered by the feasibility test in the solution implementation stage. As such, they pose side issues or constraints that do not enter into the solution generation phase but the solution implementation phase.

4. Brainstorming. Moral creativity, which involves designing non-obvious solutions, forms an essential part of solution generation. Here are some guidelines to get you started.

- Individually make out a list of solutions before the group meeting. Work quickly to realize a pre-established quota of five to ten solutions. After composing a quick first draft, revise the list for clarity only; make no substantial changes.

- Start the group brainstorming process by having the group review and assemble all the individual solutions. Do this quickly and without criticism. Beginning criticism at this stage will kill the creativity necessary for brainstorming and shut down the more timid (but creative) members of the group.
- Review the list and identify solutions that are identical or overlap. Begin the refining process by combining these solutions.
- Having reviewed all the brainstormed solutions, it is now time to bring in criticism. Begin by eliminating solutions with major ethical problems such as those that violate rights, produce injustices, or cause extensive harm.
- Identify but do not eliminate solutions that are ethical but raise serious practical problems. Do not initially eliminate an ethical solution because there are obstacles standing in the way of its implementation. Be descriptive. Identify and impartially describe the obstacles. Later, in the solution implementation stage, you may be able to design creative responses to these obstacles.
- Identify solutions that do not "fit" your problem statement. These require a decision. You can throw out the solution because it does not solve the problem or you can change the problem. If a solution does not fit the problem but, intuitively, seems good, this is a sign that you need to take another look at your problem statement.
- Don't automatically reject partial solutions. For example, sending memos through email rather than printing them out and wasting paper may not solve the entire recycling problem for your company. But it represents a good, partial solution that can be combined with other partial solutions to address the bigger problem.
- Through these different measures, you will gradually integrate criticism into your brainstorming process. This will facilitate working toward a manageable, refined list of solutions for testing in the next stage.

Exercise 3: Develop a Solution List

- Have each member of your team prepare a solution list and bring it to the next group meeting. Set a quota for this individual list, say, 5 to 10 solutions.
- Prepare a group list out of the lists of the individual members. Work to combine similar solutions. Be sure to set aside criticism until the preliminary group list is complete.
- Make use of the following table.
- Refine the group list into a manageable number of solutions for testing in the next stage. Combine overlapping solutions. Eliminate solutions that do not respond to the requirements and the problem statement that you prepared in the previous exercise. Eliminate solutions that violate important ethical considerations, i.e., solutions that violate rights, produce harms, etc.
- Check your refined solution list with your problem statement. If they do not match, eliminate the solution or redefine the problem

Refined Brainstorm List

Solution Ranking	Description of Solution	Justification (fits requirements, fits problem)
Best Solution		
Second Best Solution		
Third Best Solution		
Fourth Best Solution		
Fifth Best Solution		

Anthony Weston provides an illuminating and useful discussion of creative problem solving in the reference provided below.

Exercise Three: Solution Testing

In this section, you will test the solutions on the refined list your group produced in the previous exercise. Three ethics tests, described below, will help you to integrate ethical considerations in the problem-solving process. A global feasibility test will help to identify solutions with serious practical problems. Finally, a Solution Evaluation Matrix summarizes the results for class debriefings.

Setting up for the test.

- Identify the agent perspective from which the decision will be made
- Describe the action as concisely and clearly as possible.

- Identify the stakeholders surrounding the decision, i.e., those who will suffer strong impacts (positively or negatively) from the implementation of your decision. Stakeholders have a vital or essential interest (right, good, money, etc) in play with this decision.
- In the harm/beneficence test, identify the likely results of the action and sort these into harms and benefits.
- For the reversibility test, identify the stakeholders with whom you will reverse positions.
- For the public identification test, identify the values, virtues, or vices your action embodies. Associate these with the character of the agent.

Harm/Beneficence Test

1. **What are the harms your solution is likely to produce? What are its benefits? Does this solution produce the least harms and the most benefits when compared to the available alternatives?**
2. **Pitfall—Too much.** In this "Paralysis of Analysis" one factor in too many consequences. To avoid the fallacy restrict the analysis to the most likely consequences with the greatest magnitude (Magnitude indicates the range and severity of impact).
3. **Pitfall—Too Little.** A biased or incomplete analysis results when significant impacts are overlooked. Take time to uncover all the significant impacts, both in terms of likelihood and in terms of magnitude.
4. **Pitfall—Distribution of Impacts.** Consider, not only the overall balance of harms and benefits but also how harms and benefits are distributed among the stakeholders. If they are equally or fairly distributed, then this counts in the solution's favor. If they are unequally or unfairly distributed, then this counts against the solution. Be ready to redesign the solution to distribute better (=more equitably or fairly) the harmful and beneficial results.

Reversibility Test

1. **Would this solution alternative be acceptable to those who stand to be most affected by it? To answer this question, change places with those who are targeted by the action and ask if from this new perspective whether the action is still acceptable?**
2. **Pitfall—Too much.** When reversing with Hitler, a moral action appears immoral and an immoral action appears moral. The problem here is that the agent who projects into the immoral standpoint loses his or her moral bearings. The reversibility test requires viewing the action from the standpoint of its different targets. But understanding the action from different stakeholder views does not require that one abandon himself or herself to these views.
3. **Pitfall—Too little.** In this pitfall, moral imagination falls short, and the agent fails to view the action from another stakeholder standpoint. The key in the reversibility test is to find the middle ground between too much immersion in the viewpoint of another and too little.
4. **Pitfall—Reducing Reversibility to Harm/Beneficence.** The reversibility test requires that one assess the impacts of the action under consideration on others. But it is more than a simple listing of the consequences of the action. These are viewed from the standpoint of different stakeholders. The reversibility test also goes beyond considering impacts to considering whether the action treats different stakeholders respectfully. This especially holds when the agent disagrees with a stakeholder. In these disagreements, it is important to work out what it means to disagree with another respectfully.
5. **Pitfall—Incomplete survey of stakeholders.** Leaving out significant stakeholder perspectives skews the results of the reversibility test. Building an excellent death chamber works when one considers the action from the standpoint of Hitler; after all, it's what he wants. But treating an individual with respect does not require capitulating to his or her desires, especially when these are immoral. And considering the action from the standpoint of other stakeholders (say the possible victims of newer, more efficient gas chambers) brings out new and radically different information.
6. **Pitfall—Not Weighing and Balancing Stakeholder Positions.** This pitfall is continuous with the previous one. Different stakeholders have different interests and view events from unique perspectives. The reversibility test requires reviewing these interests and perspectives, weighing them against one another, and balancing out their differences and conflicts in an overall, global assessment.

Publicity (or Public Identification) Test

1. **Would you want to be publicly associated or identified with this action? In other words, assume that you will be judged as a person by others in terms of the moral values expressed in the action under consideration. Does this accord with how you would want to or aspire to be judged?**
2. **Pitfall—Failure to association action with character of agent.** In the publicity test, the spotlight of analysis moves from the action to the agent. Successfully carrying out this test requires identifying the agent, describing the action, and associating the

agent with the action. The moral qualities exhibited in the action are seen as expressing the moral character of the agent. The publicity test, thus, rests on the idea that an agent's responsible actions arise from and express his or her character.

3. **Pitfall—Failure to appreciate the moral color of the action.** The publicity test assumes that actions are colored by the ends or goods they pursue. This means that actions are morally colored. They can express responsibility or irresponsibility, courage or cowardice, reasonableness or unreasonableness, honesty or dishonesty, integrity or corruption, loyalty or betrayal, and so forth. An analysis can go astray by failing to bring out the moral quality (or qualities) that an action expresses.
4. **Pitfall—Reducing Publicity to Harm/Beneficence Test.** Instead of asking what the action says about the agent, many reduce this test to considering the consequences of publicizing the action. So one might argue that an action is wrong because it damages the reputation of the agent or some other stakeholder. But this doesn't go deep enough. The publicity test requires, not that one calculate the consequences of wide-spread knowledge of the action under consideration, but that one draws from the action the information it reveals about the character of the agent. The consequences of bad publicity are covered by the harm/beneficence test and do not need to be repeated in the public identification test. The publicity test provides new information by turning from the action to the agent. It focuses on what the action (its moral qualities and the goods it seeks) says about the agent.

Comparing the Test Results: Meta-Tests

1. The ethics tests will not always converge on the same solution because each test (and the ethical theories it encapsulates) covers a different dimension of the action: (1) harm/beneficence looks at the outcomes or consequences of the action, (2) reversibility focuses on the formal characteristics of the action, and (3) publicity zeros in on the moral character of the agent.
2. The meta-tests turn this surface disagreement into an advantage. The convergence or divergence between the ethics tests become indicators of solution strength and weakness.
3. **Convergence.** When the ethics tests converge on a given solution, this indicates solution strength and robustness.
4. **Divergence.** When tests diverge on a solution—a solution does well under one test but poorly under another—this signifies that it needs further development and revision. Test divergence is not a sign that one test is relevant while the others are not. Divergence indicates solution weakness and is a call to modify the solution to make it stronger.

Exercise 3: Summarize your results in a Solution Evaluation Matrix

1. Place test results in the appropriate cell.
2. Add a verbal explanation to the SEM table.
3. Conclude with a global feasibility test that asks, simply, whether or not there exist significant obstacles to the implementation of the solution in the real world.
4. Finish by looking at how the tests converge on a given solution. Convergence indicates solution strength; divergence signals solution weakness.

Solution Evaluation Matrix

Solution/Test	Harm/Beneficence	Reversibility	Publicity (public identification)	Feasibility
First Solution				
Second Solution				
Third Solution				
Fourth Solution				
Fifth Solution				

The ethics tests are discussed in Cruz and Davis. See references below. Wike and Brincat also discuss value based approaches in the two references below.

Exercise Four: Solution Implementation

In this section, you will trouble-shoot the solution implementation process by uncovering and defusing potential obstacles. These can be identified by looking at the constraints that border the action. Although constraints specify limits to what can be realized in a given situation, they are more flexible than generally thought. Promptly identifying these constraints allows for proactive planning

that can push back obstacles to solution implementation and allow for realization of at least some of the value embodied in the solution.

A **Feasibility Test** focuses on these situational constraints and poses useful questions early on in the implementation process. What conditions could arise that would hinder the implementation of a solution? Should the solution be modified to ease implementation under these constraints? Can the constraints be removed or modified through activities such as negotiation, compromise, or education? Can solution implementation be facilitated by modifying both the solution and the constraints?

Feasibility Constraints

Category	Sub-Category		
Resource	Money/Cost	Time/Deadlines	Materials
Interest	Organizational(Supervisor)	Legal (laws, regulations)	Political/Social
Technical	Technology does not exist	Technology patented	Technology needs modification

Resource Constraints:

- **Does the situation pose limits on resources that could limit the realization of the solution under consideration?**
- **Time.** Is there a deadline within which the solution has to be enacted? Is this deadline fixed or negotiable?
- **Financial.** Are there cost constraints on implementing the ethical solution? Can these be extended by raising more funds? Can they be extended by cutting existing costs? Can agents negotiate for more money for implementation?
- **Resource.** Are necessary resources available? Is it necessary to plan ahead to identify and procure resources? If key resources are not available, is it possible to substitute other, more available resources? Would any significant moral or non-moral value be lost in this substitution?

Interest Constraints

- **Does the solution threaten stakeholder interests? Could it be perceived as so threatening to a stakeholder's interests that the stakeholder would oppose its implementation?**
- **Individual Interests.** Does the solution threaten the interests of supervisors? Would they take measures to block its realization? For example, a supervisor might perceive the solution as undermining his or her authority. Or, conflicting sub-group interests could generate opposition to the implementation of the solution even though it would promote broader organizational objectives.
- **Organizational Interests.** Does the solution go against an organization's SOPs (standard operating procedures), formal objectives, or informal objectives? Could acting on this solution disrupt organization power structures? (Perhaps it is necessary to enlist the support of an individual higher up in the organizational hierarchy in order to realize a solution that threatens a supervisor or a powerful sub-group.)
- **Legal Interests.** Are there laws, statutes, regulations, or common law traditions that oppose the implementation of the solution? Is it necessary to write an impact statement, develop a legal compliance plan, or receive regulatory approval in order to implement the solution?
- **Political/Social/Historical Constraints.** Would the solution threaten or appear to threaten the status of a political party? Could it generate social opposition by threatening or appearing to threaten the interests of a public action group such as an environmental group? Are there historical traditions that conflict with the values embedded in the solution?

Technical Constraints

- **Technology does not yet exist.** Would the implementation of the solution require breaking new technological ground?
- **Technology Protected by Patent.** The technology exists but is inaccessible because it is still under a patent held by a competitor.
- **Technology Requires Modification.** The technology required to implement solution exists but needs to be modified to fit the context of the solution. Important considerations to factor in would be the extent of the modification, its cost, and how long it would take to bring about the modification.

Exercise Five: Ethical Perspective Pieces

Getting Consent to Information Transfer Customer Consent If you have followed the case so far, you see that while the money Toysmart owes to Citibank may just be a drop in the bucket, the welfare and even survival of other Toysmart creditors depends on how much money can be retrieved through the bankruptcy process. The following Ethical Perspective argues that the right of creditors for their money cannot be traded off with the right to privacy of Toysmart customers profiled in their now valuable data base. These two stakeholders and their stakes—in this case rights—need to be integrated as fully as possible. The key lies in the execution of the consumer right to be informed and to freely consent to the transfer of their data to third parties This right's execution must address three important aspects.

- Customer consent must be obtained by having them opt-in rather than opt-out of the transfer of PII. Opt-in represents a more active, opt-out a more passive mode of consent. By opting into the data transfer, Toysmart customers consent explicitly, knowingly, and freely to the transfer of their information. Opt-out is passive because unless customers expressly forbid it, the transfer of their PII to a third party will occur. The chances are that many customers will consent only if compensated. And the mechanics of obtaining positive opt-in consent are complicated. Is this done by email or snail mail? How can Toysmart customers be fully informed? What kind of timeline is necessary for their full consent? Implimentation of opt-in consent is more adequate morally speaking but much more difficult, time-consuming, and costly in its implementation.
- Any exchange of information must be in accord with TRUSTe standards which Toysmart agreed to when they solicited the right to use the TRUSTe seal. TRUSTe has its own standards (they can be found through the link above) which reinforce the above discussion of informed consent but also bring in other matters. Important here is the utilitarian concern of building and maintaining consumer trust to encourage their using the Internet for e-business. Web site certification agencies like TRUSTe exist to validate that a web site is trustworthy; but to maintain this validation, customers must know that TRUSTe will enforce its standards when websites become reluctant to follow them. TRUSTe must be aggressive and strict here in order to maintain the high level of trust they have generated with e-business customers.
- An important part of TRUSTe standards on the transfer of PII to third parties is their insistence that these third parties share the values of those who have been given the information. Toysmart cultivated a reputation as a trustworthy company devoted to producing safe, high quality, educational toys. The customer data base should be transferred only to concerns that share these goals and the accompanying values. (What are these?) Did Toysmart compromise on these goals and values when they agreed to accept Disney financing and advertising support? What are Toysmart values? What are Disney values?

In conclusion, this perspective piece is designed to get you to think about the right of informed consent, whether it can be reconciled with financial interests and rights of Toysmart creditors, and how this right can be implemented in the concrete details of this case. It has argued that customer PII can be transferred but only with the consent of the customers themselves. It has defined this consent in terms of express opting-into the transfer on the part of the customers. It has also argued that the third part must share the values and goals of Toysmart, especially those values accompanying Toysmart promises to customers.

Group Exercise

Identify the role played and the values held by each of the following participants:

1. David Lord (CEO of Toysmart)
2. Disney (as venture capitalist)
3. TRUSTe (as non-profit)
4. Toysmart Creditors (Pan Communications)
5. FTC (government regulatory agency)
6. Toysmart Customers

Toysmart's customer data base

1. Should Toysmart creditors be allowed to sell the customer data base to third parties? Respond to arguments pro and con given by participants in the case.
2. Assume Toysmart should be allowed to sell the data base to their third party. What kind of values should this third party have?
3. Assume Toysmart has to get customer consent before selling the data base. How should customer consent be obtained? (What counts as customer consent?)

What did you learn?

This section provides closure to the module for students. It may consist of a formal conclusion that summarizes the module and outlines its learning objectives. It could provide questions to help students debrief and reflect on what they have learned. Assessment forms (e.g., the “Muddiest Point” Form) could be used to evaluate the quality of the learning experience. In short, this section specifies the strategy for bringing the module to a close.

In this module, you have...

- studied a real world case that raised serious problems with intellectual property, privacy, security, and free speech. Working with these problems has helped you to develop a better “working” understanding of these key concepts,
- studied and practiced using four decision-making frameworks: (1) using socio-technical analysis to specify the problem in a complex, real world case, (2) practiced brainstorming techniques to develop and refine solutions that respond to your problem, (3) employed three ethics tests to integrate ethical considerations into your solutions and to test these solutions in terms of their ethics, and (4) applied a feasibility analysis to your solutions to identify and trouble-shoot obstacles to the implementation of your ethical solution,
- explored the analogy between solving ethical and design problems,
- practiced the skills of moral imagination, moral creativity, reasonableness, and perseverance, and...
- experienced, through key participant perspectives, the challenges of ethics advocacy “under the gun.”

Debrief on your group work before the rest of the class

1. Provide a concise statement and justification of the problem your group specified
2. Present the refined solution generation list your group developed in exercise 2.
3. Present and provide a quick summary explanation of the results of your group’s solution evaluation matrix.
4. Show your group’s feasibility matrix and summarize your assessment of the feasibility of implementing the solution alternatives you tested in exercise three.

Group Debriefing

1. Were there any problem you group had working together to carry out this case analysis? What were the problems and how did you go about solving them?
2. What problems did you have with understanding and practicing the four frameworks for solving problems? How did you go about solving these problems? Does your group have any outstanding questions or doubts?
3. Now that you have heard the other groups present their results, what differences emerged between your group’s analysis and those of the other groups? Have you modified your analysis in light of the analyses of the other groups? If so how? Do the other groups need to take into account any aspects of your group’s debriefing?

Toysmart Presentations

[Toysmart 2.pptx](#) [Toysmart 3.pptx](#)

Updated concept presentation for Spring 2011

[Review on Privacy and Property.pptx](#)

Privacy, Intellectual Property, Free and Informed Consent

[Review on Privacy Property Consent.pptx](#) [IMC V2 97.doc](#)

Appendix

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This optional section contains additional or supplementary information related to this module. It could include: assessment, background such as supporting ethical theories and frameworks, technical information, discipline specific information, and references or links.

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EAC ToolKit Project

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Funded by the National Science Foundation: "Collaborative Development of Ethics Across the Curriculum Resources and Sharing of Best Practices," NSF-SES-0551779

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CHAPTER OVERVIEW

3: Social Responsibility

[3.1: Socio-Technical Systems in Professional Decision Making](#)

[3.2: Ethics and Laptops - Identifying Social Responsibility Issues in Puerto Rico](#)

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3.1: Socio-Technical Systems in Professional Decision Making

Module Introduction

Milagro Beanfield War: Joe Mondragon has created quite a stir in Milagro, a small village in New Mexico. He has illegally diverted water from the irrigation ditch to his field to grow beans. Access to scarce water in New Mexico has created sharp political and social disputes which have reached a crises point in Milagro. Competing with traditional subsistence farmers like Joe is the profitable recreation industry. Ladd Devine, a wealthy developer, has joined with the state government in New Mexico to build a large recreational center consisting of a restaurant, travel lodge, individual cabins and a lavish golf course. Since there is not enough water to cover both recreational and agricultural uses and since Ladd Devine's project promises large tax revenues and new jobs, the state government has fallen behind him and has promised to give to the recreational facilities all the water it needs. Hence, the problem created by Mondragon's illegal act. You work for Ladd Devine. He has asked you to look into local opposition to the recreational facility. Along these lines, you attend the town meeting scheduled by Ruby Archuleta in the town's church. You are concerned about Charlie Bloom's presentation and the impact it may have on the local community. Prepare a STS analysis to test Bloom's assertions and better prepare Ladd Devine for local opposition to his facility.

Incident at Morales Fred is a chemical engineer hired by Phaust Corporation to design and make operational a new chemical plant for the manufacture of their newly redesigned paint thinner. Under financial pressure from the parent French company, Chemistre, they have decided to locate their new plant in Morales, Mexico to take advantage of lower costs and more flexible government regulations. You are well on the way toward designing this new plant when news comes from Chemistre that all budgets are being cut 20% to finance Chemistre's latest takeover acquisition. You are Fred and are now faced with a series of difficult financial-engineering decisions. Should you hold out for the more expensive Lutz and Lutz controls or use the cheaper ones produced locally? Should you continue with the current plant size or cut plant size and capacity to keep within budgetary constraints? You have also been made aware of the environmental and health risks associated with not lining the waste ponds used by the plant. Do you advocate lining the ponds or not, the latter being within compliance for Mexican environmental and health regulations. Prepare a STS analysis to help you make and justify these decisions. Make a series of recommendations to your supervisors based on this study.

Puerto Rican Projects

- Your company, Cogentrix, proposes a cogeneration plant that uses coal, produces electricity, and creates steam as a by-product of electricity generation process. Because the steam can be sold to nearby tuna canning plants, your company wishes to study the feasibility of locating its plant in or near Mayaguez, Puerto Rico. (Co-generation technology has become very popular and useful in some places.) Carry out an STS analysis to identify potential problems. Make a recommendation to your company. If your recommendation is positive, discuss how the plant should be modified to fit into the Mayaguez, Puerto Rico STS.
- Your company, Southern Gold Resources, is interested in mining different regions in central Puerto Rico for copper and gold. But you know that twenty years earlier, two proposals by two international mining companies were turned down by the PR government. Carry out a STS study to examine the feasibility of designing a different project that may be more acceptable to local groups. What does your STS analysis tell you about social and ethical impacts, financial promise, and likely local opposition. Can profitable mining operations be developed that respect the concerns of opposed groups? What is your recommendation based on your STS analysis?
- Windmar, a company that manufactures and operates windmills for electricity generation has proposed to locate a windmill farm in a location adjacent to the Bosque Seco de Guanica. They have encountered considerable local opposition. Carry out a STS analysis to understand and clarify this opposition. Can the concerns of local stakeholders be addressed and the windmill farm still remain profitable? How should the windmill project be modified to improve its chances of implementation?

Things to Know about STSs

What is a Socio-Technical System? (STS): A socio-technical system (STS) is a tool to help a business anticipate and successfully resolve interdisciplinary business problems. "Interdisciplinary business problems" refer to problems where financial values are intertwined with technical, ethical, social, political, and cultural values. (Reference: Chuck Huff, Good Computing: A Virtue Approach to Computer Ethics, draft manuscript for Jones and Bartlett Publishers)

Some Things to Know About STSs

1. Socio-Technical systems provide a tool to uncover the different environments in which business activity takes place and to articulate how these constrain and enable different business practices.
2. A STS can be divided into different components such as hardware software, physical surroundings, people/groups/roles, procedures, laws/statutes/regulations, and information systems. Other components include the natural environment, markets, and political systems.
3. But while different components can be distinguished, these are, in the final analysis, inseparable. Socio-Technical Systems are first and foremost **systems**: their components are interrelated and interact so that a change in one often produces changes that reverberate through the system.
4. Socio-Technical systems embody moral values such as justice, responsibility, respect, trust, and integrity as well as non-moral values such as efficiency, satisfaction, productivity, effectiveness, and profitability. Often these values can be located in one or more of the system components. Often they conflict with one another causing the system as a whole to change.
5. STSs change, and this change traces out a path or trajectory. The normative challenge here is to bring about and direct changes that place the STS on a value-positive trajectory. In the final analysis, we study STS to make sure that they change in a value-realizing direction.

Constituents or Sub-Environments of Business Activity

Paragraph summary of sub-environments of business followed by a table devoted to each one.

- **Technology** including hardware, software, designs, prototypes, products, or services. Examples of engineering projects in Puerto Rico are provided in the PR STS grid. In the Therac-25 case, the hardware is the double pass accelerator, in Hughes the analog-to-digital integrated circuits, and in Machado the UNIX software system and the computers in the UCI laboratories that are configured by this system. Because technologies are structured to carry out the intentions of their designers, they embed values.
- **Physical Surroundings.** Physical surroundings can also embed values. Doors, by their weight, strength, material, size, and attachments (such as locks) can promote values such as security. Physical surroundings promote, maintain, or diminish other values in that they can permit or deny access, facilitate or hinder speech, promote privacy or transparency, isolate or disseminate property, and promote equality or privilege.
- **People, Groups, and Roles.** This component of an STS has been the focus of traditional stakeholder analyses. A stakeholder is any group or individual which has an essential or vital interest in the situation at hand. Any decision made or design implemented can enhance, maintain, or diminish this interest or stake. So if we consider Frank Saia a decision-maker in the Hughes case, then the Hughes corporation, the U.S. Air Force, the Hughes sub-group that runs environmental tests on integrated circuits, and Hughes customers would all be considered stakeholders.
- **Procedures.** How does a company deal with dissenting professional opinions manifested by employees? What kind of due process procedures are in place in your university for contesting what you consider to be unfair grades? How do researchers go about getting the informed consent of those who will be the subjects of their experiments? Procedures set forth ends which embody values and legitimize means which also embody values.
- **Laws, statutes, and regulations** all form essential parts of STSs. This would include engineering codes as well as the state or professional organizations charged with developing and enforcing them
- The final category can be formulated in a variety of ways depending on the specific context. Computing systems gather, store, and disseminate information. Hence, this could be labeled **data and data storage structure**. (Consider using data mining software to collect information and encrypted and isolated files for storing it securely.) In engineering, this might include the information generated as a device is implemented, operates, and is decommissioned. This information, if fed back into refining the technology or improving the design of next generation prototypes, could lead to uncovering and preventing potential accidents. Electrical engineers have elected to rename this category, in the context of power systems, rates and rate structures.

Technological ComponentTechnological component of STS

Component	Description	Examples	Frameworks	More Frameworks

Technological	Hardware: Machines of different kinds	Door (with tasks delegated to it such as automatically shutting and being locked)	Value Discovery (identifying and locating values in STS)	Social Constructionism>: Restoring interpretive flexibility to reconstruct a technology to remove bias and realize value
	Code that configures machines around human purposes	Power generating technologies based on renewable and nonrenewable resources	Value Translation (Operationalizing and implementing values in an STS by designing and carrying out a procedure)	Identifying and mitigating complexity in the form of tightly-coupled systems and non-linear causal chains
	Technology can constrain business activity by de-skilling	Automobiles, computers, cell phones all of which have produced profound changes in our STSs	Value Verification (Using methods of participatory observation to determine how effectively values have been realized.)	De-centralizing control and authority
	Technology, especially software, can instrument human action	Microsoft Office, Firefox Browser, Google Chrome, Google Docs, Social Networking software	Transperspectivity: discovering strands of construction of current STS; identifying possibilities for reconstruction	Designing to avoid the technological imperative and reverse adaptation (where humans abandon ends and serve the ends of technologies)

Table 2: Ethical and Social Component Ethical Environments of the socio-technical system

Component	Description	Examples	Frameworks	More Frameworks
Ethical Environment	Moral Constructs: Spheres of justice where distribution takes place according to context-dependent rules (Rules)	Basic Moral Concepts: rights, duties, goods, values, virtues, responsibility, and justice	Utilitarianism: Happiness is tied to maximizing the satisfaction of aggregated preferences.	Basic Capabilities: life, bodily health, bodily integrity
	Social Constructs: Power and its distribution among groups and individuals	Intermediate Moral Concepts: Privacy, Property, Informed Consent, Free Speech, due Process, Safety/Risk	Rights: Capacities of action that are essential to autonomy, vulnerable to standard threats, and correlated with feasible duties	Cognitive Capabilities: Sense, Imagination, Thought; Emotion; Practical Reason
	Right: A right is a capacity of action, essential to autonomy, that others are obliged to recognize and respect.	Privacy: If the information is directly relevant to the relation to the holder and the seeker, then it is not private.	Virtues: Settled dispositions toward choosing the mean between extremes of excess and defect. (Courage is the mean between cowardice and recklessness)	Social Capabilities: Affiliations, Other Species

	Duty: A duty is a principle that obliges us to recognize and respect the rights of others.	Property: That with which I mix my labor is mine. Intellectual property is non-rivalrous and non-excludable.	Capabilities Approach: For Nussbaum, capabilities answer the question, “What is this person able to do or be?” For Sen, capabilities are “‘substantial freedoms,’ a set of (causally interrelated) opportunities to choose and act.”	Capabilities that address vulnerabilities: Play and Control over one's environment
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Physical Surroundings This table summarizes the physical environment of the STS and how it can constrain or enable action.

Physical Surroundings	Description	Examples	Frameworks	Frameworks
	Physical environment imposes constraints (limits) over actions that restrict possibilities and shape implementation.	Influence of rivers, mountains, and valleys on social and economic activities such as travel, trade, economic and agricultural activity, commerce, industry, and manufacturing.	Classroom environment enables or constrains different teaching and learning styles. For example, one can pair off technically enhanced and technically challenged classrooms with student-centered and teacher-centered pedagogical styles and come up with four different learning environments. Each constrains and enables a different set of activities.	The physical arrangement of objects in the classroom as well as the borders created by walls, doors, and cubicles can steer a class toward teacher-centered or student-centered pedagogical styles.

People, Groups, and Roles (Stakeholders) This table shows the social or stakeholder environment of the STS. A stakeholder is any group or individual that has a vital interest at play in the STS.

Stakeholders	Description	Examples	Frameworks	Frameworks
	Any group or individual that has a vital interest at play (at stake) in the STS.	Market Stakeholders: Employees, Stockholders	Non-Market Stakeholders: communities, activist groups and NGOs	Role: The place or station a stakeholder occupies in a given organizational system and the associated tasks or responsibilities.
		customers, suppliers retailers/wholesalers, creditors	business support groups, governments, general public (those impacted by projects who do not participate directly in their development	Interests: Goods, values, rights, interests, and preferences at play in the situation which the stakeholder will act to protect or promote.

		(Distinction between market and non-market stakeholders comes from Lawrence and Weber, Business and Society: Stakeholders, Ethics, Public Policy , 12th edition. McGraw-Hill, 14-15.	Alliances are discussed by Patricia Werhane et al., Alleviating Poverty Through Profitable Partnerships: Globalization, Markets, and Economic Well-being . Routledge (2009).	Relation: Each stakeholder is related to other stakeholders in an alliance and each relation is tied to goods and values.
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Procedural Environment

Procedural	Description	Examples	Framework	Framework
	A series of interrelated actions carried out in a particular sequence to bring about a desired result, such as the realization of a value. Procedures can schematize value by setting out a script for its realization.	Hiring a new employee: (a) settling on and publishing a job description; (b) soliciting and reviewing applications from candidates; (c) reducing candidate list and interviewing finalists; (d) selecting a candidate; (e) tendering that candidate a job offer. Other procedures: forming a corporation, filing for bankruptcy, gaining consent to transfer TGI and PII to a third party (Toysmart: opt-in and opt-out procedures).	Value Realization Process in Software Engineering: (a) Discovery: Uncovering values shared by a given community; (b) Translation: operationalizing and implementing values in a given STS; (c) Verification: using methods of participatory observation (surveys and interviews) to validate that the values in question have been discovered and translated.	Challenging the Statement of Values: (a) A stakeholder group raises a conceptual, translation, range, or development issue; (b) Group presents their challenge and response to other stakeholders; (c) If other stakeholder groups agree, then the challenge leads to a revision in the SOV; (d) Community as a whole approves the revision.

Legal Environment: Laws, Statutes, Regulations

Laws, Statutes, Regulations	Description	Examples	Frameworks	Frameworks
	Laws differ from ethical principles and concepts in that laws prescribe the minimally moral while ethical principles and concepts routinely explore higher moral "spaces."	Criminal Law: Applies to individuals; interested party in a criminal trial is society, not the victim.	Civil Law: Torts concern wrongful injury. The objective of a tort is to make the victim "whole" after an injury.	US and British law work through a common law system where current decisions are based on past decisions or precedent.

	Ethical principles challenge and criticize laws by bringing into question their normative content.	Involves proving a mens rea (guilty mind) and actus reus (guilty or law-breaking act) and that the mens rea caused the actus reus.	To prevail in a tort one must prove (in order of severity) negligence, recklessness, or intent.	The Puerto Rican system of law is based on the Napoleonic code where decisions relate directly to existing law and statute and precedent plays a weaker role.
	Laws can challenge ethical principles and concepts by raising issues of practicality. Also, as in responsibility theory, the law can structure and inform the moral discussion.	Criminal law does not apply to corporations because they "have no soul to damn and no body to kick" Baron Thurlow	Negligence involves proving that the defendant failed to meet some standard of due care .	Question: How does the statute-based Napoleonic system in PR constrain and enable business practice in relation to other systems such as the British and American common law systems?
			Contract law concerns the violation of the terms of a contract.	

Market Environment

Market Environment	Description	Examples	Frameworks	Frameworks
	Business takes place within different markets that shape supply, demand, and price. Globalization frequently requires that a business be adept at operating across different markets	Laissez- Faire: Each economic unit makes choice based on rational (enlightened) self-interest. (Private ownership of goods.)	Assumptions of a Free Market System: (a) Individual decisions are aggregated. (b) Information flows through price structure.	Recent economic studies of the limits of laissez-faire markets:
	Liberal use made here of notes from Economics class taught by CR Winegardner, University of Toledo, 1971-1972	Liberal Democratic Socialism: Limited government intervention is needed to improve upon the choice of individual economic units. (Mixture of private and public ownership)	(c) Free association. (d) Absence of force or fraud. (e) Individual agents are rational utility maximizer	(a) Information Asymmetries (as studied by Stieglitz). (b) Monopolies which, in the absence of competition, can dictate standards of price, product and service.
	Materials also take from Natural Capitalism from Lovins and Hawkings.	Communist, Authoritarian Socialism: The state is in the best position to know what choices and policies are beneficial for the economy as a whole and its component parts. (Public ownership of goods and services)	(f) Governments should adopt a hands-off stance because interference disrupts the ability of markets to produce utility-maximizing conditions.(4,4)	Animal spirits deflect economic decision-making away from perfect utility maximizing. They include confidence, fairness, corruption, money illusion, and stories.(4,5)

(5,1)	(5,2)	(5,3)	(5,4)	Ghoshal: bad management theories are destroying good management practices as they become self-fulfilling prophecies. Ghoshal is especially critical of agency theory, compliance/punitive approaches to corporate governance, and the theory of human nature he calls "Homo Economicus."(5,5)
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Information Environment: Collecting, Storing, and Transferring Information

Information Environment(1,1)	Description(1,2)	Examples(1,3)	Frameworks(1,4)	Frameworks(1,5)
(2,1)	How data and information is collected, stored, and transmitted along with ethical issues such as informed consent and privacy that accompany information management (2,2)	Informed Consent: Obtaining consent from information holder when collecting, storing, and transferring personal identifying information or transaction generated information.(2,3)	Privacy in Context (2,4)	Data Transfer and Informed Consent (2,5)
(3,1)	(3,2)	Belmont Report: (a) Principles: Respect for persons, beneficence, and justice; (b) Application 1: Informed consent as "subjects to the degree that they are capable be given the opportunity to choose what shall or shall not happen to them;" (c) Application 2: assessment of risks and benefits; (d) Application 3: Selection of subjects for experiment.(3,3)	(a) Identify individuals in groups in a context; (b) Identify the roles played by these individuals and groups. (3,4)	Opt-in: Information is not transferred unless data-holder expressly consents; Opt-out: Data will be transferred unless holder expressly refuses or withdraws consent. (3,5)

(4,1)	(4,2)	Conditions of Informed Consent Information, Comprehension, Voluntariness. (4,3)	(c) Identify context-relative norms that guide activities within context and between one context and another. (Materials on privacy in context are taken from Helen Nissenbaum in her book, Privacy in Context (4,4)	Fair Information Practices: (a) Notice: full disclosure and redress (way to resolve problems); (b) Choice: Choice about how information is to be used; (c) Access: access to stored and about to be disclosed information; (d) Security: ways that information will be kept secure and unauthorized access prevented incollection, storage, and transfer of information. (4,5)
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System of the Natural Environment

Natural Environment (1,1)	Description (1,2)	Examples (1,3)	Frameworks (1,4)	Frameworks (1,5)
(2,1)	Wicked Problems (2,2)	Principles of Sustainability according to B. Norton (2,3)	Four Theoretical Approaches to Environmental Ethics (2,4)	Environmental Value as determined by shadow markets (2,5)
(3,1)	(a) Difficulties in formulating and structuring problem; (b) Non-compatibility of solutions (several ways of stating solutions). (3,2)	Precautionary Principle: "in situations of high risk and high uncertainty,always choose the lowest risk option." (Cass Sunstein distinguishes several senses of the PP including one which makes it impossible to deviate from the status quo) (Norton 348)(3,3)	(a) Extensionism: Peter Singer's extension of Utilitarianism to cover sentient beings; (b) Tom Regan's ascription of rights to select animals. Biocentrism: Taylor's attribution of moral consideration to all teleological centers of a life.(3,4)	Willingness-to-pay: Resource in question would go to the highest bidder, that is, value is dependent on most intense preference and the disposable income to assert that preference(3,5)

(4,1)	(c) Wicked problems are "non-repeatable" in that they are context-dependent. This renders learning from previous problems and solutions much more difficult; (d) Wicked problems involve "competing values" that cannot be realized at the same time and that cannot be homogenized or plotted on a single scale; (e) Wicked problems exhibit "open-ended inter-temporal effects". Closely paraphrased from Norton, Sustainability, 133-5(4,2)	Safe Minimum Standard: "save the resource, provided the costs of doing so are bearable" (Norton 346) (4,3)	Land Ethics: A thing has value or is good insofar as it promotes the integrity, stability, and beauty of the biotic community. Biotic community includes humans, non-humans, species, and ecosystems all interacting as a system. From Aldo Leopole, Sand County Almanac; Virtue Environmental Ethics: Approach centers on virtues as habits that promote sustainable transactions with the natural environment. Hursthouse provides a provocative example with the virtue, respect for nature.(4,4)	Willingness-to-sell: Resource is owned by the public so its value is determined by its selling rather than buying price. This frees bid from disposable income. Now value becomes more reflective of the identity-conferring beliefs and attitudes of a community and its members.(4,5)
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Ethics of STS Research

- **Right of Free and Informed Consent:** This is the right of participants in a research project to know the harms and benefits of the research. It also includes the right not to be forced to participate in a project but, instead, offer or withdraw voluntarily their consent to participate. When preparing an STS analysis, it is mandatory to take active measures to facilitate participants's free and informed consent.
- Any STS analysis must take active measures to recognize potential harms and minimize or eliminate them. This is especially the case regarding the information that may be collected about different individuals. Special provisions must be taken to maintain confidentiality in collecting, storing, and using sensitive information. This includes careful disposal of information after it is no longer needed.

Participatory Observation

- As we said above, a socio-technical system (STS) is "an intellectual tool to help us recognize patterns in the way technology is used and produced." Constructing these tools requires combining modes of analysis that are ordinarily kept separate. Because STSs embed values, they are normative. These values can help to chart out trajectories of change and development because they outline values that the system needs to realize, maintain, or even enhance. In this way, the study of STSs is normative and a legitimate inquiry for practical and professional ethics. On the other hand, STS analysis requires finding out what is already there and describing it. So STS analysis is descriptive as well. In this textbox, we will talk briefly about the descriptive or empirical components of STS analysis. This material is taken from the draft manuscript of Good Computing: A Virtue Approach to Computer Ethics and has been developed by Chuck Huff.
- **Interviews:** Semi-Structured and Structured Interviews conducted with those familiar with a given STS provide an excellent source of information on the constituents of a given STS and how these fit together into an interrelated whole. For example, the STS grid on power systems was put together by experts in this area who were able to provide detailed information on power rates and protocols, software used to distribute energy through the gridlines, and different sources (representing both hard and soft technologies) of power generation.
- **Field Observation:** Those constructing a STS analysis go directly to the system and describe it in its day-to-day operation. Two books provide more information on the types and techniques of field observation: 1. David M. Fetterman, *Ethnography*; 2nd Edition, Applied Social Research Methods Series, Vol 17. London, UK.: Sage Publishers, 1998 and 2. James P. Spradley,

Participant Observation. New York, Harcourt, 1980. The data collected in this method can also be used to construct day-in-the-life scenarios that describe how a given technology functions on a typical day. These scenarios are useful for uncovering value conflicts and latent accidents. See James T. Reason, *Human Error*, Cambridge, UK.: Cambridge University Press, 1990 for information on latent accidents, how they are detected, and how they are prevented.

- **Questionnaires:** Questionnaires are useful for gathering general information from large numbers of people about a STS. Constructing good questionnaires is a difficult process that requires patience as well as trial and error. (Trying out questions on classmates and friends is the best way to identify unclear or misleading questions.) Avoiding complex, overly leading, and loaded questions represent a few of the challenges facing those who would construct useful questionnaires.
- **Archival and physical trace methods:** Looking at user manuals provides insight into how a system has been designed and how it works. Studying which keys are worn down on computer keyboards provides information on the kind of work being done. Comparing how a system is intended to work with how it is in fact being used is also illuminating, especially when one is interested in tracing the trajectory of a STS. Working with archival and physical trace methods requires critical thought and detective work.
- None of the above methods, taken in isolation, provides complete information on a STS. Triangulation represents the best way to verify data and to reconcile conflicting data. Here we generate evidence and data from a variety of sources then compare and collate. Claims made by interviewees that match direct on-site observations confirm one another and indicate data strength and veracity. Evidence collected through questionnaires that conflicts with evidence gathered through archival research highlights the need for detective work that involves further observation, comparison, interpretation, and criticism.
- Developing STS analyses bears a striking resemblance to requirements analysis. In both cases, data is collected, refined, and put together to provide an analysis. A key to success in both is the proper combination of normative and descriptive procedures.

Exercise 1: Make a Table that Describes the Socio-Technical System

Directions: Identify the constituents of the Socio-Technical System. Use the broad categories to prompt you.

1. What are the major hardware and software components?
2. Describe the physical surroundings.
3. What are the major people groups or roles involved?
4. Describe any procedures in the STS.
5. Itemize the laws, statutes, and regulations.
6. Describe the data and data structures in your STS. Use the two templates below that fill in this table for energy generation systems and for engineering ethics in Puerto Rico.

Socio Technical System Table

	Hardware	Software	Physical Surroundings	People, Groups, Roles	Procedures	Laws	Data and Data Structures

Exercise 2: Identify Value Mismatches in the STS

Directions: identify the values embedded in the STS. Use the table below to suggest possible values as well as the locations in which they are embedded.

1. **Integrity:** "Integrity refers to the attributes exhibited by those who have incorporated moral values into the core of their identities. Such integration is evident through the way values denoting moral excellence permeate and color their expressions, actions, and decisions. Characteristics include wholeness, stability, sincerity, honesty to self and others, authenticity, and striving for excellence.
2. **Justice:** Justice as fairness focuses on giving each individual what is his or her due. Three senses of justice are (1) the proper, fair, and proportionate use of sanctions, punishments and disciplinary measures to enforce ethical standards (retributive justice), (2) the objective, dispassionate, and impartial distribution of the benefits and burdens associated with a system of social cooperation (distributive justice), (3) an objectively determined and fairly administered compensation for harms and injustices suffered by individuals (compensatory justice), and (4) a fair and impartial formulation and administration of rules within a given group.

3. **Respect:** Respecting persons lies essentially in recognizing their capacity to make and execute decisions as well as to set forth their own ends and goals and integrate them into life plans and identities. Respects underlies rights essential to autonomy such as property, privacy, due process, free speech, and free and informed consent.
4. **Responsibility:** (Moral) Responsibility lies in the ability to identify the morally salient features of a situation and then develop actions and attitudes that answer to these features by bringing into play moral and professional values. Responsibility includes several senses: (1) individuals are responsible in that they can be called upon to answer for what they do; (2) individuals have responsibilities because of commitments they make to carrying out the tasks associated with social and professional roles; (3) responsibility also refers to the way in which one carries out one's obligations (This can range from indifference to others that leads to minimal effort to high care for others and commitment to excellence)
5. **Free Speech:** Free Speech is not an unlimited right. Perhaps the best place to start is Mill's argument in **On Liberty**. Completely true, partially true, and even false speech cannot be censored, the latter because censoring false speech deprives the truth of the opportunity to clarify and invigorate itself by defending itself. Mill only allows for a limitation of free speech based on harm to those at which the speech is directed. Speech that harms an individual (defamatory speech or shouting "fire" in a crowded theatre) can be censored out of a consideration of self-defense, not of the speaker, but of those who stand to be harmed by the speech.
6. **Privacy:** If an item of information is irrelevant to the relation between the person who has the information and the person who seeks it, then that information is private. Privacy is necessary to autonomy because control over information about oneself helps one to structure and shape one's relations with others.
7. **Property:** According to Locke, we own as property that with which we have mixed our labor. Thomas Jefferson argues that ideas are problematic as property because, by their very nature, they are shared once they are expressed. They are also nonrivalrous and nonexclusive.

Drawing Problems from Embedded Values

- Changes in an STS (e.g., the integration of a new technology) produce value mismatches as the values in the new component conflict with those already existing within the STS. Giving laptops to children produces a conflict between children's safety requirements and the safety features embedded in laptops as designed for adults.
- Changes within an STS can exaggerate existing value conflicts. Using digitalized textbooks on laptop computers magnifies the existing conflict concerning intellectual property; the balance between copyrights and educational dissemination is disrupted by the ease of copying and distributing digitalized media.
- Changes in STS can also lead to long term harms. Giving laptops to children threatens environmental harm as the laptops become obsolete and need to be safely disposed of.

Values Embedded in STS

	Hardware	Software	Physical Surroundings	People, Groups, Roles	Procedures	Laws	Data and Data Structures
Integrity							
Justice							
Respect							
Responsibility for Safety							
Free Speech							
Privacy							
Intellectual Property							

Using Socio-Technical System Grids for Problem Specification

The activity of framing is a central component of moral imagination. Framing a situation structures its elements into a meaningful whole. This activity of structuring suggests both problems and solutions. Framing a situation in different ways offers alternative

problem specifications and solution possibilities. Since skillful framing requires practice, this part of the module suggests how socio-technical system tables can help provide different frames for problem specification and solution generation.

Different Problem Frames

- **Technical Frame:** Engineers frame problems technically, that is, they specify a problem as raising a technical issue and requiring a technical design for its resolution. For example, in the STS grid appended below, the Burger Man corporation wishes to make its food preparation areas more safe. Framing this technically, it would be necessary to change the designs of ovens so they are more accident-proof.
- **Physical Frame:** How can the Burger Man corporation redesign its restaurants as physical facilities to make them more accessible? One way is to change the access points by, say, designing ramps to make restaurants wheelchair accessible. Framing this as a physical problem suggests solutions based on changing the physical structure and arrangement of the Burger Man STS.
- **Social Frame:** Burger Man as a corporation has stakeholders, that is, groups or individuals who have an essential interest at play in relation to the corporation. For example, framing the problem of making Burger Man more safe as a social problem might suggest the solution of integrating workplace safety into worker training programs and conducting regular safety audits to identify embedded risks.
- **Financial or Market-Based Frames:** Burger Man is a for-profit corporation which implies that it has certain financial responsibilities. Consequently, Burger Man should be concerned with how to provide safe, child-proof chairs and tables that do not cut unduly into corporate profits. But like the legal perspective, it is necessary to conduct ethical and social framing activities to compensate for the one-sidedness of financial framing.
- **Managerial Frame:** Many times ethical problems can be framed as managerial problems where the solution lies in changing managerial structures, reporting relations, and operating procedures. For example, Burger Man may develop a specific procedure when a cashier finishes a shift and turns over the cash register and its contents to another cashier. Burger Man may develop cleaning procedures and routines to minimize the possibility of serving contaminated or spoiled food to customers.
- **Legal Frame:** Burger Man may choose to frame its environmental responsibilities into developing effective procedures for complying with OSHA and EPA regulations. Framing a problem legally certainly helps to identify effective and necessary courses of action. But, because the ethical and social cannot be reduced to the legal, it is necessary to apply other frames to uncover additional risks not suggested by the legal framing.
- **Environmental Framing:** Finally, how does Burger Man look from the environmental standpoint? Does it consider environmental value (environmental health, safety, and integrity) as merely a side constraint to be addressed only insofar as it interferes with realizing supposedly more important values such as financial values? Is it a value to be traded off with other values? (For example, Burger Man may destroy the local environment by cutting down trees to make room for its latest restaurant but it offsets this destruction through its program of planting new trees in Puerto Rican tropical rain forests.) Framing a problem as an environmental problem puts the environment first and sets as a goal the integration of environmental values with other values such as worker safety and corporate profits.

Burger Man Socio-Technical System Table

[Socio
Technical System Grid for Business Ethics.docx](#)

Clicking on this figure will open as a Word file an STS table based on the fictional corporation, Burger Man. Below is a list of problems suggested by the STS analysis.

Media File Uplinks

This module consists of two attached Media Files. The first file provides background information on STSs. The second file provides two sample STS grids or tables. These grids will help you to develop specific STSs to analyze cases in engineering, business, and computer ethics without having to construct a completely new STS for each case. Instead, using the two tables as templates, you will be able to zero in on the STS that is unique to the situation posed by the case. This module also presents background constraints to problem-solving in engineering, business, and computer ethics. These constraints do not differ absolutely from the constituents of STSs. However, they pose underlying constraints that outline the feasibility of an ethical decision and help us to identify obstacles that may arise when we attempt to implement ethical decisions.

Socio-Technical Systems

[STS Background V3.doc](#)

Socio-Technical Systems: Constituents, Values, Problems, and Constraints.

STS Templates

[STS_Templates.doc](#)

Two STSs, Power Engineering and the Puerto Rican Context of Engineering Practice.

Socio-Technical Environments Table[STS2.pdf](#)

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3.2: Ethics and Laptops - Identifying Social Responsibility Issues in Puerto Rico

Introduction

While social responsibility has been recognized as one of the key areas of business ethics, much more needs to be done to develop frameworks and tools to clarify the concept itself and to implement it in business and professional practice on a day-to-day basis. This module will give students the opportunity to practice using frameworks and techniques that address these two needs.

Developing socio-technical system analyses provides an effective means to highlight issues of social responsibility. Since socio-technical systems embody values, building their descriptions allows us to read off potential problems due to harmful impacts and value conflicts. To facilitate this, you will be building socio-technical system descriptions using a grid or matrix that provides the components of socio-technical systems, levels under which they can be analyzed, and the values that they tend to embody. Building socio-technical system descriptions also requires using methods of participatory observation. These include constructing surveys and questionnaires, developing interviews, and building day-in-the-life scenarios. This module will help you frame and respond to social responsibility issues by providing a framework for socio-technical analysis and a set of methodological tools taken from participatory observation.

Case Narrative

Texas Laptop Case

1. In the late 1990s, the Texas State Board of Education proposed the ambitious plan of providing each of the state's four million public school students with their own laptop computer. This plan was devised to solve several problems confronting Texas public education.
2. Laptop computers could make educational resources more accessible to students who were faced with special challenges like deafness or blindness. Computers offer software options (such as audiobooks) that promise to reach more students than traditional printed textbooks.
3. Laptops also promised to solve the problem of obsolete textbooks. Texas purchased textbooks for their students at considerable costs. The purchasing cycle ran for six years. By the end of this cycle, textbooks were out of date. For example, in the late 1990s when the laptop plan was proposed, history textbooks still referred to the Soviet Union and to the existence of the Berlin Wall. Laptops, on the other hand, would present textbook content in digital form which would eliminate printing and shipping costs and facilitate updates through online downloads.
4. Texas business leaders were concerned about the computer literacy of the upcoming generation of students. By employing laptops in more and more teaching activities, students would learn how to interact with computers while taking advantage of the new and more effective modes of presentation offered.
1. However, adopting laptops also presented problems that critics quickly brought forth.
2. Teachers would need to learn how to use laptop computers and would have to change their teaching to accommodate them in the classroom.
3. Apparent cost savings disappeared upon further, closer examination. For example, it became clear that textbook publishers would not so easily give up the revenues they had come to depend upon that came from textbook purchases for public school students. Updates from downloads could turn out to be more expensive and educational software could be coded to restrict access and dissemination.
4. Further studies indicated that technical support costs would run two to three times initial outlays. Keeping laptop hardware and software up and running required technical support and continued investment.
5. Texas found that while some school districts--the richer ones--had already begun projects to integrate computing technology, the poorer school districts would require considerable financial support.

To deal with these problems, Texas carried out several pilot projects that examined the effectiveness of laptop integration in select school districts. While several successes were reported a series of problems arose that led the Texas Board of Education officials to postpone the laptop project. First, pilot projects depended on donations from private computing vendors. While some were forthcoming, others failed to deliver hardware on time and provided only minimal technical support. Second, teachers resisted laptop integration due to the extensive investment of time required to appropriate computing skills and the difficulty of modifying existing curricula and teaching styles to accommodate laptop hardware and software. Third, at that time the available educational software, such as digitalized textbooks, was expensive, inadequately developed, and narrowly focused on curricular areas such as writing and math practice. Teachers also began to develop more comprehensive and philosophical criticisms of laptop use.

Education specialist, Larry Cuban, argued that while laptops provided good support for vocational education, they failed to deliver on other educational goals such as teaching children how to interact with their peers and teachers and teaching children the civic virtues necessary to become active participants in a democratic form of government. Studies began to appear that argued that skills developed through computer use came at the expense of other, more social skills.

The Texas Laptop plan was never formally implemented beyond the pilot project phase. However, several computer-integration projects have been carried out in other parts of the country. For example, Larry Cuban reports on computer integration projects carried out in Silicon Valley in California. MIT has developed a cheap laptop computer for use in developing nations. You can find a link to computer integration projects that have been implemented in Philadelphia public schools through the support of the Microsoft Foundation.

Students in computer ethics classes at the University of Puerto Rico at Mayaguez have looked into the feasibility of integrating laptops in the public school socio-technical system in Puerto Rico. They began by looking at the project to provide public school teachers with laptops that was carried out in the late 1990s under the Pedro Rossello administration. The student research projects came to focus on three problem areas. First, they examined whether there were structures in laptop design that made computers unfit for use by children. Second, they studied whether social or ethical problems would arise from the disposal of spent laptops. Third, they investigated the impact on copyright law and intellectual property practices that digitalizing printed textbooks would have.

What you are going to do...

Decision Point One

You are a computer engineer and have been subcontracted by your local government to purchase new portable computers for high school teachers. Your job includes...

- Selecting the kind of computer to be used
- Identifying vendors who will sell the computers
- Overseeing the distribution of computers to high school teachers
- Developing a training program to help teachers learn to use computers
- Designing a technical support hotline to help teachers work out any technical problems that may arise

Distributing computers to high school teachers seems simple enough. You select computers, buy them, and give them to the teachers. Yet only a slight change in circumstances can bring into the open latent or potential ethical issues:

- How should you go about setting up the bidding process to determine the computers to be used?
- What should you do to determine teacher and student needs and how computers can respond to these needs? It makes very little sense to provide computers and then tell teachers and students to use them. What are they to do with these computers? How do they fit them into everyday education? This requires seeing the computer project from the standpoints of students, their parents, and teachers. The reversibility test will help here.
- Who stands to benefit from your actions? Who stands to be harmed from these actions? How will benefits and harms be distributed through the different stakeholders in this case?
- Latent ethical problems exist in this socio-technical system that can erupt into full-blown problems with small changes in circumstances
- Someone you know well--say your cousin--submits a bid. What ethical issues does this turn of events give rise to?
- The contract to provide computers is awarded to your cousin, and he provides reliable computers at a reasonable price. Then, a few weeks later, you read the following headline in the newspaper: "More Government Corruption--Computer Czar's Cousin Counts Millions in Cozy Computer Contract" What do you do now?
- A group of angry high school teachers holds a press conference in which they accuse the government of forcing them to use computing technology in their classes. They say you are violating their academic freedom. How should you respond?
- Someone in the government suggesting placing a program in each computer that allows government officials to monitor the computers and track user behavior. How would you feel if your computer use were being monitored without your knowledge or consent? Are their circumstances under which monitoring could bring about any social benefits? What are the likely harms? Do the benefits outweigh the harms? Suppose you go along with this and read the following headline in the morning newspaper: "Government Snoops Bug High School computers". Using the publicity test, what kind of person would you appear to be in the public's eye? How would you view yourself in terms of this action?

Decision Point Two

You are Dr. Negroponte from MIT. For several years now, you have been working to design laptop computers that respond to a wide range of needs of children in poor, developing nations. You have set up an incentive for people in developed nations to contribute to children in poor nations. For \$300, one can buy two laptops, keep one, and have the other donated to a child in a developing nation. This has generated computers but governments in developing nations--enthusiastic at first--have recently shown themselves reluctant to carry through on their commitments. Your goal of reducing laptop costs to \$100 per computer has also stalled. It has been difficult to generate projected economies of scale.

- The laptops employ a simple design. They use Linux as an operating system since this shareware can be freely downloaded. The computers are also designed to be used in areas where the underlying infrastructure, especially electricity, is unreliable. They are battery-driven and a hand crank allows for recharging batteries when electricity is unavailable. They employ a wireless connection to the Internet.
- An Open Education Resource movement has been started to generate educational resources directly and freely available to children using MIT laptops. This movement has generated considerable educational content of varying qualities. Reports available online provide insights into the pros and cons of the open resource educational movement. Whether this can (or should) replace traditional textbooks (which can be quite expensive and difficult to update) is still open to debate.
- There is evidence that laptops can and have contributed to an enhanced learning experience for children in developing nations. Poor attendance, a large and chronic problem, has been improved in laptop programs. Children enjoy their computers and seem better motivated in general as a result. They take their computers home for homework and share them with the rest of their family. Many teachers have successfully adapted their teaching styles to this Internet-supported, technologically enhanced educational mode.
- But recently, laptops have come under increasing critical scrutiny.
- They are more expensive than traditional educational materials such as textbooks
- They compete for scarce financial resources and may be less cost-effective in the long run than other, more traditional educational resources.
- The MIT laptop has no hard drive, a fact critically singled out by Microsoft's founder, Bill Gates. They have been designed to use the Linux operating system rather than Microsoft's more expensive and complicated one.
- Developing nation governments have recently shown "cold feet" to putting action behind their verbal commitments to laptop computers. This may, in part, be due to concerns expressed by parents and teachers.
- Defend the MIT Laptop Project in the face of these and other criticisms.
- Should their design be modified to suit better children's needs as well as the concerns of teachers and parents?
- What features do MIT laptops already display that respond to student, parent, and teacher needs?
- What are the alternatives to MIT Laptops? For example, evaluate the proposal made by a group in computer ethics to invest in and emphasize instruction in computer laboratories housed in schools themselves. What problems would this new approach avoid? What are its limitations in comparison to the laptop approach?

Decision Point Three

- You live in a developing nation. While you have work, it doesn't pay well and you are barely able to provide for your family's basic needs. One problem and things will get very difficult for you and your family.
- Your child came home with an MIT-designed laptop computer. She and her classmates have benefited from the computers donated to their school by the generosity of developed nations where concerned citizens can buy two computers and have one donated to needy children. You find this somewhat patronizing and you see these laptops as a mixed blessing.
- On the one hand, this laptop has helped you and your family to enjoy the benefits of access to the Internet, although, because of poor infrastructure, this access is limited, sporadic, and subject to frequent breakdowns. On the other hand, you question whether your child is mature enough to use and care for her computer. If anything should happen, you would be required to buy a new replacement laptop, and you simply don't have the money.
- Yet should you not replace your daughter's broken laptop, she would be excluded from the education her peers enjoy because she would no longer have a computer. You question whether you want to run on this "treadmill."
- Furthermore, you can see that laptops--even MIT laptops--are designed for adults, not children. They are made of heavy metals and other toxic materials. The batteries, especially, are dangerous because of the materials they contain. They wear out and replacing them can be expensive.

- Your child could also become a target for robbers. She walks to and from school carrying her computer, and you know of other children who have been beaten and robbed of their laptops.
- So you see these laptops as a mixed blessing fraught with risk. What should you do?

What you are going to do...

Exercise 1: Prepare an STS Grid

- Construct a socio-technical system (STS) grid for public schools in Puerto Rico
- Using the templates found in "Socio-Technical Systems in Professional Decision Making", identify the key constituents such as hardware, software, physical surroundings, etc.
- Select key levels for analysis. For example, you may want to look at the STS from the standpoint of individuals (students and teachers), small groups (public school systems), and institutions (education and business)
- Starting with a short list of values, identify the values embedded in the public school STS and, if possible, the specific components in which these values are embedded. A good place to start is to see how different physical arrangements of the classroom embody different approaches to education

Values in STSs: Values that can be used for exercise 1 include Justice (equity and access), Property, Privacy, Free Speech, Responsibility (Safety). More on these values can be found by clicking on the Computing Cases link provided in this module. Several of these values are defined in the Ethics of Team Work module, m13769.

Exercise 2: Identifying Potential or Latent Problems in STSs

- Choose one of the following three problem areas to help focus your work: (1) value problems that may arise when laptops with their current design are integrated in the PR STS; (2) value problems that may arise by the digitalization of textbooks and other educational materials; (3) value problems and potential harms that may arise during the disposal of spent laptops.
- Compare values embodied in current laptop design with those embodied in the Puerto Rican public school STS. Are there any conflicts? What are these?
- Look more closely at the Puerto Rican public school STS. Are there any conflicts that will be highlighted, exaggerated, or increased by the integration of laptop computers.
- Finally, look for potential harms that could occur in the short, middle, and long term future.

Exercise 3: Develop Counter-Measures to Problems

- Generate 5 to 10 options to respond to the problems you have identified. Make sure that you include the status quo among your options.
- Check each option against the problems you have identified. Does the option solve the problems identified in your STS analysis? Does it integrate the conflicting values and avoid untoward results? Does it give rise to new problems?
- Prepare a short presentation for the class (5 to 10 minutes) where you outline your problem, set forth the range of solutions you have identified, and describe and justify your solution. Be sure to address issues that may arise when you turn to implementing your solution.
- Provide a one or two-sentence argument that your solution is best for delivering on social responsibility.

Exercise 4: Evaluate the Microsoft Philadelphia Public Schools Project

- Listen to/read the news report on the Microsoft Foundation's project to integrate computing technology in Philadelphia. (You can find it by clicking on the link in this module.)
- Is this an example of a corporation carrying out its social responsibility to the surrounding community?.
- Evaluate Microsoft generally in terms of its social responsibility.

Presentations

Social Justice and Responsible Technology: [Social Justice and Resp Tech.pptx](#)

Educational Laptops Presentation: [Educational Laptops.pptx](#)

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CHAPTER OVERVIEW

4: Changing Organizational

[4.1: Pirate Code for Engineering Ethics](#)

[4.2: Developing Ethics Codes and Statements of Values](#)

[4.3: Gray Matters for the Hughes Aircraft Case](#)

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4.1: Pirate Code for Engineering Ethics

Module Introduction

In this module, you will learn about professional and occupational codes of ethics by looking at a bad code, writing your own code, and then critically examine a professional code of ethics, the engineering code for the Colegio de Ingenieros y Agrimensores de Puerto Rico. Three exercises will take you through the process of examining the Pirate Creed, writing your own code, and examining the Colegio's code. Text boxes will provide helpful background information on purposes served by professional codes, philosophical objections, and a framework for working your way through a stakeholder-based code like that of the CIAPR or the National Society of Professional Engineers. This module provides a Spanish translation of the Pirate Creed prepared by Dr. Dana Livingston Collins of the Department of Humanities in the University of Puerto Rico at Mayagüez.

Concluding this module are two word documents uploaded as media files. One provides the exercises that are presented in this module in XML format. The other provides the background information that has been presented in this module as Textboxes.

Module Activities

1. You will analyze the Pirate Creed in terms of (a) its different functions, (b) the community values it embodies, and (c) how it stands toward nonmembers of the pirate community as well as members.
2. You will write a code of ethics for an occupational or professional area such as business or engineering.
3. You will debrief the rest of the class on your group's code, clarify its functions and values, and defend it if necessary.
4. This module will conclude with a look at the code of ethics of the Puerto Rico State Society of Professional Engineers and Land surveyors or Colegio de Ingenieros y Agrimensores de Puerto Rico.

Pirates Creed of Ethics (translated into Spanish by Dana Collins)

1. El capitán tendrá comando total durante una batalla y tendrá la autoridad para dirigir el barco. El que no siga al capitán podrá ser castigado si la tripulación no vota en contra del castigo.
2. Si el barco naufraga, la tripulación permanecerá unidos hasta el capitán consigue otra nave. Si la nave es propiedad común de la tripulación, la primera nave capturada pertenecerá al capitán con una (1) parte de botín.
3. El cirujano del barco recibirá doscientas (200) coronas para el mantenimiento de su equipo médico y recibirá una (1) parte del botín.
4. Los otros oficiales recibirán una (1) parte cada uno, y si se distinguen, la tripulación determinará cuanto recibirán como recompensa.
5. El botín de una nave capturada será distribuido en partes iguales.
6. El primero que señale la aparición de un barco que sea capturado recibirá cien (100) coronas.
7. El que pierda un ojo, una mano, o una pierna mientras está en servicio, recibirá hasta seis esclavos o seiscientos (600) coronas.
8. Los suministros y raciones serán compartidos por igual.
9. La penalidad por traer una mujer disfrazada a bordo es la muerte.
10. Si un hermano roba de otro, perderá su nariz u orejas. Se peca de nuevo, se le darán un mosquete, municiones, plomo y una botella de agua y será abandonado en una isla.
11. Si hay duda en una disputa entre hermanos, una corte de honor determinará el veredicto. Si un hermano es encontrado culpable, la primera vez será perdonado, pero al ofender de nuevo, será atado a un cañón y recibirá un latigazo de cada miembro de la tripulación. El mismo castigo será dado a todos, incluyendo oficiales, quienes se emborrachen al punto de perder sus sentidos mientras estén en el barco.
12. El que se duerma mientras está trabajando como centinela, recibirán latigazos por todos los miembros de la tripulación. Se repite el crimen, su cabeza será rajada.
13. A todos quienes conspiren para desertar, o lo que hayan desertado y sean capturados, sus cabezas serán rajadas.
14. Pelas entre varios hermanos mientras estén a bordo será resueltos en tierra con pistolas y espadas. El que saque primera sangre será el vencedor. No pueden golpear a otro mientras estén a bordo de la nave.

Exercise 1: Pirate Creed

- What is good about the Pirate Creed of Ethics?
- what is bad about the Pirate Creed of Ethics?
- What is the purpose of the Creed for the Pirate Community?

- What values are embedded in the Pirate Creed
- How does the Pirate Creed deal with nonmembers?

Exercise 2: Writing a Code of Ethics for Engineers

- Step One: Identify the purpose of your engineering code of ethics. For example, is it to punish wrongful behavior, provide a set of guidelines, educate the community, support ethical behavior, or create an ethics dialogue?
- Step Two: Identify the contributions that engineering makes to society.
- Step Three: Identify the stakeholders of the engineering profession. A stakeholder is any group or individual with a vital or essential interest tied to what engineers do. along with these stakeholders, identify their stakes, that is, the goods, rights, interests or values that are maintained, promoted, or diminished by what engineers do?
- Step Four: Enumerate the obligations or duties that engineers have toward each of these stakeholders. In other words, what can engineers do to maintain, promote, or diminish the stakes of each stakeholder?
- Step Five: Identify the conflicting obligations that arise from the fact that engineers have different stakeholders who hold conflicting stakes? Do any of these stakeholders or stakes have obvious priority over the others?
- Step Six: Step back and reflect on what you have written. For example, look for different kinds of provisions. Does your code use ideals of the profession which set forth the profession's central or cardinal objectives? Does your code contain principles of professional conduct which set forth minimal levels of behavior and prescribe sanctions and punishments for compliance failures? In the CIAPR (Colegio de Ingenieros y Agrimensores de Puerto Rico) code of ethics, the fundamental principles and basic canons set forth the ideals of the profession. The principles of professional conduct fall in the section on practical norms.
- Step Seven: The Final Audit. Submit your code to an overall audit to see if anything has been left out. Have you included all the stakeholders and their stakes? Have you left out any ethical considerations such as rights and duties? Compare your code to the law. Are your code's provisions legal? Do they overlap with existing law? Do they imply criticisms of existing laws? If they imply punishments or sanctions, what measures does your code prescribe to administer justly and properly these sanctions? Finally, be sure to guard against the equal but opposite sins of over-specificity and too much generality. Overly specific codes try to provide a rule for every possible situation. Because this is impossible, these codes tend toward rigidity, inflexibility, and irrelevance. Codes that are too general fail because they can be interpreted to rationalize any kind of claim and, thus, mask immoral actions and intentions.

Exercise 3: Studying the code of Ethics of the Colegio de Ingenieros y Agrimensores de Puerto Rico

- Identify the provisions that touch upon the relation of the engineer to the public. What goods are at stake in this relation? What can engineers do to preserve or promote these goods?
- Identify provisions that touch upon the relation of the engineer to the client. What goods are at stake in this relation? What can engineers do to preserve or promote these goods?
- Identify provisions that touch upon the relation of the engineer to the CIAPR (professional engineering society) What goods are at stake in this relation? what can engineers do to preserve or promote these goods?.
- Finally, identify provisions that touch upon the relation of the engineer to other engineers (peer relations). What goods are at stake in this relation? What can engineers do to preserve or promote these goods?

Textbox 1: Code of Ethics of Colegio de Ingenieros y Agrimensores de Puerto Rico (Puerto Rico State society of Professional Eng

- The CIAPR code of ethics has three parts:
- Part One: Three Fundamental Principles which express cardinal objectives for engineering practice in Puerto Rico
- Part Two: Ten Canons which set forth general rules for ethical engineering practice
- Part Three: Each canon is repeated followed by several practical norms. by setting forth detailed rules, practical norms specify and interpret the basic canons. They also set forth specific and concrete rules for professional and ethical conduct
- The CIAPR code of ethics is a stakeholder code. This means it identifies engineering stakeholders, the goods they depend upon, and the duties engineers have in protecting or promoting these goods.

Key Engineer Relations

- The relation between engineer and public is founded on the goods of health, safety and welfare.
- The relation between engineer and client is founded on the good of faithful agency (trust).

- The relation between the individual engineer and the profession is founded on the engineer working to maintain the good reputation and integrity of the profession.
- The peer relation between practicing engineers is founded on the good of collegiality.

Engineer and Public

- Duties arising in this relation are tied to maintaining or promoting the goods of health, safety, and welfare. They include minimizing harm, avoiding paternalism (making decisions for others who have the right and ability to make these for themselves), free and informed consent (the right of those taking a risk to consent to that risk).
- FP1: Deberán considerar su principal función como profesionales la de servir a la humanidad. Su relación como profesional y cliente, y como profesional y patrono, deberá estar sujeta a su función fundamental de promover el bienestar de la humanidad y la de proteger el interés público.
- Canon 1: Velar por sobre toda otra consideración por la seguridad, el ambiente, la salud y el bienestar de la comunidad en la ejecución de sus responsabilidades profesionales.
- Practical Norm 1d: Cuando tengan conocimiento o suficiente razón para creer que otro ingeniero o agrimensor viola las disposiciones de este Código, o que una persona o firma pone en peligro la seguridad, el ambiente, la salud o el bienestar de la comunidad, presentarán tal información por escrito a las autoridades concernidas y cooperarán con dichas autoridades proveyendo aquella información o asistencia que les sea requerida.

Engineer to Client

- Duties stemming from this relation arise out of faithful agency, that is, the responsibility of an engineer to remain true to the client's interests. Positively this includes exercising due care for the client by carrying out the client's interests through the exercise of sound, competent engineering professional judgment. Negatively this entails avoiding conflicts of interest and revealing the client's confidential information.
- Faithful Agency: Canon 4—Actuar en asuntos profesionales para cada patrono o cliente como agentes fieles o fiduciarios, y evitar conflictos de intereses o la mera apariencia de éstos, manteniendo siempre la independencia de criterio como base del profesionalismo.
- Conflict of Interest: 4a—Evitarán todo conflicto de intereses conocido o potencial con sus patronos o clientes e informarán con prontitud a sus patronos o clientes sobre cualquier relación de negocios, intereses o circunstancias que pudieran influenciar su juicio o la calidad de sus servicios.
- Confidentiality: 4i—Tratarán toda información, que les llegue en el curso de sus encomiendas profesionales, como confidencial y no usarán tal información como medio para lograr beneficio personal si tal acción es adversa a los intereses de sus clientes, de sus patronos, de las comisiones o juntas a las que pudiera pertenecer o del público.

Engineer to Profession

- This includes working to promote the profession's autonomy and independence as well as maintaining its good reputation. Moreover it requires that engineers participate in their professional society, work to advance engineering, be objective and impartial in their work, and associate only with persons of good reputation.
- Canon 3: Emitir declaraciones públicas únicamente en una forma veraz y objetiva.
- Practical Norm 3a: Serán objetivos y veraces en informes profesionales, declaraciones o testimonios. Incluirán toda la información relevante y pertinente en tales informes, declaraciones o testimonios.

Engineer to Engineer

- This relation is based on the good of Collegiality. It requires that engineers work to maintain friendly and collaborative relations with other engineers by avoiding disloyal competition and comparative advertising and by always giving peers due credit for their contributions to engineering projects and designs.
- Practical Norm 4l: Antes de realizar trabajos para otros, en los cuales puedan hacer mejoras, planos, diseños, inventos, u otros registros, que puedan justificar la obtención de derechos de autor o patentes, llegarán a un acuerdo en relación con los derechos de las respectivas partes. (Give due credit to colleagues for their work).
- Canon 5: Edificar su reputación profesional en el mérito de sus servicios y no competir deslealmente con otros. (Avoid disloyal competition)
- Practical Norm 6b: Anunciarán sus servicios profesionales sin auto-alabanza y sin lenguaje engañoso y de una manera en que no se menoscabe la dignidad de sus profesiones. (Non-comparative advertising)

- Practical Norm 5h: No tratarán de suplantar, ni suplantarán otro ingeniero o agrimensor, después de que una gestión profesional le haya sido ofrecida o confiada a éste, ni tampoco competirá injustamente con él. (Avoid disloyal competition)

Professional Codes as Social Contracts

- What some have said about defining ethics could also be applied to defining a profession: it's a bit like "nailing jello to a tree." Nevertheless, we can make reasonable claims about professions: they can be treated as social contracts, and they have something to do with specialized knowledge. If these two claims hold, then a third claim can be made, namely, that professions have an ineliminable ethical dimension.
- A legitimate contract between two parties requires a quid pro quo (a mutually beneficial exchange) and free consent (consent that includes full information and excludes force or deception). The social contract between engineering and society can be pictured in the following way:

Profession as Social Contract

Society grants to Profession	Profession grants to Society
Autonomy	Self-Regulation
Prestige	Primacy of public health, safety, and welfare
Monopoly	Developing and enforcing ethical and professional standards

Society grants autonomy, prestige, and monopoly control to the profession of engineering.

1. Autonomy includes freedom from regulation and control from the outside through cumbersome laws, regulations, and statutes.
2. Prestige includes high social status and generous pay.
3. Monopoly status implies that the profession of engineering itself determines who can practice engineering and how it should be practiced.
4. The profession promises to use its autonomy responsibly by regulating itself. It does this by developing and enforcing professional and ethical standards. By granting prestige to the profession, society has removed the need for the profession to collectively bargain for its self-interest.
5. Not having to worry about its collective self-interest, the profession is now free to hold paramount the health, safety, and welfare of the public.
6. This contract explains why professions develop codes of ethics. Codes document to the public the profession's commitment to carry out its side of the social contract, namely, to hold paramount public welfare. They can do this because society will honor its side of the contract, namely, to remove from the profession the need to fight for its self-interest

This social contract is more symbolic and explanatory than real.

- Codes allow the profession to document to society that it has developed proper standards and intends to enforce them. They express the profession's trust in society to keep its side of the bargain by granting autonomy, prestige, and monopoly. Of course this contract has never been explicitly enacted at a point in historical time. But the notion of a social contract with a mutually beneficial exchange (a quid pro quo) provides a useful device for modeling the relation that has actually evolved between society and its professions.

Professions and Responsibility

- Professions have been created to exercise stewardship over knowledge and skill domains.
- Exercising stewardship over X generally means watching over, preserving, protecting, and even improving X. Stewardship is a forward-looking kind of responsibility similar to the responsibility that a parent exercises toward his or her children. The steward is a trusted servant or agent of the landowner who acts in the owner's place while the latter is absent or incapacitated.
- "Stewardship," thus, refers to the profession's responsibility to safeguard its specific domain of knowledge and skill. This domain is essential to society in some way (it provides society with a basic, common good) and society delegates responsibility for this domain to its members who are specially suited to exercise it.
- So, generally speaking, professions can be characterized in terms of epistemological and ethical responsibilities.
- The epistemological responsibility refers to stewardship over the knowledge and skills that characterizes the profession. The profession preserves, transmits, and advances this domain of knowledge and skill. (Epistemology = study of knowledge.)

- The ethical dimension refers to the responsibility of the profession to safeguard knowledge and skill for the good of society. Society trusts the profession to do this for the sake of the common good. Society also trusts the profession to regulate its own activities by developing and enforcing ethical and professional standards.

Objections to and Mischievous Side Effects of Codes of Ethics

These objections are taken from John Ladd, "The Quest for a Code of Professional Ethics: An Intellectual and Moral Confusion." This article can be found in Deborah G. Johnson, editor, (1991) *Ethical Issues in Engineering*, New Jersey: Prentice Hall: 130-136. The author of this module has taken some liberties in this presentation.

- Codes "confuse ethics with law-making" (Ladd, 130). Ethics is deliberative and argumentative while law-making focuses on activities such as making and enforcing rules and policies.
- A code of ethics is an oxymoron. Ethics requires autonomy of the individual while a code assumes the legitimacy of an external authority imposing rule and order on that individual.
- Obedience to moral law for autonomous individuals is motivated by respect for the moral law. On the other hand, obedience to civil law is motivated by fear of punishment. Thus, Ladd informs us that when one attaches "disciplinary procedures, methods of adjudication and sanctions, formal and informal, to the principles that one calls 'ethical' one automatically converts them into legal rules or some other kind of authoritative rules of conduct..." (Ladd 131) Accompanying code provisions with punishments replaces obedience based on respect for the (moral) law with conformity based on fear of punishment.
- Codes lead to the dangerous tendency to reduce the ethical to the legal. Ethical principles can be used to judge or evaluate a disciplinary or legal code. But the reverse is not true; existing laws cannot trump ethical principles in debates over ethical issues and ethical decisions. As Ladd puts it, "That is not to say that ethics has no relevance for projects involving the creation, certification and enforcement of rules of conduct for members of certain groups....[I]ts [ethics's] role in connection with these projects is to appraise, criticize and perhaps even defend (or condemn) the projects themselves, the rules, regulations and procedures they prescribe, and the social and political goals and institutions they represent." (Ladd 130)
- Codes have been used to justify immoral actions. Professional codes have been misused by individuals to justify actions that go against common morality. For example, lawyers may use the fact that the law is an adversarial system to justify lying. Ladd responds in the following way to this dodge: "{T}here is no special ethics belonging to professionals. Professionals are not, simply because they are professionals, exempt from the common obligations, duties and responsibilities that are binding on ordinary people. They do not have a special moral status that allows them to do things that no one else can." (Ladd 131)

Mischievous Side-Effects of Codes (from John Ladd)

- Codes make professionals complacent. (Ladd 135) First, they reduce the ethical to the minimally acceptable. Second, they cover up wrongful actions or policies by calling them--within the context of the code--"ethical". For example, the NSPE code of ethics used to prohibit competitive bidding. Enshrining it in their code of ethics gave it the appearance of being ethical when in fact it was motivated primarily by self interest. This provision was removed when it was declared unconstitutional by the U.S. Supreme Court for violating the Anti-Trust law.
- Because codes focus on micro-ethical problems, "they tend to divert attention from macro-ethical problems of a profession." (Ladd 135) For example, in Puerto Rico, the actions of the Disciplinary Tribunal of the Colegio de Ingenieros y Agrimensores de Puerto Rico tend to focus on individual engineers who violate code provisions concerned with individual acts of corruption; these include conflicts of interest, failing to serve as faithful agents or trustees, and participating in corrupt actions such as taking or giving bribes. On the other hand, the CIAPR does not place equal attention on macro-ethical problems such as "the social responsibilities of professionals as a group" (Ladd 132), the role of the profession and its members in society (Ladd 135), and the "role professions play in determining the use of technology, its development and expansion, and the distribution of the costs." (Ladd 135)

Exercise: Questions for Reflection

1. Which of Ladd's criticisms apply to the Pirate Creed?
2. How does your group's code of ethics stand in relation to Ladd's criticisms?
3. Do Ladd's objections apply to the ABET, NSPE, or CIAPR codes?

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4.2: Developing Ethics Codes and Statements of Values

Module Introduction

Codes of ethics evoke opposite reactions from people who teach, do research in, or are practitioners of occupational and professional ethics. Some hold that teaching codes of ethics is essential to preparing students for their future careers. Corporations, for example, have come to view codes as the cornerstone of a successful compliance program. Professional societies, such as the Puerto Rico State Society of Professional Engineers and Land Surveyors, also make the drafting, revising, and disseminating professional codes of ethics a central part of practicing professional engineering ethics. But many strongly oppose codes because they promote the wrong sorts of attitudes in those who would be influenced by them. As you will see below, philosophical ethicists raise objections to codes because they undermine moral autonomy, lead to uncritical acceptance of authority, and replace moral motives with fear of punishment. These polar stances are grounded in the very different perspectives from which different groups approach codes. But they are also grounded in the fact that codes take many different forms and serve distinct functions. For example, consider the introductory considerations presented in the following:

Different Uses for Codes

Kinds of Codes:

- Professional Codes of Ethics. Professions such as engineering and accounting have developed codes of ethics. These set forth the ideals of the profession as well as more mundane challenges faced by members. Engineering codes, for example, set forth service to humanity as an ideal of the profession. But they also provide detailed provisions to help members recognize conflicts of interest, issues of collegiality, and confidentiality responsibilities.
- Corporate Codes of Ethics. Corporate codes are adopted by many companies to respond better to the Federal Sentencing Guidelines. These codes provide guidelines on particularly sticky issues (When does a gift become a bribe?) They also set forth provisions that express the core values of the corporation. These lengthy codes with detailed provisions support a compliance approach to organizational discipline.
- Corporate Credos. Some companies have shortened their lengthy codes into a few general provisions that form a creed. Johnson and Johnson's Credo is famous in this respect and can be found by clicking on the Business Ethics Library link provided above.
- Statements of Values. Finally, more mature companies find it useful to express and disseminate their core value commitments in Statements of Values. These form the basis of values-based decision-making. While codes of ethics clearly establish minimum standards of acceptable conduct, Statements of Values outline the aspirations that can drive companies toward continuous improvement.

Functions or Purposes Served by Codes:

- Discipline. This function gets all the attention. Most codes are set forth to establish clearly and forcefully an organization's standards, especially its minimum standards of acceptable conduct. Having established the limits, organizations can then punish those who exceed them.
- Educate. This can range from disseminating standards to enlightening members. Company A's employees learned that anything over \$100 was a bribe and should not be accepted. But engineers learn that their fundamental responsibility is to hold paramount public safety, health, and welfare. Codes certainly teach minimum standards of conduct, but they can help a community to articulate and understand their highest shared values and aspirations.
- Inspire. Codes can set forth ideals in a way that inspires a community's members to strive for excellence. They can be written to set forth the aspirations and value commitments that express a community's ideals. They can point a community toward moral excellence.
- Stimulate Dialogue. Engineering professional codes of ethics have changed greatly over the last 150 years. This has been brought about by a vigorous internal debate stimulated by these very codes. Members debate controversial claims and work to refine more basic statements. Johnson and Johnson credits their credo for their proactive and successful response to the Tylenol crisis. Regularly, employees "challenge the credo" by bringing up difficult cases and testing how effectively the credo guides decision-making and problem-solving. The CIAPR's Disciplinary Tribunal cases have served as a focus for discussions on how to interpret key provisions of the organization's code of ethics. The NSPE Board of Ethical Review decisions have also provided an excellent forum for clarifying ethical concepts (public safety, conflict of interest) in the context of cases brought to the board by NSPE members. The BER discusses cases in terms of relevant provisions of the NSPE code. Over the years, the

NSPE BER has established a firm foundation for the resolution of difficult ethical cases by developing analogies with cases it has already discussed and clarified.

- Empower and Protect. Codes empower and protect those who are committed to doing the right thing. If an employer orders an employee to do something that violates that employee's ethical or professional standards, the code provides a basis for saying, "No!". Engineers have refused to carry out directives that place in jeopardy the health and safety of the public based on statements like canon 1 of the CIAPR code. (The NSPE code has similar provisions.) Because codes establish and disseminate moral standards, they can provide the structure to convert personal opinion into reasoned professional judgment. To reiterate, they provide support to those who would do the right thing, even under when there is considerable pressure to do the opposite.
- Codes capture or express a community's identity. They provide the occasion to identify, foster commitment, and disseminate the values with which an organization wants to be identified publicly. These values enter into an organization's core beliefs and commitments forming an identify-conferring system. By studying the values embedded in a company's code of ethics, observing the values actually displayed in the company's conduct, and looking for inconsistencies, the observer can gain insight into the core commitments of that company. Codes express values that, in turn, reveal a company's core commitments, or (in the case of a hypocritical organization) those values that have fallen to the wayside as the company has turned to other value pursuits.

Difficulties with Codes:

- The following objections lead philosophers to argue that presenting codes of ethics in ethics classes undermines several key moral attitudes and practices.
- Codes can undermine moral autonomy by habituating us to act from motives like deference to external authority and fear of punishment. We get out of the habit of making decisions for ourselves and fall into the habit of deferring to outside authority.
- Codes often fail to guide us through complex situations. Inevitably, gaps arise between general rules and the specific situations to which they are applied; concrete situations often present new and unexpected challenges that rules, because of their generality, cannot anticipate. Arguing that codes should provide action recipes for all situations neglects the fact that effective moral action requires more than just blind obedience to rules.
- Codes of ethics can encourage a legalistic attitude that turns us away from the pursuit of moral excellence and toward just getting by or staying out of trouble. For example, compliance codes habituate us to striving only to maintain minimum standards of conduct. They fail to motivate and direct action toward aspirations. Relying exclusively on compliance codes conveys the idea that morality is nothing but staying above the moral minimum.

This module is designed to steer you through these complex issues by having you draft a Statement of Values for students at your university. As you work through your Statement of Values, you will learn that codes have strengths and weaknesses, serve different functions, and embody values. To get you started in this process, you will study a defective code, the Pirate Credo. A quick glance is all that is needed to see that codes are "all too human" and need to be approached critically. In a second activity, you will identify the values embedded in professional, corporate, and academic codes. Working with these values, you will develop a list upon which your group will build its own Statement of Values in a third activity. Finally, you will construct value profiles that include a general description, sample provisions, value-based challenges, and value principles. These will all contribute to motivating those in your community to commit to and work in concert to realize these values.

How an academic community developed a Statement of Values

A False Start: The faculty of the Arts and Sciences College of University X decided to form a committee to write a code of ethics. This committee met several times during the course of an academic semester to prepare the first draft. When they finished, they circulated copies throughout the college. Then they held a series of public hearings where interested members of the College could criticize the code draft. These were lightly attended and those attending had only a few suggestions for minor changes. However, when the code was placed before the faculty for approval, considerable opposition emerged. For example, a provision discouraging faculty from gossiping was characterized by opponents as an attempt by a hostile College administration, working through the committee, to eliminate faculty free speech. Several opponents expressed opposition to the very idea of a code of ethics. "Does the administration think that our faculty is so corrupt," they asked, "that the only hope for improvement is to impose upon them a set of rules to be mindlessly followed and ruthlessly enforced?" At the end of this debate, the faculty overwhelmingly rejected the code.

Reflections on "A False Start"

- Should codes of ethics be democratically developed from the "bottom-up" or should they be authoritatively imposed from the "top-down?" Or does this depend on certain characteristics of the community? Maybe corporate managers should have lawyers

draft their codes to meet the Federal Sentencing Guidelines; these completed codes should then be implemented throughout the company at all levels. Maybe academic communities should democratically determine their own codes, and if they are unable to do so, then so much the worse for the "very idea" of a code of ethics.

- The Ethics of Team Work module presents three ways that lead groups to go off the tracks: Group Polarization, Groupthink, and "Going to Abilene." Do you think that any of these would explain false starts in developing a code of ethics? How can these group pitfalls be overcome?
- Groups are often polarized around different and conflicting ideologies or paradigms. Thomas Kuhn discusses paradigms in the context of scientific debates. When these debates are fueled by conflicting and incompatible paradigms, they can turn acrimonious and prove extraordinarily difficult to resolve. For Kuhn, paradigms articulate and encapsulate different world views; the meanings and experiences shared by one group operating under one paradigm are often not shared by those operating under different paradigms. Members of the Arts and Sciences faculty of University X may have disagreed about the provisions proscribing gossiping because they were operating under different conceptual systems brought about by incommensurable paradigms. If faculty members assumed different meanings for 'gossiping', 'code', and 'discipline', then this would fuel the polarization of non-agreement like that which occurred at University X.
- Cass Sunstein proposes that communities work around ideological or paradigm-driven disputes by developing, in special circumstances, "incompletely theorized agreements." These agreements are brought about by bracketing commitments to a given ideology or paradigm. This allows one side to work on understanding the other instead of marshaling arguments to defend the set of views entailed by its paradigm. So Sunstein's recommendation to the College of Arts and Sciences of University X would be to suspend commitment to defending the core beliefs of the conflicting ideologies and try to hold discussions at a more concrete, incompletely theorized level. This makes finding common ground easier. When shared understandings are forged, then they can serve as bridges to more complex, more completely theorized positions.
- Looking at this problem from a completely different angle, do codes of ethics require a background of trust? If so, how can trust be built up from within highly diverse and highly polarized communities or groups?
- Finally, can codes of ethics be abused by more ruthless groups and individuals? For example, as those in the College of Arts and Sciences claimed, can codes of ethics be used by those in positions of power to strengthen that power and extend control over others?

A Success Story:

- Three years later at the same university, another faculty group set out to construct a code of ethics in order to respond to accreditation requirements. They began with the idea of constructing a stakeholder code.
- First, they identified the stakeholders of the college's activities, that is, groups or individuals who had a vital interest in that community's actions, decisions, and policies.
- Second, they identified the goods held by each of these stakeholders which could be vitally impacted by the actions of the college. For example, education represented the key good held by students that could be vitally impacted by the activities and decisions of the College.
- Working from each stakeholder relation and the good that characterized that relation, members of the college began crafting code provisions. Some set forth faculty duties such as keeping regular office hours, grading fairly, and keeping up to date in teaching and research. Others emphasized student duties such as working responsibly and effectively in work teams, adhering to standards of academic honesty, and attending classes regularly.

Because stakeholder codes embody a community's values, the individuals in charge of drafting the code decided that a more direct approach would be to identify the embodied values and refine them into a Statement of Values. This formal statement could later be developed in different directions including a more detailed compliance code.

Turning their efforts toward preparing a Statement of Value Process, the Business Administration community went through the following steps:

1. They discussed a flawed document, the Pirate Credo. This brought about three positive results: participants came to see how codes embody values, that codes serve different functions, and that codes clarify relations between the insiders and outsiders of a community.
2. Participants examined "bona fide" codes of ethics such as academic codes, codes of honor, corporate codes, and professional codes. Since codes embody values, they developed lists of the values these codes embodied.
3. The sample provisions crafted in the earlier stakeholder code effort were presented so that participants could identify the values these embodied. Previous efforts in developing a stakeholder code could be benchmarked against the codes studied in the

previous step. Convergences and divergences were noted and used to further characterize the college's community in terms of its similarities and differences with other communities.

4. In this step, faculty members were asked to reduce the values list to a manageable number of five to seven. This led to the most contentious part of the process. Participants disagreed on the conception of value, the meaning of particular values like justice, and on whether rights could be treated as values.
5. To resolve this disagreement, discussion leaders proposed using ballots to allow participants to vote on values. This process was more than a simple up or down vote. Participants also ranked the values under consideration.
6. After the top five values were identified, efforts were made, in describing each of the remaining values, to find places to include at least components of the values left out. For example, while confidentiality was not included in the final value list, it was reintegrated as a component of the more general value of respect. Thus, the final values list could be made more comprehensive and more acceptable to the faculty community by reintegrating some values as parts of other, more general values. Another way of picking up values left behind in the voting process was to combine values that shared significant content. Values that did not make it into the final list were still noted with the provision that they could be integrated into subsequent drafts of the Statement of Values.
7. A committee was formed to take each value through a value template. After describing the value, they formulated a principle summarizing the ethical obligations it entailed, crafted sample provisions applying the value, and posed different challenges the value presented to help guide a process of continuous improvement.
8. The committee presented its results to the faculty who approved this first draft Statement of Values
9. The faculty then developed a schedule whereby the Statement of Values would be revisited, expanded, revised, and improved.

Textbox 1: Responding to the Federal Sentencing Guidelines

Recent efforts to develop ethics codes in the academic context for both students and faculty may, in part, stem from the success of ethics compliance programs developed in business and industry in response to the Federal Sentencing Guidelines. Organizational codes of ethics have been integrated alongside other compliance structure and activities to prevent criminal behavior, to detect criminal behavior, and to ensure prompt and effective organizational response once such behavior has been detected.

The following section contains short excerpts from the Federal Sentencing Guidelines. For more details consult the materials referenced in note 5 below.

- "The hallmark of an effective program to prevent and detect violations of law is that the organization exercised due diligence in seeking to prevent and detect criminal conduct by its employees and other agents. Due diligence requires at a minimum that the organization must have taken the following types of steps:
- The organization must have established compliance standards and procedures to be followed by its employees and other agents that are reasonably capable of reducing the prospect of criminal conduct.
- Specific individual(s) within high-level personnel of the organization must have been assigned overall responsibility to oversee compliance with such standards and procedures.
- The organization must have used due care not to delegate substantial discretionary authority to individuals whom the organization knew, or should have known through the exercise of due diligence, had a propensity to engage in illegal activities.
- The organization must have taken steps to communicate effectively its standards and procedures to all employees and other agents, e.g., by requiring participation in training programs or by disseminating publications that explain in a practical manner what is required.
- The organization must have taken reasonable steps to achieve compliance with its standards, e.g., by utilizing monitoring and auditing systems reasonably designed to detect criminal conduct by its employees and other agents and by having in place and publicizing a reporting system whereby employees and other agents could report criminal conduct by others within the organization without fear of retribution.

Recommendations by the Federal Sentencing Guidelines for an Effective Compliance Program

- Appointing individuals to serve as ethics or compliance officers
- Developing corporate credos and codes of ethics that effectively communicate an organization's ethical standards and expectations to employees.
- Designing ethics training programs for all employees
- Designing and implementing monitoring and auditing systems
- Designing and implementing an effective system of punishments and sanctions. These must be accompanied by investigative procedures that respect employee due process rights.

Textbox 2: Compliance Oriented Codes and Programs Versus Values Oriented Codes and Programs

Compliance Strategy

1. The initial and still probably the most prevalent method for responding to the Federal Sentencing Guidelines is the compliance strategy. This strategy is based on three interrelated components:
2. Rules: Compliance strategies are centered around strict codes of ethics composed of rules that set forth minimum thresholds of acceptable behavior. The use of rules to structure employee action does run into problems due to the gap between rule and application, the appearance of novel situations, and the impression that it gives to employees that obedience is based on conformity to authority.
3. Monitoring: The second component consists of monitoring activities designed to ensure that employees are conforming to rules and to identify instances of non-compliance. Monitoring is certainly effective but it requires that the organization expend time, money, and energy. Monitoring also places stress upon employees in that they are aware of constantly being watched. Those under observation tend either to rebel or to automatically adopt behaviors they believe those doing the monitoring want. This considerably dampens creativity, legitimate criticism, and innovation.
4. Disciplining Misconduct: The last key component to a compliance strategy is punishment. Punishment can be effective especially when establishing and enforcing conduct that remains above the criminal level. But reliance on punishment for control tends to impose solidarity on an organization rather than elicit it. Employees conform because they fear sanction. Organizations based on this fear are never really free to pursue excellence.

Values Orientation

1. To facilitate comparison, three correlative but different elements to Values-Based or aspirational approaches will be identified.
2. Development of Shared Values: Using a process similar to the one described above, a company develops a Statement of Shared Values. These provide guidelines that replace the hard and fast rules of a compliance code. Statements in values-oriented codes play a different logical function than statements in compliance codes. "Principles of Professional/Organizational Conduct" in compliance codes specify circumstances of compliance: time, agent, place, purpose, manner, etc. These provide sufficient content to set forth principles of professional conduct as rules that can be violated. This, in turn, allows them to be backed by punishment for violation. "Ideals of the Profession" (or organization) set forth a community's shared aspirations. These are pitched at a level well above and beyond the minimum. Communities can and should define themselves as much by their aspirations as by their threshold standards.
3. Support for Employees: Since Statements of Values set forth excellences or aspirations, the role of the organization changes from monitoring and then punishing misbehavior to finding ways of opening avenues for employees to realize key values in their day to day activity. Excellence is not something to be reached overnight. It requires rethinking basic motivations, attitudes, beliefs, and goals. Companies need to identify obstacles to achieving ideals and then develop support structures to help those who seek to realize ideals. Values-based approaches change from punishing conduct that falls below the minimum to providing collective support to those who strive for the excellent.
4. Locking in on Continual Improvement: The philosopher, John Dewey, characterizes moral responsibility as the drive to better ourselves. The particular twist in Dewey's approach is to find ways of folding what has been learned from the past into meeting new challenges that arise in the future. This involves changing habits and, ultimately, changing character. Continual improvement is the ultimate goal of corporations oriented toward excellence. The values these "moral ecologies" identify structure and channel this endeavor. What is needed at this stage is to develop concrete programs and strategies for identifying obstacles to excellence, removing them, and remaining on track for excellence.
5. To summarize, some companies identify a compliance strategy where they set forth rules that establish minimum levels of acceptable conduct, monitor compliance, and punish non-compliance. Others, value-oriented or aspiration-oriented companies, identify core values or aspirations (by reflecting on community values and finding them embedded in extant codes of ethics), develop programs and structures to support those who strive for these values, and work to lock in a program of continual improvement or betterment.
6. Something to think about. Compliance approaches work best in what of company, organization or moral ecology. (Think about this in terms of the central or core commitments such as those in finance-, customer-, and quality-driven companies.) Values-based approaches work best in what kind of company, organization or moral ecology? How does one transition from compliance to values-based approaches? How does one integrate the two?

Exercise 1: Evaluating the Pirate Credo

Read the Pirate Credo. Then answer the following questions individually...

- What is good about the Pirate Credo?
- What is bad about the Pirate Credo?
- What is the purpose served by the Pirate Credo? For the Pirate Community? For non-members?

Exercise 2: Statement of Value Challenge

- Is the SOV comprehensive? (For example, can you think of a case that it does not adequately cover? Are there values that it leaves out in the sense that they cannot be subsumed by one or more SOVs?
- Are the value descriptions clear? For example, if you have confused values on the multiple-choice or matching sections of your exams, is this because the descriptions need reworking and clarifying?
- Last year, an ADEM stakeholder group suggested that values should be paired with one another. For example, because integrity is a meta-value it should be paired with other values like trust. Or should trust and responsibility be paired with one another? In this case, should the SOV be expanded to explore the relations between different values?
- When ADEM stakeholders identified their values in 2005, they prioritized and ranked them. Justice was ranked highest followed by responsibility, respect, trust, and integrity. Should this hierarchy or ranking be changed? For example, last year stakeholders suggested that integrity should be ranked first because it is a meta-value that talks about the relationship between other values.

Exercise 3: Developing Corporate Codes of Ethics

1. Ethics Bowl Corporations. You have been assigned corporations corresponding to two of the six ethics bowl cases. For your presenting corporation, you will be developing a partial code of ethics. For the commenting corporation, you need to familiarize yourself with the moral ecology of the corporation, its needs, and be ready to comment on the code offered by another group.
2. What kind of moral ecology is predominant in your corporation? Is it financial-, customer-, or quality-driven. Look at how the type of moral ecology structures other organizational activities: allocation of praise and blame, exchange of information, treatment of dissenting opinions, and central of moral concerns. All of these issues need to be addressed directly or indirectly in your code.
3. What is the ethical challenge that is highlighted in the ethics bowl scenario based on your case? For this information, see the "Ethics Bowl in the Environment of the Organization" module
4. What functions are you addressing in your code outline? Looking above, these would include educate, inspire, create dialogue, discipline, empower, secure and express identity.
5. Develop within the time available a sketch of a code. This could be a section of a compliance code, a corporate credo, or a statement of values. In choosing your form, think carefully about the function(s) of your code. Have something that you can present, informally, for around 3 to 5 minutes.

Exercise 4: Evaluating Bona Fide Codes of Ethics

Form small work teams of four to five individuals. Carry out the following four steps and report your results to the rest of the group...

1. Review a few sample codes per team
2. List the values you identify in the codes. Express each value as a word or in as few words as possible
3. Identify any recurring values
4. Record and post the list of values

Exercise 5: Do a Statement of Values for Students at Your University

In this exercise, work with your group to develop a refined list of five to seven values. You can refine your list by integrating or synthesizing values, grouping specific values under more general ones, and integrating values into others as parts. Do your best to make your list comprehensive and representative.

1. Brainstorm: list the values for your group. Keep in mind that values are multi-dimensional. For example, in the academic context, the values will break down into dimensions corresponding to stakeholder: faculty, students, administration, and other academic stakeholders
2. Refine: reduce your list to a manageable size (5-7). Do this by rewording, synthesizing, combining, and eliminating
3. Post: share your list with the entire group
4. Revise: make any last-minute changes
5. Combine: a moderator will organize the lists into a ballot
6. Vote: Each person ranks the top five values

Exercise 6--Conveying Our Values: Crafting a Values-Based Code

Each value in your Statement of Values needs to be accompanied by a Value Profile. Give a description of the value in everyday, non-technical terms. Think concretely. For example, those who exemplify your value behave in a certain fashion, exhibit certain commitments, pursue certain projects, and show certain attitudes and emotions. Try to think of general guidelines to keep in mind when working to realize your value. Finally, values challenge us because they portray our aspirations. Think of specific ways values challenge us. For example, students may set for themselves the challenge of working responsibly in teams. They can further spell out what kinds of actions and attitudes this might require. Faculty members might set for themselves the challenge of grading more fairly. This could require actions like developing rubrics and refining exams to make them clearer. The purpose of this fourth exercise is to provide content to your statement of values and begin its implementation in your community. The following steps enumerated below will help.

1. Value: Responsibility
2. Description: A responsible person is a person who...
3. Principle: The faculty, students, and staff of the College of Business Administration will...
4. Commitments: Keep office hours, do your fair share in work teams, divide work into clear and coordinated tasks, etc.

Exercise 7: Creating Awareness of the UPRM College of Business Administration Statement of Values

This exercise provides you an opportunity to study and discuss the UPRM College of Business Administration Statement of Values (available via the PREREQUISITE LINKS). Your task consists of the following tasks:

- Read the entire UPRM CBA Statement of Values (individually)
- Discuss the particular section/value assigned to your group and briefly describe what commitments or challenges does this value present for the students, faculty and/or staff of the CBA
- List the most important commitments or challenges as precise and concise principles

Exercise 8: Assessing the UPRM College of Business Administration Statement of Values

This exercise offers four scenarios in academic integrity. Your job is to discuss each scenario in terms of the values listed in the UPRM College of Business Administration Statement of Values (available via the PREREQUISITE LINKS).

Marta Acevedo, a business administration student, has a report due tomorrow. She has been overwhelmed for the last few weeks with assignments from other classes and doesn't really have time to complete this exercise. She discovers that her roommate took this same class the previous semester and has a complete report on disk. She considers using her roommate's report. Should she? What would you do if you were her?

- Is Marta threatening any of the values listed in the ADEM SOV? Which ones?
- What can be done prevent this kind of problem from arising in the first place? Should Marta have planned her course load better when registering? Can teachers coordinate to prevent overloading students with the same deadlines? Whose fault is this? The students? The teachers? The system?
- Can this problem be posed as a conflict between ADEM values and other values held by students and teachers? If so, what are values that are in conflict? How can these conflicts be addressed?
- Do you think the ADEM SOV adequately addresses this problem? If not, how can it be improved?

You are head of your department. A recent study has revealed that plagiarism, which is a university-wide problem, is especially bad in your department. Imagine your relief when a member of your faculty brings you his latest software project, a super-effective and comprehensive anti-plagiarism software program. This program does everything. It detects subtle changes in

style in student papers. Its new search engine quickly connects to existing online paper data bases, greatly expanding the ability of a professor to detect the sources from which their students have copied. Furthermore, it allows professors to upload papers and projects from past semesters and provides fast and flexible indexing to help them identify recycled student work. Professors can zero in on students using recycled papers, and the former students who have become their suppliers. Following the recent lead of Ohio State University, you can now revoke the degrees of past students who participate in this version of academic dishonesty. In short, this new and exciting software package allows you to monitor the work of present and past students to a degree thought impossible even in the recent past. "Plagiarism," your colleague tells you, "will now become a thing of the past."

- Does this anti-plagiarism program threaten any of the values in the ADEM SOV? If so, which values?
- Is the department chairperson treating students disrespectfully by adopting and implementing the anti-plagiarism software? Can faculty treat students disrespectfully as "justifiable" retaliation for student cheating and plagiarizing? Do two wrongs make a right?
- What is the cause of plagiarism? Do students do it out of ignorance of standards and practices of documentation and acknowledgment? Do they do it because they procrastinate until they do not have time to do the assignment properly? Do students resort to plagiarism because they have too many conflicting obligations such as family, job, large course loads, etc.?

You teach an advanced course in Engineering Economics that has both graduate and undergraduate students. At the end of the semester the students turn in a group project that comprises 40% of their grade. One of the groups complains to you that only 4 out of the 5 members have done any work. The fifth student, the one who allegedly has done no work, is an undergraduate. The others are graduate students. You talk with the undergraduate who claimed that she tried to involve herself in the group activities but was excluded because she was an undergraduate. What should you do?

- ADEM faculty have identified students not working together effectively in groups as a major concern. Do you find this a problem? What do you think are the causes of students not participating effectively in work groups?
- Assume that the teacher in this case is committed to implementing the ADEM SOV. Which values are at play in this case? Design an action for the teacher that realizes these values?
- Assume you are a member of this student work group. What can groups do to ensure that every member is able to participate fully? What do group members do to exclude individuals from participating?

You are studying frantically for your exam in a computer engineering course. It will be very difficult. But your roommate, who is also taking the course and has the exam tomorrow, seems unconcerned. When you ask why, he tells you that he has a copy of the exam. Apparently, a group of students in the class found out how to hack into the professor's computer and download the exam. (They installed a Trojan horse called Sub-Seven into the professor's computer which allows unauthorized access; then they searched through the professor's files, found the exam and downloaded it.) Your roommate has the exam in his hand and asks you if you would like to look at it. What should you do?

- A group of students in a computer ethics class created a survey that asked students if they would avail themselves of exams obtained through means such as that described in the scenario above. Sixty percent of the respondents said that they would. Compare this to the value commitments expressed in the ADEM SOV? Is there a gap between aspiration and behavior? What can be done to reduce this gap?
- Suppose you took the exam. Would this have any long term effects on your character? Would acting dishonestly this time make it easier to do so in the future?
- Suppose you wish to uphold standards of academic integrity in this case and not take the exam. Should you turn your roommate in to the teacher? Would keeping this exam theft a secret undermine any of the UPRM ADEM values? If so, which ones?

You have now discussed some or all of the above cases in terms of the ADEM Statement of Values. What do you think are the strengths of this document? What are its weaknesses? Do you recommend any changes? What are these?

Sources for Cases:

- Case 1 has been developed by William Frey, Chuck Huff, and José Cruz for their book, Good Computing: A Virtue Approach to Computer Ethics. This book is currently in draft stage and is under contract with Jones and Bartlett Publishing Company.
- Cases 2 and 3 were developed by UPRM faculty teams from the College of Engineering during workshops held for the ABET 2001 Steering Committee and the Department of Industrial Engineering. These workshops took place April 6, 2001 and May

14, 2001.

- Case 4 has been modified from “The Plagiarism Detector” written by Moshe Kam. It can be found at the beginning of the ethics chapter in Practical Engineering Design, edited by Maja Bystrom and Bruce Eisenstein. Moshe Kam. “The Plagiarism Detector”, in Practical Engineering Design, edited by Maja Bystrom and Bruce Eisenstein. Boca Raton, FLA: CFC Press, 2005: 27-28.

Assessment Tools

Ethics Across the Curriculum Matrix

[EACMatrix Template ADEM Feb 17.doc](#)

This table will help you document your class discussion of the ADEM Statement of Values.

Muddy Point Exercise

[MP.doc](#)

Clicking on this media file will open a word format for the Muddiest Point Exercise. Students are invited to discuss the strongest and weakest facets of the ADEM Statement of Values.

Module Assessment Form

[MAP.doc](#)

Clicking on this media file will open a general module assessment form taken from Michael Davis' IIT EAC workshop. This form will help you assess the SOV activity as well as other EAC modules.

This presentation is composed of slides previously given before the AACSB, ADEM faculty at UPRM, and material published by the authors in Technology and Society Magazine. (See bibliography below)

[SOV_Development.pptx](#)

Bibliography

1. Lynn Sharp Paine (1994) "Managing for Organizational Integrity," in Harvard business review, March-April: 106-117
2. Gary R. Weaver and Linda Klebe Trevino (1999) "Compliance and Values Oriented Ethics Programs: Influences on Employees' Attitudes and Behavior," in Business Ethics Quarterly 9(2): 315-335
3. Stuart C. Gilman (2003) "Government Ethics: If Only Angels Were to Govern," in Professional Ethics, edited by Neil R. Luecke in Phi Kappa Phi Forum, Spring 2003: 29-33.
4. Stephen H. Unger (1994) Controlling Technology: Ethics and the Responsible Engineer, 2nd Edition. New York: John Wiley and Sons: 106-135.
5. "Federal Sentencing Guidelines--Sentencing of Organizations," in Ethical Theory and Business, 5th Edition, edited by Tom L Beauchamp and Norman E. Bowie, New Jersey: Prentice Hall: 182-187. This article was reprinted with permission from The United States Law Week, Vol. 50 pp. 4226-29 (March 26, 1991) (Bureau of National Affairs, Inc.

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4.3: Gray Matters for the Hughes Aircraft Case

Introduction

I. Introduction: The Hughes Aircraft Case involves a group of employees in charge of testing chips for weapons systems. Because of the lengthy testing procedure required by the U.S. Defense Department, Hughes soon fell behind schedule in delivering chips to customers. To get chips out faster, some Hughes middle-level managers began to put pressure on employees to pass chips that had failed tests or to pass them without testing. The scenarios below consist of narratives that stop at the point of decision. Your job is to complete the narrative by making a decision. Alternatives are provided to get the process started, but you may find it necessary to design your own solution. Ethics and feasibility tests help you to evaluate these alternatives and even design new ones more to your liking. This format superficially resembles the Gray Matters exercise used at Boeing Corporation. (More information on the history of Gray Matters can be found by consulting Carolyn Whitbeck, *Ethics in Engineering Practice*, 1998, 176-182.) This version differs in being more open-ended and more oriented toward giving you the opportunity to practice using ethical theory (which has been encapsulated into ethics tests).

Directions

- Read the following scenarios and the accompanying solutions
- Evaluate the alternatives in terms of the tests described below.
- Choose the one you think best or design your own solution if you believe you can do better.
- Summarize your results by filling in the solution evaluation matrix that appears on the page following the scenario. Notice that the first column repeats the solution alternatives.
- Be prepared to present your matrix to the class. You will also provide the other groups in the class with a copy of your matrix for their ethics portfolios

Bibliographical Note: The six scenarios below were developed by Chuck Huff as Participant Perspectives. They were first published online through the Computing Cases website. (Computing Cases was developed through two National Science Foundation grants, DUE-9972280 and DUE-9980768.) A revised version of these participant perspectives has been published in the anthology, *Whistleblowing: Perspectives and Experiences*, edited by Reena Raj and published in 2008 by the Icfai University Press, Nagarjuna Hills, Punjagutta, Hyderabad, India. These materials can be found on pages 75-80.

Scenario One: Responding to Organizational Pressure: Frank Saia has worked at Hughes Aircraft for a long time. Now he is faced with the most difficult decisions of his career. He has been having problems in the environmental testing phase of his microchip manufacturing plant; the detailed nature of these tests has caused Hughes to be consistently late in delivering the chips to customers. Because of the time pressure to deliver chips, Saia has been working to make the production of chips more efficient without losing the quality of the product. Chips are manufactured and then tested, and this provides two places where the process can bottle up. Even though you might have a perfectly fine chip on the floor of the plant, it cannot be shipped without testing. And, since there are several thousand other chips waiting to be tested, it can sit in line for a long time. Saia has devised a method that allows testers to put the important chips, the “hot parts,” ahead of the others without disrupting the flow and without losing the chips in the shuffle. He has also added a “gross leak” test that quickly tells if a chip in a sealed container is actually sealed or not. Adding this test early in the testing sequence allows environmental testing to avoid wasting time by quickly eliminating chips that would fail a more fine-grained leak test later in the sequence. Because environmental testing is still falling behind, Saia’s supervisors and Hughes customers are getting angry and have begun to apply pressure. Karl Reismueller, the director of the Division of Microelectronics at Hughes, has given Saia’s telephone number to several customers, whose own production lines were shut down awaiting the parts that Saia has had trouble delivering. His customers are now calling him directly to say “we’re dying out here” for need of parts. Frank Saia has discovered that an employee under his supervision, Donald LaRue, has been skipping tests on the computer chips. Since LaRue began this practice, they have certainly been more on time in their shipments. Besides, both LaRue and Saia know that many of the “hot” parts are actually for systems in the testing phase, rather than for ones that will be put into active use. So testing the chips for long-term durability that go into these systems seems unnecessary. Still, LaRue was caught by Quality Control skipping a test, and now Saia needs to make a decision. Upper management has provided no guidance; they simply told him to “handle it” and to keep the parts on time. He can’t let LaRue continue skipping tests, or at least he shouldn’t let this skipping go unsupervised. LaRue is a good employee, but he doesn’t have the science background to know which tests would do the least damage if they were skipped. He could work with LaRue and help him figure out the best tests to skip so the least harm is done. But getting directly involved in skipping the tests would mean violating company policy and federal law.

Alternatives:

1. Do nothing. LaRue has started skipping tests on his own initiative. If any problems arise, then LaRue will have to take responsibility, not Saia, because LaRue was acting independently of and even against Saia's orders.
2. Call LaRue in and tell him to stop skipping tests immediately. Then call the customers and explain that the parts cannot be shipped until the tests are carried out.
3. Consult with LaRue and identify nonessential chips or chips that will not be used in systems critical to safety. Skipping tests on these chips will do the least damage.
4. Your solution...

Scenario Two: Responding to Wrongdoing Margaret Goodearl works in a supervisory position in the environmental testing group at Hughes Aircraft. Her supervisor, Donald LaRue, is also the current supervisor for environmental testing. The group that LaRue and Goodearl together oversee test the chips that Hughes makes in order to determine that they would survive under the drastic environmental conditions they will likely face. Rigorous testing of the chips is the ideal, but some chips (the hot chips) get in line ahead of others. Goodearl has found out that over the last several months, many of these tests are being skipped. The reason: Hughes has fallen behind in the production schedule and Hughes upper management and Hughes customers have been applying pressure to get chip production and testing back on schedule. Moreover, LaRue and others feel that skipping certain tests doesn't matter, since many of these chips are being used in systems that are in the testing phase, rather than ones that will be put into active use. A few months after Margaret Goodearl started her new position, she was presented with a difficult problem. One of the "girls" (the women and men in Environmental Testing at Hughes), Lisa Lightner, came to her desk crying. She was in tears and trembling because Donald LaRue had forcefully insisted that she pass a chip that she was sure had failed the test she was running. Lightner ran the hermeticity test on the chips. The chips are enclosed in a metal container, and one of the questions is whether the seal to that container leaks. From her test, she is sure that the chip is a "leaker"—the seal is not airtight so that water and corrosion will seep in over time and damage the chip. She has come to Goodearl for advice. Should she do what LaRue wants and pass a chip she knows is a leaker?

Alternatives:

1. Goodearl should advise Lightner to go along with LaRue. He is her supervisor. If he orders to pass the chip, then she should do so.
2. Goodearl should go to Human Resources with Lightner and file a harassment complaint against LaRue. Skipping tests is clearly illegal and ordering an employee to commit an illegal act is harassment.
3. Goodearl and Lightner should blow the whistle. They should go to the U.S. defense department and inform them of the fact that Hughes Aircraft is delivering chips that have either failed tests or have not been tested.
4. Your solution...

Scenario 3: Goodearl, Ibarra, and the AMRAAM Incident: Now that Goodearl had few sympathizers among upper management, she increasingly turned to Ruth Ibarra in Quality assurance for support in her concerns about test skipping and the falsification of paperwork. One day, Goodearl noticed that some AMRAAM chips with leak stickers were left on her project desk in the environmental testing area. The leak stickers meant that the seal on the chips' supposedly airtight enclosure had failed a test to see if they leaked. AMRAAM meant that the chips were destined to be a part of an Advanced Medium Range Air-to-Air Missile. Goodearl knew that these parts could not be retested and needed to be simply thrown away. So why was someone keeping them? She also knew that these were officially "hot parts" and that the company was behind schedule in shipping these parts. After consulting with Ruth Ibarra, the two of them decided to do some sleuthing. They took the chips and their lot travelers to a photocopy machine and made copies of the travelers with "failed" noted on the leak test. They then replaced the chips and their travelers on the desk. Later that day, as Don LaRue passed the desk, Goodearl asked Don LaRue if he knew anything about the chips. "None of your business," he replied. The chips disappeared, and later the travelers showed up in company files with the "failed" altered to "passed." So, Goodearl and Ibarra had clear evidence (in their photocopy of the "failed" on the traveler) that someone was passing off failed chips to their customers. And these were important chips, part of the guidance system of an air-to-air missile.

Alternatives: Since they have clear evidence, Goodearl and Ibarra should blow the whistle. Evaluate each of the following ways in which they could blow the whistle

1. Blow the whistle to Hughes' Board of Directors. In this way they can stop the test skipping but will also be able to keep the whole affair "in house."
2. Blow the whistle to the local news media. In this way they will shame Hughes into compliance with the testing requirements.

3. Take the evidence to the U.S. Department of Defense, since they are the client and are being negatively impacted by Hughes' illegal actions.
4. Some other mode of blowing the whistle....

Solution Evaluation Matrix

Alternatives/Tests	Reversibility/Rights Test	Harm/Benefits Test	Virtue/Value Test (Also Publicity)	Global Feasibility Test (Implementation Obstacles)
Alternative One (Worst Alternative)	Evaluate Alt 1 using reversibility/rights test			
Alternative Two (Best among those given)		Weigh harms against benefits for alt 2		
Alternative Three			What values/disvalues are realized in alt 3?	
Your Solution				What obstacles could hinder implementation of solution?

Ethics Tests: Set Up and Pitfalls

III. Solution Evaluation Tests

- REVERSIBILITY: Would I think this is a good choice if I were among those affected by it?
- PUBLICITY: Would I want to be publicly associated with this action through, say, its publication in the newspaper?
- HARM/BENEFICENCE: Does this action do less harm than any of the available alternatives?
- FEASIBILITY: Can this solution be implemented given time, technical, economic, legal, and political constraints?

Harm Test Set-Up

- Identify the agent (=the person who will perform the action). Describe the action (=what the agent is about to do).
- Identify the stakeholders (individuals who have a vital interest at risk) and their stakes.
- Identify, sort out, and weight the expected results or consequences.

Harm Test Pitfalls

- Paralysis of Action--considering too many consequences
- Incomplete analysis--considering too few results
- Failure to weigh harms against benefits
- Failure to compare different alternatives
- Justice failures--ignoring the fairness of the distribution of harms and benefits

Reversibility Test Set-Up

- Identify the agent
- Describe the action
- Identify the stakeholders and their stakes
- Use the stakeholder analysis to select the relations to be reversed.
- Reverse roles between the agent (you) and each stakeholder: put them in your place (as the agent) and yourself in their place (as the target of the action)
- If you were in their place, would you still find the action acceptable?

Reversibility Pitfalls

- Leaving out a key stakeholder relation
- Failing to recognize and address conflicts between stakeholders and their conflicting stakes
- Confusing treating others with respect with capitulating to their demands (Reversing with Hitler)

- Failing to reach closure, i.e., an overall global reversal assessment that takes into account all the stakeholders the agent has reversed with

Public Identification Set-Up

- Set up the analysis by identifying the agent, describing the action under consideration, and listing the key values or virtues at play in the situation
- Associate the action with the agent
- Identify what the action says about the agent as a person. Does it reveal him or her as someone associated with a virtue/value or a vice?

Public Identification Pitfalls

1. Action is not associated with the agent. The most common pitfall is failure to associate the agent and the action. The action may have bad consequences and it may treat individuals with disrespect but these points are not as important in the context of this test as what they imply about the agent as a person who deliberately performs such an action
2. Failure to specify the moral quality, virtue, or value of the action that is imputed to the agent in the test. To say, for example, that willfully harming the public is bad fails to zero in on precisely what moral quality this attributes to the agent. Does it render him or her unjust, irresponsible, corrupt, dishonest, or unreasonable?

Gray Matters in Hughes Exercises

[GM Hughes V2.doc](#)

These exercises present three decision points from Hughes, solution alternatives, summaries of ethics and feasibility tests, and a solution evaluation matrix. Carry out the exercise by filling in the solution evaluation matrix.

This timeline is taken from the Computing Cases website developed and maintained by Dr. Charles Huff at St. Olaf College. Computing Cases is funded by the National Science Foundation, NSF DUE-9972280 and DUE 9980768.

Time Line

1979	Ruth Ibarra begins working for Hughes Aircraft company's Microelectronic Circuit Division (Hughes MCD) in Newport Beach, CA
1981	Margaret Goodearl begins working for Hughes MCD as a supervisor for assembly on the hybrid production floor and as a supervisor in the hybrid engineering lab
1984	Ibarra becomes supervisor for hybrid quality assurance
1985	Goodearl asks Ibarra to look at errors in paperwork, Ibarra brings errors to the attention of her supervisors and was told to keep quiet. This begins time period where Goodearl/Ibarra become aware of problems in hybrid chip testing and paperwork.
1986	Goodearl becomes supervisor for seals processing in the environmental testing area.
1986	False Claims Act (31 U.S. C 3729-3733) becomes False Claims Reform Act of 1986 making it stronger and easier to apply.
Oct. 1986	Goodearl/Ibarra report problems of Hughes management, and, after the problems were not fixed, Goodearl/Ibarra reported the allegations of faulty testing to the United States Department of Defense.
Jan 9, 1987	Earliest date that Hughes may have stopped neglecting environmental screening tests.
1988	Ibarra leaves Hughes feeling that her job had been stripped of all real responsibility.
March 1989	Goodearl is laid off from Hughes.

1995	Goodearl and her husband are divorced.
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Civil Suit Timeline

1990-1996	United States of America, ex rel. Taxpayers Against Fraud, Ruth Aldred (was Ibarra), and Margaret Goodearl v. Hughes Aircraft Company, Inc.
1990	Goodearl files wrongful discharge suit against Hughes and a number of individual managers, which was eventually dropped in favor of the civil suit.
May 29, 1990	Thinking the government investigation was taking too much time, Goodearl/Aldred file civil suit against Hughes under False Claims Reform Act of 1986 with the help of Taxpayers Against Fraud and Washington law firm Phillips and Cohen.
December 1992	Under provisions of the FCA, the U.S. Department of Justice Civil Division takes over the civil case.
Sep. 10, 1996	Hughes found guilty in civil trial. Pays U.S. Government 4,050,00 dollars and each relator 891,000 dollars plus a separate payment of 450,000 dollars to cover attorney's fees, costs, and expenses.

Criminal Suit Timeline

1991-1993	United States of America v. Hughes Aircraft Co., and Donald LaRue
December 13, 1991	After a lengthy investigation, the U.S. Department of Defense charges Hughes and Donald A. LaRue with a 51-count indictment accusing it of falsifying tests of microelectronic circuits (criminal suit).
June 15, 1992	Hughes found guilty of conspiring to defraud the U.S. Government in criminal case, co-defendent LaRUE acquitted following 4-week trial. Goodearl/Aldred called as witnesses in trial. Hughes appeals.
Oct. 29, 1992	Hughes fined 3.5 million in criminal trial decision.
December 2, 1993	Appellate court upholds 1992 criminal conviction and sentence. Hughes appeals.

Hughes Case Socio Technical System

Hughes Socio Technical System

	Hardware/Software	Physical Surroundings	People, Roles, Structures	Procedures	Laws and Regulations	Data and Data Structures
Description	Hybrid Chips (circuitry hermetically sealed in metal or ceramic packages in inert gas atmosphere)	Battle conditions under which chips might be used	Hughes Microelectric Circuit Division	Chip Testing: Temperature Cycle, Constant Acceleration, Mechanical Shock, Hermeticity (Fine and Gross Leak), P.I.N.D.	Legally Mandated Tests	Lot Travelors to document chips

	Analogue to Digital Conversion Chips	E-1000 at Hughes (Clean Room)	Department of Defense (Office of Inspector General)	Hughes Human Resources Procedures for Complaints	Whistle Blower Protection Legislation	
	Radar and Missile Guidance Systems		Hughes Quality Control	Dissenting Professional Opinions	Qui Tam Lawsuit, Civil Suit, Criminal Suit	
			Individuals: Reismueller, Temple, Saia, LaRue, Goodearl, Ibarra/Aldren			

Responsible Dissent

Sources:

- Computing Cases is the primary source for the material below on responsible dissent. It is based on the materials for responsibly carrying out dissent and disagreement that was formerly posted at the IEEE website. The IEEE has since taken this material down.
- The Online Ethics Center has also posted the IEEE material on responsible dissent. The origin of this material as well as a thorough discussion of its content can be found in Carolyn Whitbek, *Ethics in Engineering Practice and Research: 2nd Edition*, Cambridge, UK: Cambridge University Press, 2011. Chapter 7, "Workplace Rights and Responsibilities, pp. 227-269.
- Much of this material (IEEE Guidelines and a discussion of Dissenting Professional Opinion Guidelines) can be found in Chapter 7 ("Averting the Conflict at the Source") in the following: Stephen H., Unger, *Controlling Technology: Ethics and the Responsible Engineer: 2nd Edition*, New York: John Wiley and Sons, INC.

Generic Forms of Dissent:

- Gather more information
- Nolo Contendere (Don't fight it. Go along.)
- Oppose diplomatically: Offer your supervisor alternatives to the wrong he or she has ordered.
- Oppose by confronting: Threaten to go over your supervisor's head or threaten to blow the whistle
- Distance yourself: Ask to be transferred to another section to avoid being implicated in the wrongdoing
- Exit: Quit and do nothing or quit and blow the whistle
- Document your position: If your company has a Dissenting Professional Opinion process. If it doesn't, work to have one implemented. By establishing your opposition, you distance yourself morally from the wrong you have documented
- Which one is right? Use your tests. Which does the best job of satisfying the three ethics tests of reversibility, harm, and publicity? Which does the best job with the ADEM values: justice, responsibility, respect, trust, and integrity?

Introduction to Circumstances of Compromise: The following presents the circumstances of compromise as laid down by Martin Benjamin in *Splitting the Difference*. (See below for complete reference.) Benjamin provides five conditions that indicate when a compromise may be necessary. But he also argues that integrity helps draw a line beyond which compromise must not go. One should not sacrifice basic beliefs that constitute one's personal identity or self-system. A good example of using integrity to draw the line on compromise can be found in the characterization of Thomas More in Robert Bolt's *A Man for All Seasons*. (See also the movie of the same name.) In the preface to the play, Bolt explicitly presents what follows as an exercise in articulating and testing integrity.

Circumstances of Compromise:

- Under these conditions, it may be necessary to "split the difference."
- Factual uncertainty
- Moral complexity

- Continuing Cooperative Relationship
- Decision cannot be deferred
- Scarcity of Resources

More on Moral Complexity: Martin Benjamin in Splitting the Difference quotes John Rawls on moral complexity: “Diversity naturally arises from our limited powers and distinct perspectives; it is unrealistic to suppose that all our differences are rooted in ignorance and perversity, or else in the rivalries that result from scarcity....Deep and unresolvable differences on matters of fundamental significance...[must be acknowledged] as a permanent condition of human life.”

Application of Circumstances of Compromise:

- Factual Uncertainty: Where are the chips under consideration going? If they go to an essential system in an operative technology, then their malfunctioning could lead to loss of life. If they go to a non-essential system (like a prototype being tested) then maybe the testing process can be streamlined. This may require compromise between Hughes management, chip-testing team, and customers.
- Moral Complexity: How should an employee like LaRue weigh his loyalty to supervisors and company and his obligation to the public and client? Setting aside his harassment of Gooderal, is Saia's position (or at least a part of it) morally defensible?
- Continuing cooperative relationship: How important should it be to Gooderal that she needs to sustain her relationship with her supervisor, LaRue, for the long term? How important is it that Hughes managers respond to difficult messages rather than attempt to "shoot the messenger." (Again, thinking in terms of continuing cooperative relationship?)
- Decision cannot be deferred: Why is it impossible to defer the decision on whether to respond to test skipping? This case poses several difficult constraints. How many of these can be "pushed back" through negotiation? Could Saia use his newly found accessibility to customers to negotiate with them an extension on the delivery deadlines?
- Scarcity of resources: How are the resources of time, personnel, and money scarce in this case? Is there any way to push back these constraints by negotiating more time (extending deadlines for delivering chips), personnel (bringing in additional people to test chips), and resources (developing better tools to test chips more quickly). Could, for example, it be possible to transfer Hughes employees from other areas to help out, temporarily, on chip testing?

Ethical Dissent:

1. Establish a clear technical foundation.
2. Keep your arguments on a high professional plane, as impersonal and objective as possible, avoiding extraneous issues and emotional outbursts.
3. Try to catch problems early, and keep the argument at the lowest managerial level possible.
4. Before going out on a limb, make sure that the issue is sufficiently important.
5. Use (and help establish) organizational dispute resolution mechanisms.
6. Keep records and collect paper.
7. These items originate with the IEEE which has dropped them from their website. They can be accessed through the link above with the Online Ethics Center; the list there is more complete. The above is quoted from the Computing Cases website:
http://computingcases.org/case_mater...l_dissent.html.

Before Going Public:

1. Make sure of your motivation.
2. Count your costs.
3. Obtain all the necessary background materials and evidence.
4. Organize to protect your own interests.
5. Choose the right avenue for your disclosure.
6. Make your disclosure in the right spirit.
7. These items come from the IEEE (see onlineethics link) and from the manuscript of Good Computing by Chuck Huff, William Frey, and Jose Cruz.

Places to Go:

1. Government Agencies
2. Judicial Systems
3. Legislators
4. Advocacy Groups

5. News Media
6. In Puerto Rico, laws 14 and 426 have been passed to protect those who would blow the whistle on government corruption. The Oficina de Etica Gubernamental de Puerto Rico has a whistle blower's hotline. See link above.

When to Blow the Whistle:

1. Serious and Considerable Harm
2. Notification of immediate supervisor.
3. Exhaustion of internal channels of communication/appeal.
4. Documented Evidence.
5. Likelihood of successful resolution.
6. When the first three conditions are satisfied, whistle-blowing is morally permissible. (You may do it but you are not required or obligated to do it.) This is because you have brought your concerns before decision-makers, given them a chance to respond, and, in the face of their unwillingness to do so, still find the issue of great importance.
7. When all five conditions are satisfied, then whistle-blowing becomes morally obligatory. In this case, you have a moral duty to blow the whistle. Here, your duty is grounded in your responsibility to inform those who are likely to be harmed by the wrongdoing.

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1. Richard T. De George, "Ethical Responsibilities of Engineers in Large Organizations: The Pinto Case," in *Ethical Issues in Engineering*, ed. Deborah G. Johnson (1991) New Jersey: Prentice-Hall: 175-186.
2. Carolyn Whitbeck (1998) *Ethics in Engineering Practice and Research*. U.K. Cambridge University Press: 55-72 and 176-181.
3. Charles Harris, Michael Pritchard and Michael Rabins (2005) *Engineering Ethics: Concepts and Cases*, 3rd Ed. Belmont, CA: Thomson/Wadsworth: 203-206.

Hughes Dramatic Rehearsals

A note on dramatic rehearsals:

- The notion of dramatic rehearsal comes from John Dewey's *Human Nature and Moral Conduct*. An agent works through a solution alternative in the imagination before executing it in the real world. The dramatic rehearsal tests the idea in a mental laboratory created by the moral imagination. Steven Fesmire in his book, *John Dewey and Moral Imagination: Pragmatism in Ethics* (Indiana University Press, 2003), provides a comprehensive interpretation of Dewey's suggestive idea.
- The scenarios portrayed below reflect events in the case but some changes have been made to create six focused decision points. For a more accurate portrayal of the case events, see *Computing Cases* (computingcases.org)

Decision Point One

- Frank Saia has worked at Hughes Aircraft for a long time. Now he is faced with the most difficult decisions of his career. He has been having problems in the environmental testing phase of his microchip manufacturing plant; the detailed nature of these tests has caused Hughes to be consistently late in delivering the chips to customers.
- Because of the time pressure to deliver chips, Saia has been working to make the production of chips more efficient without losing the quality of the product. Chips are manufactured and then tested, and this provides two places where the process can bottle up. Even though you might have a perfectly fine chip on the floor of the plant, it cannot be shipped without testing. And, since there are several thousand other chips waiting to be tested, it can sit in line for a long time. Saia has devised a method that allows testers to put the important chips, the "hot parts," ahead of the others without disrupting the flow and without losing the chips in the shuffle. He has also added a "gross leak" test that quickly tells if a chip in a sealed container is actually sealed or not. Adding this test early in the testing sequence allows environmental testing to avoid wasting time by quickly eliminating chips that would fail a more fine-grained leak test later in the sequence.
- Because environmental testing is still falling behind, Saia's supervisors and Hughes customers are getting angry and have begun to apply pressure. Karl Reismueller, the director of the Division of Microelectronics at Hughes, has given Saia's telephone number to several customers, whose own production lines were shut down awaiting the parts that Saia has had trouble delivering. His customers are now calling him directly to say "we're dying out here" for need of parts.

Dialogue for Decision Point One

- Construct a dialogue in which Saia responds to the pressure from his supervisor, Karl Reismueller
- Be sure to address the customer complaints

Decision Point Two

- Frank Saia has discovered that an employee under his supervision, Donald LaRue, has been skipping tests on the computer chips. Since LaRue began this practice, they have certainly been more on time in their shipments. Besides, both LaRue and Saia know that many of the “hot” parts are actually for systems in the testing phase, rather than for ones that will be put into active use. So testing the chips for long-term durability that go into these systems seems unnecessary. Still, LaRue was caught by Quality Control skipping a test, and now Saia needs to make a decision. Upper management has provided no guidance; they simply told him to “handle it” and to keep the parts on time.
- He can’t let LaRue continue skipping tests, or at least he shouldn’t let this skipping go unsupervised. LaRue is a good employee, but he doesn’t have the science background to know which tests would do the least damage if they were skipped. He could work with LaRue and help him figure out the best tests to skip so the least harm is done. But getting directly involved in skipping the tests would mean violating company policy and federal law.

Dialogue

- Construct a dialogue in which Saia confronts LaRue about skipping the tests
- Address the following issues:
- Should Saia work with LaRue to identify tests that are not necessary and then have LaRue skip these?
- How should Saia and LaRue deal with the concerns that Quality Control has expressed about skipping the tests?

Decision Point Three

- Margaret Goodearl works in a supervisory position in the environmental testing group at Hughes Aircraft. Her supervisor, Donald LaRue, is also the current supervisor for environmental testing. The group that LaRue and Goodearl together oversee test the chips that Hughes makes in order to determine that they would survive under the drastic environmental conditions they will likely face. Rigorous testing of the chips is the ideal, but some chips (the hot chips) get in line ahead of others. Goodearl has found out that over the last several months, many of these tests are being skipped. The reason: Hughes has fallen behind in the production schedule and Hughes upper management and Hughes customers have been applying pressure to get chip production and testing back on schedule. Moreover, LaRue and others feel that skipping certain tests doesn’t matter, since many of these chips are being used in systems that are in the testing phase, rather than ones that will be put into active use.
- A few months after Margaret Goodearl started her new position, she was presented with a difficult problem. One of the “girls” (the women and men in Environmental Testing at Hughes), Lisa Lightner, came to her desk crying. She was in tears and trembling because Donald LaRue had forcefully insisted that she pass a chip that she was sure had failed the test she was running.
- Lightner ran the hermeticity test on the chips. The chips are enclosed in a metal container, and one of the questions is whether the seal to that container leaks. From her test, she is sure that the chip is a “leaker”—the seal is not airtight so that water and corrosion will seep in over time and damage the chip. She has come to Goodearl for advice. Should she do what LaRue wants and pass a chip she knows is a leaker?

Dialogue

- Construct a dialogue that acts out Goodearl’s response to her knowledge that LaRue is regularly skipping tests
- Should Goodearl first talk directly to LaRue? What if he responds defensively?
- Should Goodearl go over LaRue’s head and discuss his skipping the tests with one of his supervisors? To whom should she go? How could she prepare for possible retaliation by LaRue? What should she know before doing this?
- If LaRue or another supervisor should fail to respond to the test skipping, should Goodearl continue responsible dissent or drop the issue (=nolo contendere)
- Could Goodearl not contend the issue but distance herself? (What if Hughes has no DPO procedure?)

Decision Point Four

Ruth Ibarra (from Quality Assurance) has seen Shirley Reddick resealing chips without the authorization stamp. Ibarra has asked Goodearl to find out what’s going on. When Goodearl asks LaRue, he replies, “None of your damn business.” Shortly after this, Goodearl receives a phone call from Jim Temple, one of her superiors, telling her to come to his office. Temple informs Goodearl in no uncertain terms that she needs to back down. “You are doing it again. You are not part of the team, running to Quality with every little problem.” When Goodearl insisted she did not “run to Quality” but Quality came to her, Temple replies, “Shape up and be part of the team if you want your job.”

Dialogue

- Construct a dialogue in which Gooderal responds to this latest test skipping issue
- Consider these issues in constructing your dialogue:
- Goodearl had already confronted LaRue about test skipping when Lisa Lightner came to her. After failing to get results, she had decided to drop the issue
- How should Goodearl respond to Temple? Should she continue pushing responsible dissent or give way to Temple's threats?

Decision Point Five

- After her conversation with Temple, Goodearl goes to the Personnel Department to inquire into filing a harassment complaint against her supervisors at Hughes
- After her discussion she sees the personnel official leave his office and turn toward Frank Saia's office, one of Goodearl's supervisors.
- Goodearl is then called to Saia's office. An angry Saia throws his glasses at her and threatens to fire her if she persists. He also asks her where she gets off filing a harassment charge against him.

Dialogue

- Construct a dialogue in which Gooderal reacts to Saia both during Saia's outburst and after it.
- Is Saia harassing Gooderal? (How do we define "harassing" in this context?)
- How should Goodearl respond given that Saia's latest outburst was caused by the personnel official reporting to him the confidential meeting he had with Goodearl?
- What are Goodearl's options at this point? Are any of the strategies for responsible dissent we have studied so far relevant or of use?

Decision Point Six

- Margaret Goodearl and Ruth Ibarra have made several attempts to get their supervisors to respond to the problem of skipping the environmental tests. The general response has been to shoot the messenger rather than respond to the message. Both Goodearl and Ibarra have been branded trouble makers and told to mind their own business. They have been threatened with dismissal if they persist.
- So they have decided to blow the whistle, having exhausted all the other options. They initiated contact with officials in the U.S. government's Office of the Inspector General. These officials are interested but have told Goodearl and Ibarra that they need to document their case.
- One day they find two hybrids (chips that combine two different kinds of semiconductor devices on a common substrate) on LaRue's desk. These chips which are destined for an air-to-air missile have failed the leak test. It is obvious that LaRue plans on passing them without further testing during the evening shift after Goodearl has gone home. Goodearl and Ibarra discuss whether this presents a good opportunity to document their case for the Office of the Inspector General.

Dialogue

- Construct an imaginary conversation between Goodearl and Ibarra where they discuss different strategies for documenting their concerns to the Office of the Inspector General?
- Have them consider the following:
- By looking for documented evidence against their employer, have Goodearl and Ibarra violated their duties of trust and confidentiality?
- Some argue that before blowing the whistle, an employee should exhaust internal channels. Have Goodearl and Ibarra discuss whether they can do anything more inside Hughes before taking evidence outside

Questions on Dramatic Reflections

Directions: After you have acted out your decision point in the Hughes case, you and your group have two further activities. First, you will answer the questions below to help you reflect generally on the nature of dramatic rehearsals and specifically on recent dramatization. These five questions (outlined in detail just below) ask you to discuss your dramatic form, the form of responsible dissent you used, how the action you dramatized fared with the three ethics tests, the value and interest conflicts you dealt with, and the constraints that bordered your decision point. Second, you will provide a storyboard that summarizes the drama you acted out before class. This is also detailed just below.

As was said above, John Dewey suggested the idea that underlies these dramatic rehearsals or What-if dramas. As Dewey puts it "[d]eliberation is actually an imaginative rehearsal of various courses of conduct. We give way, in our mind, to some impulse; we try, in our mind, some plan. Following its career through various steps, we find ourselves in imagination in the presence of the consequences that would follow: and as we then like and approve, or dislike and disapprove, these consequences, we find the original impulse or plan good or bad. Deliberation [becomes] dramatic and active...." (Dewey, 1960, p. 135) Think of your "dramatic rehearsal" as an experiment carried out in your imagination. The hypothesis is the alternative course of action decided upon by your group that forms the basis of your "What-if" rehearsal. Imagine that you carry out your alternative in the real world. What are its consequences? How are these distributed? How would each of the stakeholders in your case view the action? How does this fit in with your conception of a moral or professional career? Your imagination is the laboratory in which you test the action your group devises as a hypothesis.

This quote will also help you understand the concept of dramatic rehearsal. It comes from John Gardner, a famous novelist, and Mark Johnson, a theorist in moral imagination. John Gardner has argued that fiction is a laboratory in which we can explore in imagination the probable implications of people's character and choices. He describes what he calls "moral fiction" as a "philosophical method" in which art "controls the argument and gives it its rigor, forces the writer to intense yet dispassionate and unprejudiced watchfulness, drives him—in ways abstract logic cannot match—to unexpected discoveries and, frequently, a change of mind." (Johnson, p. 197; Gardner, p. 108).

There are different versions of what form dramatic form takes. In general there is plot, character, agon (struggle or confronting obstacles), resolution, and closure. A drama is a narrative, an unfolding of related events in time. One event arises to give way to another and so forth. Dramas can be driven by the ends of the characters and the unfolding can be the realization or frustration of these activities. Dramatic rehearsals take isolated actions and restore them to this context of dramatic form.

1. What is the dramatic form taken on by your enactment?

- Perhaps your drama is a comedy. Many groups have chosen this form but have found it hard to explain why. How does comedy help your group to get its message across? What is its message?
- Some groups have approached this rehearsal as a tragedy where the good intentions and goals of those involved all turned out bad. In many ways, this is how the case played out in reality. So, if your group chose tragedy, then it is important to state why there were no viable alternatives to the choices actually made by the participants in the Hughes case. What constraints prevented the agents from achieving their ends? (Look for more than just bad people here.)
- Some groups decided to frame their rehearsal as a documentary. Here a narrator describes and frames the activities carried out by the different participants offering commentary and analysis.
- Continuing with the documentary line, some groups have presented their drama as proceedings in a trial where a judge presides over attorneys presenting the arguments from both sides. This approach has the advantage of laying out the different perspectives but when the judge reaches a decision, it takes on the risk of oversimplifying the case by making one side completely right and the other completely wrong. The "winner takes all" interpretation of a court trial (guilty-innocent) often leaves out moral complexity.
- Some groups convert their dramas into Quixotic ventures where they "tilt at windmills." Here they try to present scenarios where idealistic participants strive to realize their values over difficult, constraining and harsh realities. The advantage of this approach is that it does not compromise on values and ideals. The disadvantage is that it may underestimate elements in the real world that oppose acting on the ideal. Realizing the "intermediate possible" may be the best route here.
- Some groups approach their dramatizations as cautionary tales where they act out the harsh consequences that attend immoral, greedy, selfish, or corruption action. Here the world is constrained by justice. Those who hubristically try to exceed these constraints are punished for their transgressions. Cautionary tales are more moralistic than tragedies but, at some point, converge on this other dramatic form.
- You are, of course, encouraged to go beyond this list by inventing your own dramatic form or combining those listed above to produce a new, synthetic form. The point here is that dramatic forms both filter and structure elements of this complicated case. I am asking that you be deliberate and thoughtful about how you work your way through the Hughes case. What did your dramatic form leave out? How did it structure the drama differently than other forms? Why did you choose the form you chose?

2. Your dramatic rehearsal also should test the three forms of responsible dissent we studied this semester.

- Generic Forms of Dissent. Did your rehearsal test any of the generic forms of dissent such as gather more information, nolo contendere, oppose diplomatically, oppose confrontationally, distance yourself, or exit?

- Moral Compromise. Did your rehearsal deploy any of the strategies of moral compromise? For example, referring to the Ethics of Teamwork, did it deploy bridging, logrolling, expanding the pie, or non-specific compensation? Were you able to get things moving by negotiating interests rather than person-based positions? What were the circumstances that elicited compromise? For example, does the Hughes case display any "moral complexity?"
 - Blowing the Whistle. If your drama followed the case and advocated blowing the whistle, provide a justification using the class framework. For example, argue that whistleblowing was permissible or that it was obligatory. To whom do you recommend blowing the whistle given the problems Ibarra and Goodearl had with the Inspector General's office? How would you recommend they go about gathering documented evidence? What should they do before blowing the whistle? In short, do more, both in your drama and in this reflection, than just advocate the action. Describes the means, complexities, and circumstances surrounding blowing the whistle on Hughes.
3. Outline your ethics experiment by examining the action you advocate using the three ethics tests.
- Reversibility. How does your action look when you reverse with the key stakeholders? Project into their shoes avoiding the extremes of too much identification and too little identification.
 - Harm. What harms have you envisioned through your dramatic rehearsal? Are these harms less quantitatively and qualitatively than the action actually taken in the case?
 - Publicity. Finally, project the action taken in your rehearsal into the career of a moral professional. Is it consistent with this career or does it embrace (or neglect) values out of place in such a career. In other words, carry out the publicity test by associating the values embedded in the action you portray with the character of a good or moral agent carrying out a moral, professional career.
4. Value and interest conflicts in your drama.
- All these decision points involve some kind of conflict. How did you characterize this conflict in your dramatization? Pose your conflict in terms of values. How did your drama "resolve" this value conflict?
5. Recognizing and dealing with the constraints you found in your decision point. These drama/decision points had different kinds and degrees of constraints. Early decision points have fewer constraints than later because the earlier decisions both condition and constrain those that follow. Here is another issue you may need to address. Your feasibility test from the "Three Frameworks" module outlines three kinds of constraints: resource, interest (social or personality), and technical. Did any of these apply? Outline these and other constraints and describe how they were dealt with in your drama.

Story Boards

Suggestions for Story Boards

- Divide your dramatization into four to six frames. Now draw a picture in each frame, one that captures a key moment of your dramatization.
- Check for continuity. Each frame should present elements that show how it emerges from the previous frame and how it transitions into the subsequent frame. The first frame should help the reader find the context in which your drama takes place. The last frame should provide as much closure as your drama permits.
- In general, your storyboard should summarize the dramatization you acted out in front of the rest of the class. But while acting through your drama, you received feedback from the class and, perhaps, began to rethink things. So feel free to make changes in your storyboard to reflect your deeper understanding of your decision point. If you make changes in your storyboard, discuss this in your dramatic reflections. Explain why you decided to change things.

Hughes Case Media Files

Hughes Case and Dialogue Points: [Hughes_V3a.pptx](#), [Responsible Dissent.pptx](#)

What If Dramatic Rehearsals: [Hughes_Drama_V2.pptx](#)

Debating Topics for ADMI 4016, Spring 2011: [Reflections on debate.pptx](#)

Jeopardy: Responsible Dissent: [Jeopardy6.pptx](#)

Jeopardy for codes of ethics: [Jeo_Codes.pptx](#) [Hughes 2014.docx](#)

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CHAPTER OVERVIEW

5: Business Ethics Bowl

[5.1: Ethics Bowl Rules and Procedures](#)

[5.2: Practical and Professional Ethics Bowl Activity - Follow-Up In-Depth Case Analysis](#)

[5.3: Ethics Bowl - Cases and Score Sheets](#)

[5.4: EAC Toolkit - UPRM Ethics Bowl - IIT Summer Institute Follow-up](#)

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5.1: Ethics Bowl Rules and Procedures

This media file describes the rules and procedures for the UPRM version of the ethics bowl competition. Included is a timeline for the competition and a rubric that identifies the four scoring categories. Both have been adopted from the national ethics bowl competition developed by Robert Ladenson and held yearly at the meetings of the Association for Practical and Professional Ethics.

Downloadable MS Word File

[EBRules_CNX.doc](#)

Ethics Bowl Rules and Procedures.

This media file has a powerpoint presentation delivered by Jose Cruz, Halley Sanchez, and William Frey at the 2004 meeting of the Association for Practical and Professional Ethics. The presentation describes activities that help prepare students for the competition, shows how the cases used in the competition are selected, breaks down the competition into its constituent parts, and describes how students are debriefed after the competition. The activities used to prepare students for the competition are crucial; they provide opportunities to practice skills in moral imagination. Debriefing activities are equally important since students frequently fail to see how they have developed skills in preparing for and participating in the competition.

Ethics Bowl at UPRM

[APPE 2004 EB 8.ppt](#)

This figure describes preparatory activities, debate structure, and debriefing exercises for an adaption of the Ethics Bowl held in Engineering Ethics classes at the University of Puerto Rico at Mayaguez. It was presented at APPE in 2004.

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5.2: Practical and Professional Ethics Bowl Activity - Follow-Up In-Depth Case Analysis

Module Introduction

This module provides students with a structure for preparing an in-depth case study analysis based on feedback they have received through their participation in an **Ethics Bowl** competition as part of the requirements for courses in Practical and Professional Ethics taught at the University of Puerto Rico at Mayaguez. Students viewing this module will find formats for analyzing decision making cases and position cases such as the decisions published by the National Society of Professional Engineers **Board of Ethical Review**. They will receive information pertinent to preparing in-depth case analyses, short summaries of the case pool for the Ethics Bowl competition, and a summary of procedures for carrying out a group self-evaluation. More information on the Engineering Ethics Bowl carried out at UPRM can be found in Jose A Cruz-Cruz, William J. Frey, and Halley D. Sanchez, "The Ethics Bowl in Engineering Ethics at the University of Puerto Rico - Mayaguez" in Teaching Ethics 4(3): 15-32.

Choosing Your Case

1. You must choose one of the two cases you presented on in the Ethics Bowl. (This means the case on which you gave your initial presentation.
2. You may choose either the first round decision-making case or the NSPE Board of Ethical Review Case

How should you choose your case?

1. Which case did you find the most interesting, challenging, or fruitful?
2. On which case did you receive the most interesting feedback from the other team and the judges?
3. Do you want to make, defend, and implement a decision or analyze a BER decision?

Once you choose your case, you need to analyze it according to the following steps:

Decision-Making Cases

Worksheets	Decision-Making Case
	Identify and state the (ethically) relevant facts
STS Table (Table + Verbal Explanation)	Prepare a Socio-Technical Analysis. Fill in the STS table (see below) and then verbally describe each component.
Value Table (Table + Written Problem Statement)	Fill out a Value Table (see below) Use it to identify the ethical problem or problems. Summarize this by providing a concise problem statement that is explicitly tied to the Value Table.
Brainstorm Lists (initial and refined lists)	4. Brainstorm solution to the problem or problems. Be sure to discuss how list was generated and how it was refined. Describe value integration and interest negotiating strategies used.
Solution Evaluation Matrix (Matrix + Verbal Explanation and Justification)	5. Compare, evaluate, and rank the solutions
	6. Choose the best available solution. Provide a justification summarizing ethical and feasibility considerations highlighted in Solution Evaluation Matrix.
Feasibility Matrix (Matrix + Verbal Explanation)	7. Develop a plan for implementing your solution. Discuss and justify this plan explicitly in terms of the specific feasibility considerations in the Feasibility Matrix.
	Develop and discuss preventive measures (if applicable)

NSPE-BER Case

Worksheets	
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	1. Identify and state the (ethically) relevant facts
Stakeholders (Matrix + Verbal Explanation)	2. Identify the stakeholders and their stakes.
Problem Classification (Matrix + Concise Verbal Problem Statement)	3. Identify the ethical problem or problems
	4. State the BER decision and summarize their code-based justification (cite code provisions, summarize principles, and list relevant precedents)
Solution Evaluation (Matrix + detailed verbal explanation and justification)	5. Evaluate the BER decision using the three ethics tests, code test, and global feasibility test.
	6. Construct a strong counter-position and counter-argument to the BER decision
Solution Evaluation (Matrix + detailed verbal explanation and justification)	7. Evaluate counter-position and counter-argument using the 3 ethics tests, feasibility test, and code test
Solution Implementation (Feasibility Matrix + Verbal Explanation)	8. Evaluate counter-position and counter-argument in terms of relevant feasibility considerations. Provide a matrix/table + verbal explanation.

In-Depth Analysis: Step by Step

Description of In-Depth Case Analysis

Title of Assignment: "In-Depth Case Analysis"

Due Date for Written Projects: One week after the last class of the semester

What is required?

1. Participation in at least two ethics bowl competitions
2. Each group will choose from the two cases it debated in the Ethics Bowl a case for a more extended analysis carrying out the seven-step decision-making framework. They will prepare an extended analysis of this case (10 to 20 pages)
3. Each group will prepare summaries of the 15 cases assigned for the ethics bowl. These summaries (a minimum of one page for each case) will be handed in with the extended case study analysis. These summaries should include a problem statement, a solution evaluation matrix, and a feasibility matrix
4. Each final submission will also include a group self-evaluation. This evaluation will include:
 - ____ a list of the goals each group set for itself
 - ____ a careful, justified and documented assessment of your success in reaching these goals
 - ____ a careful assessment of what you did and did not learn in this activity
 - ____ a discussion of obstacles you encountered and measures your group took to overcome these
 - ____ a discussion of member participation and contribution including the member contribution forms
 - ____ in general what worked and what didn't work for you and your group in this activity
5. A group portfolio consisting of the materials prepared by your group during the group class activities:
 - ____ Virtue Chart (Responsibility)
 - ____ Gray Matters Solution Evaluation Matrix
 - ____ Rights Chart: Free & Informed Consent
 - ____ Group Code of Ethics

Structure of Written Analysis

1. A brief summary of the case focusing on the ethically relevant facts.
2. A Socio-Technical System Table + Short paragraph on each of the seven categories.

3. A Value Table + a short paragraph on the embedded values you have identified and where they occur in the STS. Then state whether you have found any value mismatches, magnified existing value conflicts, and remote/harmful consequences.
4. On the basis of your STS analysis and value conflict analysis, provide a short, concise problem statement. Make sure the problem you have identified is grounded in your STS and value analysis. If not, one or the other (or both) needs to be changed.
5. A brainstorm list in which you record the solutions your group has designed to solve the problem stated above. The rough unrefined list should include around 10 solutions. Then refine this list into three. Spend time detailing how you reached your refined list. Did you synthesize rough solutions? On what basis did you leave a solution out all together? Did you find other ways of relating or combining solutions? Spend time documenting your brainstorming and refining process. Show in detail how you came up with the refined list.
6. Do a comparative evaluation of three of the refined solutions you developed in the previous step. First, prepare a solution evaluation matrix that summarizes your comparative evaluation. Use the table provided below. Second, provide a verbal account of the solution evaluation and comparison process you present in the solution evaluation matrix.
7. Reach a final decision. Defend your decision using the ethics and feasibility tests. If the decision situation in which you are working is a dynamic one, then propose a series of solutions that you will pursue simultaneously, including how you would respond to contingencies that might arise. (You could express this in the form of a decision tree.)
8. Fill out a Feasibility Matrix. See matrix below
9. Present an implementation plan based on your Feasibility Matrix. This plan should list the obstacles that might arise and how you plan to overcome them. (For example, don't just say, "Blow the whistle." Discuss when, how, where, to whom, and in what manner. How would you deal with reprisals? Would your action seriously disrupt internal relations of trust and loyalty? How would you deal with this?) Work out a detailed plan to implement your decision using the feasibility constraints to "suggest" obstacles and impediments.
10. Finally, discuss preventive measures you can take to prevent this type of problem from arising again in the future.

Socio-Technical System Table

Hardware	Software	Physical Surroundings	People, Groups, Roles	Procedures	Laws, Statutes, Regulations	Data and Data Structures

STS Value Table

	Hardware	Software	Physical Surroundings	People, Groups, Roles	Procedures	Laws	Data and Data Structures
Integrity							
Justice							
Respect							
Responsibility for Safety							
Free Speech							
Privacy							
Property							

Solution Evaluation Matrix

Solution/Test	Reversibility or Rights	Harms/Benefit or Net Utility	Virtue	Value	Code	Global Feasibility

Description	Is the solution reversible with stakeholders? Does it honor basic rights?	Does the solution produce the best benefit/harm ratio? Does the solution maximize utility?	Does the solution express and integrate key virtues?	Moral values realized? Moral values frustrated? Value conflicts resolved or exacerbated?	Does the solution violate any code provisions?	What are the resource, technical, and interest constraints that could impede implementation?
Best solution						
Best alternate solution						
Worst solution						

Feasibility Matrix								
Resource Constraints			Technical Constraints		Interest Constraints			
Time	Cost	Available materials, labor, etc	Applicable technology	Manufacturability	Personalities	Organizational	Legal	Social, Political, Cultural

Format

1. Group, team-written projects are to be 10-20 pages in length, double spaced, with standard 1-inch margins, and typewritten. This does not include documentation, appendices, and other notes

2. It is essential that you carefully and fully document the resources that you have consulted. The most direct way to do this is to include numbered entries in a concluding section entitled, "Works Cited". These entries should provide complete bibliographical information according to standard form (Chicago Manual of Style or the MLA Manual of Style). Then insert the number of the entry in parenthesis in the text next to the passage that is based on it. (Example: "The self is a relation that relates itself to its own self...." (4) The number "4" refers to the fourth item in the "Works Cited" section at the end of your paper)

3. Practical norm 5j of the CIAPR code of ethics sets forth the obligation of the professional engineer to give others due credit for their work. For this reason, plagiarism will not be tolerated in any form. Possible forms of plagiarism include but are not limited to the following:

- Quoting directly from other sources without documenting (footnote or bibliography) and/or without using quotation marks. Claiming that this is an appendix will not excuse this action. Claiming ignorance will not excuse this action.
- Using the ideas or work of others without giving due credit or proper acknowledgment. "Proper acknowledgment", in this context, requires a standard bibliographical reference and the use of quotation marks if the material is being directly quoted.
- If your paper relies exclusively or primarily on extensively quoted materials or materials closely paraphrased from the work of others, then it will not be credited as your work even if you document it. To make it your own, you have to summarize it in your own words, analyze it, justify it, or criticize it.
- You will not be credited for material that you translate from English to Spanish unless you add to it something substantial of your own.
- In general, what you appropriate from another source must be properly digested, analyzed, and expressed in your own words. If you have any questions on this, please ask me.
- Any plagiarized document—one which violates the above rules—will be given a zero. You will be given a chance to make this up, and the grade on the make-up project will be averaged in with the zero given to the plagiarized document. Since this is a group grade, everyone in the group will be treated the same, even though the plagiarizer may be only one person. Each member of the group is responsible to assure that other members do not plagiarize in the name of the group. (Since the due date for the written project is late in the semester, this will probably require that I give the entire group, i.e., all members, an Incomplete.) Each member of the group will be held individually responsible in the above-described manner for the final content of the written report.

4. This is not a research project but an exercise in integrating ethics into real world cases. In Chapters 2 and 3 of Engineering Ethics: Concepts and Cases, the authors present a thorough discussion of the case study analysis/problem solving method discussed in class. You also have supporting handouts in your file folders from Magic Copy Center as well as materials I have presented directly in class. Engineering Ethics: Concepts and Cases also contains several sample case studies that can help guide you in constructing your own presentation. What I am looking for is a discussion of the case in terms of the ethical approaches and decision-making frameworks we have discussed this semester. You do not need to "wow" me with research into other areas peripherally related to the case; you need to show me that you have practiced decision-making and made a serious effort to integrate ethical considerations into the practice of engineering.
5. The usual criteria concerning formal presentations apply when competing in the Ethics Bowl. Dress professionally.
6. You may write your group, team-written project in either Spanish or English.
7. All competitions will take place in the regular classroom.

Media Files Beginning Spring 2007

These media files provide information on the ethics bowl and the follow-up activities including individual decision point summaries, in-depth case analysis, and group self-evaluation. They have been integrated into the Business Ethics course during the Spring semester, 2008 and will apply from this date on into the future.

Team Member Evaluation Form

[TEAM MEMBER RATING SHEET.doc](#)

This file contains the team member rating sheet which each group member must fill out and turn in with his or her group project.

Final Project and Group Self-Evaluation Rubrics

[Be Rubric S07.doc](#)

This rubric will be used to grade the in-depth case analysis, the group self-evaluation, and the Ethics Bowl case summaries.

Basic Moral Concepts for Ethics Bowl

[BME V2 97.doc](#)

Clicking on this figure will download the basic moral concepts that you will be integrating into the ethics bowl and your final in-depth case analysis. You will be asked to show how you worked to integrate these concepts in your group self-evaluation.

Intermediate Moral Concepts for Ethics Bowl

[IMC V2 97.doc](#)

Clicking on this figure will open a table that summarizes the intermediate moral concepts that are at play in the four cases that are being used in the Ethics Bowl: Hughes, Therac, Toysmart, and Biomatrix.

Ethics Bowl Cases for ADMI 4016: Environment of the Organization

[Ethics Bowl Cases.docx](#)

Check List

Breakdown of Project Grade:

Group Team-Written Project: 200 points, group grade.

- This is your group's in-depth case analysis
- It will analyze the decision scenario your group presented on in the ethics bowl
- Your task is to give a full and comprehensive analysis of a decision point using the tables presented above, accompanying verbal descriptions, and carrying out the four-stage problem-solving framework of specifying the problem, generating solutions, testing solutions in terms of their ethics, and implementing these solutions.

Nota Bene

- After the Ethics Bowl, I will provide the class with general feedback and presentations on how to prepare the final project. When you submit your final report, I will be looking for how you responded to my comments and suggestions and to the comments and suggestions of the judges and the class.

- Attendance is mandatory for all Ethics Bowl competitions. This is important because you will help one another by the comments and discussions that are generated by the presentations. Students not competing need to listen actively and respectfully to the presenting group. Keep in mind the twin standards of respect and professionalism. I will deduct points from the grades of groups and/or individuals who do not listen courteously to the presentations of others or who do not attend class during the presentation cycle.

Nota Bene:

Check List

- **Each group will turn in this checklist, fully filled out and signed. Checking signifies that your group has completed and turned in the item checked. Failure to submit this form will cost your group 20 points**
- ____ One page summaries of the 10 Ethics Bowl decision points taken from the Therac-25, Biomatrix, Toysmart, and Hughes cases.
- ____ Group, in-depth analysis of the case your team presented on in the Ethics Bowl.
- ____ List of Ethically Relevant Facts
- ____ Socio-Technical System Table + Verbal Explanation
- ____ Value Table + Problem Statement + Justification
- ____ List of Brainstormed Solutions + Description of Refining Process + Refined list
- ____ Solution Evaluation Matrix + Verbal Comparison of Three Alternatives from refined solution list
- ____ Chosen Solution + Verbal Justification
- ____ Feasibility Matrix + Solution Implementation Plan concretely described and based on feasibility matrix
- ____ Preventive Measures (if applicable)

Materials Required from Ethics Bowl

- ____ Ethics Bowl Score Sheets
- ____ The decision point your team **presented** on in the competition
- ____ The decision point your team **commented** on in the competition

____ Group Self-Evaluation Form including...

- ____ a list of the goals your group set for itself
- ____ a carefully prepared, justified, and documented assessment of your group's success in reaching these goals
- ____ a careful assessment of what you did and did not learn in this activity
- ____ a discussion of obstacles you encountered and the measures your group took to overcome these
- ____ a discussion of member participation and contribution including the member contribution forms
- ____ a general discussion of what worked and what did not work for you and your group in this activity

____ Each member will turn in a filled out Team Member Evaluation Form. This form can be accessed through the media file listed above. It is suggested that you do this anonymously by turning in your Team Member Evaluation Form in a sealed envelope with the rest of these materials. You are to evaluate yourself along with your teammates on the criteria mentioned in the form. Use the scale suggested in the form.

Group Portfolios Include...

- ____ Virtue Tables including the moral exemplar profile your group prepared and presented.
- ____ The justification using the rights framework of the right assigned to your group. This was one of the rights asserted by engineers against their corporate employers.
- ____ A one page summary of how you developed your role in the Incident at Morales "**Vista Publica.**"
- ____ The code or statement of values summary prepared by your group as a part of the Pirate Code of Ethics module. This summary focused on one of six organizations: East Texas Cancer Center, Biomatrix, Toysmart, Hughes Aircraft, CIAPR, or AECL (in the Therac case).

Copy-paste this checklist, examine the assembled materials prepared by your group, and check the items your group has completed. Then read, copy-paste, and sign the following pledge.

Group Pledge

- I certify that these materials have been prepared by those who have signed below, and no one else. I certify that the above items have been checked and that those items with checkmarks indicate materials that we have turned in. I also certify that we have not plagiarized any material but have given due acknowledgment to all sources used. All who sign below and whose names are included on the title page of this report have participated fully in the preparation of this project and are equally and fully responsible for its results.

- Member signature here _____
- Member signature here _____
- Member signature here _____
- Member signature here _____
- Member signature here _____
- Member signature here _____

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5.3: Ethics Bowl - Cases and Score Sheets

Module Introduction

This module is designed to give you a brief orientation in the Ethics Bowl competition. It is designed to complement and complete other modules concerning the ethics bowl that you will find in the Corporate Governance course.

Ethics Bowl Rules (briefly)

- The moderator will begin the competition by flipping a coin to determine which team will present first. If the team that calls wins the toss, they choose whether they or the other team go first.
- Monday: (1) Team 1 will have one minute to consult and seven minutes to give its initial presentation. The presentation must be tied to the question/task given to it by the moderator. (2) Team 2 has a minute to consult and seven minutes to make its Commentary on Team 1's presentation. Team 2 can close its commentary by posing a question to Team 1. (3) Team 1 then has a minute to consult and five minutes to respond to Team 2's Commentary. (4) Team 1 will then answer questions posed by the two peer-review teams. Each peer review team will ask a question. A quick follow-up is allowed. The peer-review question and answer session will go for 15 minutes. (5) The peer-review teams will score the first half of the competition but not announce the results.
- Wednesday: The same procedure will occur while reversing the roles between Teams 1 and 2. Thus, team 2 will present, team 1 comment, team 2 respond, and then team 2 will answer questions from the peer review panels. The peer-review panels will add the scores for the second part of the competition but will hold off on announcing the results until Friday's class.
- Friday: The two peer-review teams will present and explain their scores. Peer Review teams will take note: you're objective is not to criticize or evaluate the debating teams but to provide them feedback in terms of the four categories.
- Debating teams may trade minutes from consulting to presenting. For example, Team 1 may decide to take two minutes to consult when given their case and task. This means that they will have 6 minutes, instead of 7, to present.
- Nota Bene: Debating teams and Peer Review teams are not allowed to bring notes into the competition. You will be provided with paper to take notes once the competition starts.
- Even though the national Ethics Bowl competition allows only one presenter, debating teams will be allowed to "pass the baton." When one person finishes speaking, another can step in his or her place. It is absolutely forbidden that more than one person speaks at a time. Also, the competing team's speaking time is limited to its commentary. Once that is over, they are instructed to quietly listen. Infractions will be followed first by a warning. Second infractions will result in points being taken away.

Competition Timeline

1. Team 1 Presentation: One minute to consult, seven minutes to present.
2. Team 2 Commentary: One minute to consult, seven minutes to present.
3. Team 1 Response to Commentary: One minute to consult, five minutes to respond.
4. The question and answer session between Team 1 and the Peer Review teams will last 15 minutes (running clock). The first peer review team will have 7 minutes 30 seconds for its questions and the second will have roughly the same time.
5. In the second round, the timeline is the same while the debating teams change roles.

Advice to Debating Teams

- Tell us what you are going to do, do it, and then tell us what you have done. In other words, start your presentation with a summary, then launch into the main body of your presentation, and then conclude with another summary. This will help the listening audience understand what you are trying to do.
- Be professional, formal, and courteous. Address yourself to the other team and the peer review team. It is a good idea to stand when you are giving your initial presentation.
- Be sure to communicate your understanding of the scoring criteria. What do you and your team understand by intelligibility, ethical integration, feasibility, and moral imagination/creativity? Take time to listen to the other team and the peer review teams to gain insights into their understanding. During the commentary and the question and answer session, you will get crucial clues into what others think you have achieved and where you need further work. Use this feedback.
- Be sure to thank the peer review teams, moderators, and your opponents during and after the competition. Such formalities make it possible to penetrate to the deeper practices that underlie the virtue of reasonableness.

- Relax and have fun! You may not have the opportunity to say everything you want to say. One of the purposes behind this competition is to help you see just how hard it is to advocate for ethical positions. We almost always have to do so under serious constraints such as time limits. Also, remember that you have other forums for "getting it said," namely, your group self-evaluation and your in-depth case analysis. In these places, you will be able to discuss these issues in the kind of depth you think necessary.

Advice to the Peer Review Teams on Scoring

- Remember that all three scoring events of the competition are worth 20 points. The initial presentation, the response to the commentary and questions, and the commentary on the other team's presentation all count for the same 20 points.
- Although you have the complete rubric only for the initial presentation, you will score the other parts of the presentation based on the four criteria: intelligibility, ethical integration, feasibility, and moral imagination/creativity. You will score 1 to 5 on each criterion for a total of 20.
- Three is the middle of the road score. In other words, three is a good, average score. It is not a C--don't think of scoring as grading. Start each team off from a default of three. Then move off that default only when something exceptionally good or not so good happens. If your scores deviate much from straight twelves (36), then you are scoring too high or too low.

Ethics Bowl Scoring Criteria

1. Intelligibility includes three skills or abilities: (1) the ability to construct and compare multiple arguments representing multiple viewpoints; (2) the ability to construct arguments and provide reasons that are clear, coherent, and factually correct; (3) evidence of realizing the virtue of reasonableness by formulating and presenting value integrative solutions?
2. Integrating Ethical Concerns includes three skills: (1) presenting positions that are clearly reversible between stakeholders; (2) identifying and weighing key consequences of positions considered; (3) developing positions that integrate values like integrity, responsibility, reasonableness, honesty, humility, and justice.
3. Feasibility implies that the positions taken and the arguments formulated demonstrate full recognition and integration of interest, resource, and technical constraints. While solutions are designed with constraints in mind, these do not serve to trump ethical considerations.
4. Moral Imagination and Creativity demonstrate four skill sets: (1) ability to clearly formulate and frame ethical issues and problems; (2) ability to provide multiple framings of a given situation; (3) ability to identify and integrate conflicting stakeholders and stakes; (4) ability to generate solutions and positions that are non-obvious, i.e., go beyond what is given in the situation.

Peer Review Team Responsibilities

- Attend the debate sessions and the feedback session on Friday after the competition. Remember this is the capstone event of the course. It looks bad if you do not bother to attend.
- Your team will ask questions during the debate. This will constitute, at a minimum, one question and a quick follow up if necessary. You are not to debate with the presenting team. So your questions should not be designed to trap them. Rather, seek through your questions to explore seeming weak points, unclear statements, and incomplete thoughts. Use your questions to help you line up the debating team against the four criteria.
- Fill out the score sheet and assess the debating teams in terms of intelligibility, integrating ethics, feasibility and moral imagination/creativity.
- Lead, with the other Peer Review team, the feedback sessions. This requires that you prepare a short, informal presentation that shows your scoring and then explains it.
- Always, always, always be courteous in your feedback comments. Try to present things positively and proactively. This is difficult but practice now will serve you well later when you are trying to explain to a supervisor how he or she has made a mistake.

Media Files with Cases and Score Sheets

Score Sheet Team One: [Revised ScoreSheet T1 V2.doc](#)

Score Sheet Team Two: [Revised ScoreSheet T2 V2.doc](#)

Word file containing the 12 Ethics Bowl classes for Business Ethics Spring 2007: [Ethics Bowl Cases for Spring 2007.doc](#)

These are the cases for the Ethics Bowl Competition for the Fall Semester in the year 2007. These scenarios or decision points are taken from Incident at Morales, Hughes Aircraft Case, Biomatrix Case, and Toysmart Case: [EB_Fall07_W97.doc](#)

This presentation was given Friday, April 27 to the Ethics Bowl teams that debated on the Therac-25 case and the Inkjet case: [Debriefing_Round_2.ppt](#)

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5.4: EAC Toolkit - UPRM Ethics Bowl - IIT Summer Institute Follow-up

Module Introduction

The "Prerequisite link" included in the upper right-hand corner of this module opens the module content located at the IIT Center for the Study of Ethics in the Professions. This file, "Report on Ethics Integration Projects," was prepared by Dr. Jose Cruz-Cruz as the follow-up to a workshop he attended at the Illinois Institute of Technology on ethics across the curriculum. Directed by Michael Davis (Senior Fellow at the Center for the Study of Ethics in the Professions), the IIT EAC workshop was funded by the National Science Foundation.

Module Activities

1. Open the link to the IIT Ethics Bowl Packet
2. Read the section beginning on page 2, "The Ethics Bowl at UPR - Mayaguez"
3. Read the cases in the Appendix from page 8 to page 12
4. Prepare a position paper on each case. Since the cases terminate at a decision point, make a decision and justify it in terms of reversibility, harm/beneficence, and publicity. Then carry out a global feasibility analysis. For more on the tests and a decision-making framework, consult the module, "Three Frameworks in Ethical Decision-Making." See the link above
5. Prepare for the Ethics Bowl debate by studying the procedures and scoring criteria presented in the report at IIT

The Ethics Bowl can be divided into eight stages

1. Team 1 receives its case and gives an initial presentation taking an ethical position and providing an ethical justification
2. Team 2 makes a commentary that critically analyzes Team 1's presentation
3. Team 1 responds to Team 2's commentary
4. Fifteen minutes are allotted for the judges in the peer review teams to ask Team 1 questions. After this, the judges/peer review teams score the first half of the competition without announcing the results
5. Team 2 receives its case and makes an initial presentation in which it states and justifies its decision or position
6. Team 1 gives a commentary to Team 2's presentation. They can take a counter-position as well as reveal weaknesses in Team 2's position and justification
7. Team 2 responds to Team 1's commentary
8. Team 2 answers questions from the judges for 15 minutes

media files

The four media files open key documents for the Peer-Reviewed Ethics Bowl, held in Corporate Governance classes at UPRM. The first file provides a presentation that will help to orient you to the Ethics Bowl. The second and third files contain the score sheets, which also serve as rubrics assessing your achievements in the debating criteria of (1) Intelligibility, (2) Integrating Ethical Concerns, (3) Feasibility, and (4) Moral Imagination and Creativity. The final media file provides Ethics Bowl rules modified to fit the peer review format.

This presentation helps orient students and faculty on the Professional Ethics Bowl held at the University of Puerto Rico at Mayaguez: [APPE 2004 EB 8.ppt](#)

Scoring sheet and rubric for Team 1 in UPRM Professional Ethics Bowl: [Revised ScoreSheet T1.doc](#)

This is the revised score sheet and rubric for Team 2 in the UPRM Professional Ethics bowl: [Revised ScoreSheet T2.doc](#)

The attached document briefly describes the UPRM Ethics Bowl competition in its current Peer-Review format: [EBRules CNX 2.doc](#)

SUPPLEMENTARY INFORMATION

Summary of Scoring Criteria

- Intelligibility includes three skills or abilities: (1) the ability to construct and compare multiple arguments representing multiple viewpoints; (2) the ability to construct arguments and provide reasons that are clear, coherent, and factually correct; (3) evidence of realizing the virtue of reasonableness by formulating and presenting value integrative solutions.

- Integrating Ethical Concerns includes three skills: (1) presenting positions that are clearly reversible between stakeholders; (2) identifying and weighing key consequences of positions considered; (3) developing positions that integrate values like integrity, responsibility, reasonableness, honesty, humility, and justice.
- Feasibility implies that the positions taken and the arguments formulated demonstrate full recognition and integration of interest, resource, and technical constraints. While solutions are designed with constraints in mind, these do not serve to trump ethical considerations.
- Moral Imagination and Moral Creativity demonstrate four skill sets: (1) ability to clearly formulate and frame ethical issues and problems; (2) ability to provide multiple framings of a given situation; (3) ability to identify and integrate conflicting stakeholders and stakes; (4) ability to generate solutions and positions that are non-obvious, i.e., go beyond what is given in the situation.

Learning Objectives

The learning objectives for this module conveniently divide into content areas and skills. The content objectives can be found in the AACSB ethics criteria of ethical leadership, ethical decision-making, social responsibility, and corporate governance. The skills objectives include the skills emphasized at the University of Puerto Rico at Mayaguez: ethical awareness, ethical evaluation, ethical integration, ethical prevention, and value realization. In addition, there are the criteria of moral creativity and moral imagination.

Content Objectives:

- **Ethical Leadership:** You have examined ethical leadership by looking at the moral exemplars portrayed in the module of that name. What skills and virtues do moral exemplars exhibit? How do these skills and virtues "cluster"? What can you do to exhibit moral leadership? In making and defending your decisions in the Ethics Bowl, spend time showing the peer review teams how your decisions exhibit moral leadership.
- **Ethical Decision-Making:** We are using a decision making framework this semester that emphasizes four stages: (1) problem specification, (2) solution generation, (3) solution testing, (4) solution implementation. Spend time during the debate to show that you know what the problem is you are trying to solve. In preparing for the debate, you have carried out a brainstorming process to generate a solution list; you will be able to show evidence of this when you do your in-depth case analysis. Solution testing you carry out when you evaluate and rank alternatives in terms of their ethics. Try not to neglect the final stage where you show the feasibility of the solution you are advocating. Show that you have thought through implementation carefully, even to the extent of uncovering the most likely obstacles to your solution.
- **Social Responsibility:** The Socio-Technical System grids we have worked on in class will help to uncover issues of social responsibility in the cases of the Ethics Bowl. Social responsibility requires that you step back from your decision point to look at the broader social and political implications of what you are doing.
- **Corporate Governance:** Many of you will quickly determine that the participant perspectives from which you are asked to make your decisions are tightly constrained by organizational problems. Companies that discourage communication, seek to pass blame down to those low on the hierarchy, and pressure employees to take legal and ethical short cuts bear much of the blame for creating the ethical problems you are required to solve. But stay focused on your agent's perspective. Formulate concrete strategies for leading organizational change from that perspective. You can talk about changing the organizational culture. Solving the problem may require reforming the "system." But do not fall into the trap of blaming the system.

Skill Objectives:

- **Ethical Awareness:** You will demonstrate ethical awareness by how well you identify and frame the ethical issues and problem that arise in the case you debate. If you spend time in your presentation framing the problem raised in your case and making sure the peer review team understands how you see the problem, you will do well in this category. A helpful hint: many of the cases you will be debating can be sharply specified as value conflict problems. Show the values that are in conflict and how you will go about integrating them.
- **Ethical Evaluation:** You have already spent time practicing ethical evaluation by using the ethics tests to assess and rank solution alternatives in the Hughes case. The tests help you to hone in on the ethical strengths and weakness of solution alternatives. When the tests converge on a solution, this is a strong sign of its ethical strength. When they diverge, this signals to you the need to reformulate the solution to cover the "ethics gaps" raised by the tests.

- **Ethical Integration:** You have examined the analogy between design and ethics problems. In ethics problems, we create solutions that realize, balance, and integrate the ethical specifications. We also implement these solutions over situational constraints like resource, interest, and technical constraints. Ethical Integration requires that you make clear the solution formulation process that your solution demonstrates. Make it crystal clear to the peer review teams that you have designed your solution to realize ethical considerations while respecting situational constraints.
- **Ethical Prevention:** This is not the prevention of the ethical but the anticipation of potential problems and the development of counter-measures to prevent these problems from arising or to minimize their impact. The earlier we address ethical problems the easier they are to solve. Taking a preventive stance toward ethical problems is the best way to promote ethics in the real world.
- **Value Realization:** Finally, make the move from asking how to fix things when they go wrong to how to make things continually better. As professionals, you are in the position to use your knowledge and skills to realize values of all types. Now you can put this to work to identify ethical value "gaps" and develop strategies for eliminating them.

A quick word on two additional objectives. Moral imagination requires examining a situation from multiple framings. As we have already seen in class, some of you approach problems from a social perspective. You see effective solutions lying in leading opposition, forming coalitions among co-workers, and leading organizational charges to resolve injustices. But others seek to formulate problems in technical terms. Changing the manufacturing process, pressing for technically innovative designs, and formulating situations as technical puzzles. The point here is that the one does not exclude the other, and moral imagination requires working through these and other possible framings.

As we have seen in the reversibility test, moral imagination also requires projecting ourselves into the positions of others and viewing the situation from their standpoint. This does not require abandoning ourselves to this perspective, especially when there are moral problems with doing so. But showing during the course of the debate that you have taken time to explore the situation from the standpoint of the different stakeholders, that you have taken the time to listen to and understand the objections of the other team, and that you have carefully considered the issues raised by the peer review teams is the best way to show moral imagination in the Ethics Bowl.

Moral creativity requires showing that you have taken the effort to design non-obvious solutions to the problems at hand. Going beyond the obvious requires re-framing so moral creativity requires moral imagination. But moral creativity also requires the exercise of the virtue of reasonableness. If you are confronted with a solution where values are in conflict, have you considered creative, out-of-the-box methods for integrating them? When one way of framing the problem and the situation fails to produce helpful answers, have you tried reformulating the problem? If you cannot solve the entire problem, have you tried solving a part and setting the rest aside for a more productive time? Moral creativity requires demonstration of out-of-the-box thinking on how to solve moral problems.

Additional Activities

Activities Before and After the Ethics Bowl:

- Work with and practice your ethical approaches, ethics tests, and other frameworks. They will help structure your presentation, responses to the other teams, and answers to the peer review judges' questions.
- Prepare your cases. This requires developing a format or template that makes it possible for one person to specialize on the case but facilitates disseminating the case to the rest of the team. Solution evaluation matrices help. So do concise problem statements.
- After the Ethics Bowl, you will be asked to do an in-depth analysis of the case you debated during the competition. You will find a format for this analysis in the Engineering Ethics Bowl: Follow-up In-Depth Case Analysis module, m13759.
- Finally, what did you learn while working together as a team? What kind of cooperative problems developed? How did you solve them? Did they correspond to the problems raised by the "Ethics of Team Work" module or were they different? In fact, go back over that module and see how well it prepared you for the issues that arose as you interacted with your team.

Alternate or optional activities related to this EAC module

Uploaded below are suggested or optional assessment activities for students to carry out.

Muddiest Point Assessment Activity - This assessment activity provides a global of the strongest and weakest points of the Professional Ethics Bowl: [MP.doc](#)

Module Assessment Form - This assessment form has been adapted from one disseminated by Michael Davis in the Illinois Institute of Technology Ethics Across the Curriculum Workshops. It provides a global assessment of a given module: [MAP.doc](#)

Module-Background Information

Information about the source or history of this module that may be interesting for students or instructors.

[This link](#) will take you to the official home of the Intercollegiate Ethics Bowl. It appears as a part of the web page of the Center for the Study of Ethics in the Professions at the Illinois Institute of Technology.

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CHAPTER OVERVIEW

6: Course Procedures

[6.1: Rubrics for Exams and Group Projects in Ethics](#)

[6.2: Realizing Responsibility Through Class Participation](#)

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6.1: Rubrics for Exams and Group Projects in Ethics

Introduction

This module provides a range of assessment rubrics used in classes on engineering and computer ethics. Rubrics will help you understand the standards that will be used to assess your writing in essay exams and group projects. They also help your instructor stay focused on the same set of standards when assessing the work of the class. Each rubric describes what counts as exceptional writing, writing that meets expectations, and writing that falls short of expectations in a series of explicit ways. The midterm rubrics break this down for each question. The final project rubrics describe the major parts of the assignment and then break down each part according to exceptional, adequate, and less than adequate. These rubrics will help you to understand what is expected of you as you carry out the assignment, provide a useful study guide for the activity, and familiarize you with how your instructor has assessed your work.

Course Syllabi

Syllabus for Environments of the Organization: [ADMI4016_F10.docx](#)

Syllabus for Business, Society, and Government: [ADMI6055_F10.docx](#)

Syllabus for Business Ethics: [Business Ethics Spring 2007.doc](#)

Syllabus for Business Ethics, Spring 2008: [Syllabus_S08_W97.doc](#)

Business Ethics Syllabus Presentation - This presentation was given on the first day of class in Business Ethics, Fall 2007. It summarizes the course objectives, grading events, and also provides a PowerPoint slide of the College of Business Administration's Statement of Values: [BE_Intro_F07.ppt](#)

Rubrics Used in Connexions Modules Published by Author

Ethical Theory Rubric

This first rubric assesses essays that seek to integrate ethical theory into problem-solving. It looks at a rights-based approach consistent with deontology, a consequentialist approach consistent with utilitarianism, and virtue ethics. The overall context is a question presenting a decision scenario followed by possible solutions. The point of the essay is to evaluate a solution in terms of a given ethical theory.

Ethical Theory Integration Rubric - This rubric breaks down the assessment of an essay designed to integrate the ethical theories of deontology, utilitarianism, and virtue into a decision-making scenario: [EE_Midterm_S05_Rubric.doc](#)

Decision-Making / Problem-Solving Rubric

This next rubric assesses essays that integrate ethical considerations into decision making by means of three tests, reversibility, harm/beneficence, and public identification. The tests can be used as guides in designing ethical solutions or they can be used to evaluate decision alternatives to the problem raised in an ethics case or scenario. Each theory partially encapsulates an ethical approach: reversibility encapsulates deontology, harm/beneficence utilitarianism, and public identification virtue ethics. The rubric provides students with pitfalls associated with using each test and also assesses their set up of the test, i.e., how well they build a context for analysis.

Integrating Ethics into Decision-Making through Ethics Tests - Attached is a rubric in MSWord that assesses essays that seek to integrate ethical considerations into decision-making by means of the ethics tests of reversibility, harm/beneficence, and public identification: [CE_Rubric_S06.doc](#)

Ethics Bowl Follow-Up Exercise Rubric

Student teams in Engineering Ethics at UPRM compete in two Ethics Bowls where they are required to make a decision or defend an ethical stance evoked by a case study. Following the Ethics Bowl, each group is responsible for preparing an in-depth case analysis on one of the two cases they debated in the competition. The following rubric identifies ten components of this assignment, assigns points to each, and provides feedback on what is less than adequate, adequate, and exceptional. This rubric has been used for several years to evaluate these group projects.

In-Depth Case Analysis Rubric - This rubric will be used to assess a final, group written, in-depth case analysis. It includes the three frameworks referenced in the supplemental link provided above: [EE_FinalRubric_S06.doc](#)

Rubric for Good Computing / Social Impact Statements Reports

This rubric provides assessment criteria for the Good Computing Report activity that is based on the Social Impact Statement Analysis described by Chuck Huff at www.computingcases.org. (See link) Students take a major computing system, construct the socio-technical system which forms its context, and look for potential problems that stem from value mismatches between the computing system and its surrounding socio-technical context. The rubric characterizes less than adequate, adequate, and exceptional student Good Computing Reports.

Good Computing Report Rubric - This figure provides the rubric used to assess Good Computing Reports in Computer Ethics classes: [CE_FinalRubric_S06.doc](#)

[Computing Cases](#) provides a description of a Social Impact Statement report that is closely related to the Good Computing Report. Value material can be accessed by looking at the components of a Socio-Technical System and how to construct a Socio-Technical System Analysis.

Business Ethics Midterm Rubric Spring 2008 - Clicking on this link will open the rubric for the business ethics midterm exam for spring 2008: [Midterm Rubric Spring 2008.doc](#)

Study Materials for Business Ethics

This section provides models for those who would find the Jeopardy game format useful for helping students learn concepts in business ethics and the environments of the organization. It incorporates material from modules in the Business Course and from Business Ethics and Society, a textbook written by Anne Lawrence and James Weber and published by McGraw-Hill. Thanks to elainefitzgerald.com for the Jeopardy template.

Jeopardy: Business Concepts and Frameworks: [Jeopardy1Template.pptx](#) [Jeopardy2.pptx](#)

Privacy, Property, Free Speech, Responsibility: [Jeopardy_3.pptx](#)

Jeopardy for EO Second Exam: [Jeopardy4a.pptx](#)

Jeopardy 5: [Jeopardy5.pptx](#)

Jeopardy 6: [Jeopardy6.pptx](#)

Jeopardy 7: [Jeopardy7.pptx](#)

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6.2: Realizing Responsibility Through Class Participation

Module Introduction

Class attendance is a normal part of every college course. In the past, attendance was left up to the individual student. Now universities, adopting the responsibility of being local parents, require that teachers monitor class attendance closely by taking attendance each class and reporting students who are chronically absent. This makes use of what are termed "compliance systems": minimum standards of acceptable attendance are established and communicated to students, behavior is regularly monitored, and non-compliance is punished. In compliance approaches, the focus is placed on maintaining the minimum level of behavior necessary to avoid punishment. But this leaves unmentioned higher levels and standards of conduct. Students who miss more than X number of classes are punished by having points subtracted from their overall grade. But what constitutes outstanding attendance or, more positively, excellent participation? This module uses class attendance as an occasion to teach the different concepts of moral responsibility. After outlining blame responsibility and excuse-making, it explores responsibility as a virtue or excellence. Being absent creates its own responsibilities (1) to the teacher (you are responsible for finding out the material covered and learning it on your own), (2) to your classmates (what did your class group do in your absence and how will you reintegrate yourself into the group as an equal participant), and (3) to yourself (what habits will you change to improve your participation in class).

Where excuses come from

Understanding Morally Legitimate Excuses:

- The table below lists characteristics of what ethicists call "capacity responsibility." These conditions--presented by F.H. Bradley--describe when we can associate an agent with an action for the purposes of moral evaluation. They consist of (1) self-sameness, (2) moral sense, and (3) ownership.
- Self-sameness bases responsibility on the ability to maintain an identity over time; you must be the same person at the moment of accountability that you were when you performed the action. You cannot be blamed for actions performed by somebody else. So Jorge cannot be blamed for classes missed by Jose. Your professor should be held responsible for taking accurate attendance and not marking you absent when you are actually in class.
- The moral sense condition requires that you have the capacity to appreciate and comply with moral directives. This includes certain perceptual sensitivities (the ability to recognize elements of a situation that are morally relevant), emotional responses (that you respond to moral elements with the appropriate emotion), and the ability to shape action in accordance with moral standards. Those who lack moral sense, whether temporarily as with children or because of psychological limitations as with psychopaths are non-responsible rather than guilty or innocent. They simply lack the general capacity to be held accountable.
- Ownership gets down to the specifics of a given situation. Did factors in the situation compel you to miss class? Did you miss class because you lacked certain crucial bits of knowledge? Why were you unable to attend class and can this "why" be translated into a morally legitimate excuse. In excusing an action, you "disown" it. There are three ways to do this: a) by showing unavoidable and conflicting obligations, b) by pointing to compelling circumstances, or c) by citing excusable ignorance.
- Formally defined, compulsion is the production in an individual of a state of mind or body against the actual will. Sickness is a state of mind and body that could compel you to stay at home even though you want to come to class and take the test. Having a flat tire on the way to school could also produce a state of body (being stuck at the side of the road) against actual will (driving to class in order to take the test). With compulsion, the key test is whether the compelling circumstances were under your control. Did your tire go flat because you postponed getting a new set of tires, even when it was clear that you needed them? Are you sick and in bed now because you overdid it at the party last night? If the compelling circumstances resulted from actions that you performed voluntarily in the past, then you are still responsible.
- You also need to have the knowledge necessary to act responsibly in a given situation. Imagine that your class was being taught by a professor who claimed to be a CIA agent. He would repeatedly change the times and locations of class meetings at the last minute to keep from being discovered by enemy spies. Not knowing where (or when) the next class would be held would make it impossible to attend. Here you would get off the hook for missing class because of excusable ignorance. But suppose changes in the class schedule were announced during class by the professor, but you were absent on that day. You are now responsible for your ignorance because you should have found out what was covered while you were absent in the past. In other words, your ignorance in the present was caused by your neglecting to find things out in the past. You are responsible because voluntary actions in the past (and inaction) caused the state of ignorance in the present.

- The table below provides sample excuses given by students for absences. These are correlated with conditions of capacity responsibility such as ignorance and compulsion. Correlating excuses with conditions of imputability is one thing. Validating them is something else, and none of these excuses have been validated.
- Here are some more typical excuses offered by students for missing class. Try correlating them with the conditions of imputability to which they tacitly appeal: (1) I missed your class because I needed the time for studying for a test in another class. (2) I missed class because the electricity went out during the night and my electric alarm clock didn't go off on time. (3) I planned on going to class but got called into work at the last minute by my boss. In all these cases, you have missed class and have a reason. Can your reason be correlated with ignorance or compulsion? Were you negligent, careless, or reckless in allowing these conditions of ignorance and compulsion to develop?
- Excuses (and blame) emerge out of a nuanced process of negotiation. Much depends on trust. Your professor might excuse you for missing a class at the end of the semester if your attendance up to that point had been exemplary. He could, on this basis, treat the absence as an exception to an otherwise exemplary pattern of attendance and participation.
- But you may have trouble getting off the hook this time, if there have been several previous absences, because the new absence falls into a pattern of poor participation accompanied by lame excuses. Excuse negotiation (and blame responsibility) occur over the background of other values such as trust and honesty.

Retroactive Responsibility Table Correlation of condition of imputability with common excuses.

Retroactive Responsibility	Excuse	Excuse Statement (Some Examples)
	1. Conflicts within a role responsibility and between different role responsibilities.	I have a special project due in another class and finishing it conflicts with attending your class.
	2. Overly determining situational constraints: conflicting interests.	I am interviewing for a position after I graduate, and I must be off the island for a few days.
	3. Overly determining situational constraints: resource constraints	My car had a flat tire. My babysitter couldn't come so I had to stay home with my child. My alarm clock didn't go off because of a power outage.
	4. Knowledge limitations	The class was rescheduled, and I was unaware of the change.
	5. Knowledge limitations	I didn't know the assignment for class so I came unprepared. (Not an excuse for missing class)

Exercise 1: Provide a Morally Justifiable Excuse for Missing Class

- Offer an honest and responsible ethical assessment of the reason you were unable to carry out your role responsibility for coming to class. Note that the default here is attending class and any departure from the default (i.e., missing class) requires a moral justification
- Begin by examining whether your action can be classified as an excuse arising out of compulsion or ignorance
- Your absence may not be morally excusable. In this case, you cannot excuse your absence but still must explain it
- Remember that, following Aristotle, you must show that your action was done under and because of compulsion or under and because of ignorance. In other words, you must show that it did not arise from past negligence or recklessness

Proactive/Prospective Responsibility

Principle of Responsive Adjustment:

- Responsibility for both good and bad things often emerges as a pattern exhibited by a series of actions. If you miss one class after establishing a pattern of good attendance and active participation, then your teacher will look for something exceptional that prevented you from doing what you habitually do. But if one absence falls into a series with other absences, then this reveals a pattern and your teacher begins to classify you as someone who is chronically absent.

- So, it is not enough to offer a moral excuse to get "off the hook" for your absence. Expressing remorse, guilt, and regret does not substitute for taking active measures to avoid repeating the wrongful act. These changes or responsive adjustments clue others into whether you have learned from your past mistakes. What happened in the past was bad and you regret it; but are you willing to make the necessary changes in your future conduct to avoid repetition of the bad act?
- This is expressed by the "Principle of Responsive Adjustment" (or PRA). Stated negatively, failure to take measures to prevent past excusable wrongs from reoccurring in the future leads to a reevaluation of these past actions. Failure to responsively adjust shows that the past action belongs to context of similar bad actions indicating a bad habit or bad character. This, in turn, leads to a reevaluation of the past act; what when taken in isolation was not blameworthy becomes blameworthy when inserted into this broader context. Showing an unwillingness to learn from the past betrays entrenched attitudes of negligence, carelessness, or recklessness. (See Peter A. French, Corporate and Collective Responsibility)

Responsibility as a Virtue:

- Responsibility can be reconfigured as a virtue or excellence
 - The table below describes the characteristics of a preventive stance where we begin by identifying potential wrongs and harms. Once we identify these then we take serious measures to prevent them from occurring
 - Finally, responsibility as a virtue opens up the horizon of the exemplary. Pursuing excellence requires our identifying opportunities to go beyond preventing harm to realizing value
 - In this context, class attendance becomes class participation. As was said in the introduction, missing a class creates a series of new tasks that arise out of your commitment to excellence in participation. These include the following:
1. What was covered while you were absent? Or better, if you know in advance that you are going to miss a class, what will be covered? How can you cover this material on your own? What can you do, proactively, to stay with the class during your absence?
 2. How will your absence impact the rest of the class (especially those in your class group), and what can you do to minimize any harmful effects? Here you should notify your team members that you are going to miss class and develop plans for maintaining your equal participation in the group and class during and after your absence
 3. In accordance with the Principle of Responsive Adjustment, what changes are you making to avoid absences in the future or--putting it as positively as possible--to achieve a level of excellence in class participation?
 4. Note how all these items focus on improvement or betterment rather than "making up." As Dewey recognizes, the real function of moral responsibility is to take the lessons we learn from the past and use them to improve ourselves

Responsibility as a Virtue or Proactive Responsibility	Characteristic	Proactive Response
	Diffuse blame avoidance strategies	Avoid trying to diffuse the blame for missing class on some other person or situation. For example, "I couldn't come to class because I had a project due in another class" is not a morally legitimate excuse because it places the blame on the other class. You have not taken responsibility for your absence.
	Design responsibilities with overlapping domains	If you fail to participate in a group activity, describe the group's "Plan B," i.e., how they worked around your absence.
	Extend the scope and depth of knowledge.	Describe how you found out what was covered in class and document how you have learned this material
	Extend power and control	Describe the measures you have taken to eliminate the "responsibility gap" between you and your workgroup. For example, how did you "make up" for not participating in the activity held in the class you missed.
	Adopt a proactive problem solving/preventive approach for the future	Describe what measures you have taken to avoid missing classes in the future.

Guidelines for Avoiding Absences:

1. Build redundancy into your schedule. Many students develop schedules that are "tightly-coupled." This means that failures or breakdowns cannot be isolated; they tend to flow over into other areas producing a cascading disaster. A co-worker calls in sick, and your boss calls you in during the time you have a class. You miss one class and fail to study for another. (The time you set aside for study has been taken up by this unexpected job demand.) You have been working so hard to catch up that you catch a cold. Now everything becomes that much harder because you are not working to full capacity. The lesson here is to set up your schedule from the beginning with a certain amount of flexibility built-in. This could be as simple as taking four instead of five classes or working 10 instead of 20 hours per week.
2. Look for incentives or motives to come to class. One important incentive is that you may get a better grade. Teachers tend to know students who come to class better; they consider them more responsible and more committed.
3. Get proactive when you return. Instead of asking the professor, "Did we do anything important while I was absent?" consult the syllabus and a classmate to find out what you missed. Then check your understanding with the professor. "My understanding is that you discussed moral responsibility with the class and applied the framework to a case. Is this correct?" Instead of asking the professor, "What should I do to make up for what I missed?" come with your own plan. Show that you have taken responsibility for your absence by getting proactive and planning the future around realizing value.
4. Absences have an impact on your fellow students as much as on you or your instructor. If you are working in groups, find out from your peers what was covered. If your group is depending on your completing a task for the class you are missing, try to develop a "work-around." ("I won't be in class tomorrow but I am sending you my part of the group assignment via email attachment.") Let your team know what is happening with you and make sure that you keep up on all your commitment and responsibilities to the group.

Exercise 2: Getting Proactive about your absence

- Develop a plan for "getting back into the loop." What are you going to do to cover the material and activities you have missed?
- Get Preventive. Describe what you are going to do now to avoid absences in the future
- Shoot for the ideal. What can you do--above and beyond class attendance--to realize exemplary participation in your ethics class

Conclusion

Exercise #3: Getting and Staying Honest

- Below is a template that you need to duplicate, fill out, and place in the class attendance file that will be on the desk in front of the class
- Duplicate and sign the honesty pledge at the end of this module
- Students often wish to provide evidence documenting their claims regarding their absences. You may do this, but remember that this is neither required nor in the spirit of prospective responsibility
- Furthermore, be aware that you are not to provide confidential information such as personal health information or student ID numbers or social security numbers. Health issues are to be referred to generically by saying something like, “I was unable to come to class Tuesday because of health reasons”

1. Class missed (day of week and date):
2. The material covered during class:
3. Reason for missing class (please do not provide confidential information):
4. Action plan for absence: how you intend to take responsibility for the material covered while you were absent; how you intend to make reparations to your group for not participating in group learning activities for the class you missed:
5. How you plan to avoid absences in the future:

Honesty Pledge

- To realize the value of honesty, you will make the following affirmation:
- The information I have provided above is truthful, the excuses I have enumerated rigorously examined from a moral point of view, and the responsive commitments I have made above are serious, and I will take active and realistic efforts to carry them out.

Signature: _____

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Index

Glossary

Sample Word 1 | Sample Definition 1

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 - [5.1: Ethics Bowl Rules and Procedures](#) - [CC BY 3.0](#)
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 - [6: Course Procedures](#) - [CC BY 3.0](#)
 - [6.1: Rubrics for Exams and Group Projects in Ethics](#) - [CC BY 3.0](#)
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 - [Index](#) - [Undeclared](#)
 - [Glossary](#) - [Undeclared](#)
 - [Detailed Licensing](#) - [Undeclared](#)